



ALPHA

User Manual

3rd version 2012, publication date August 2012

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This manual is the original documentation for the ALPHA spectrometer. The ALPHA spectrometer is covered by one or more of the following patents: DE 102004025448; DE 19940981; US 5923422; DE 19704598. Other US and international patents are pending.

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1 Introduction

1.1 About this manual

This manual is a complete documentation about the ALPHA spectrometer.

Depending on the degree of hazard important safety information and safety instructions are classified as follows:



DANGER

- Indicates a hazardous situation which, if not avoided, will result in death or serious (possibly irreversible) injury.



WARNING

- Indicates a hazardous situation which, if not avoided, could result in death or serious (possibly irreversible) injury.



CAUTION

- Indicates a hazardous situation which, if not avoided, may result in minor or moderate (reversible) injury or major material damage.

NOTE

- Hazard, which could result in material damage if the appropriate safety instructions are not observed.



Information given to the user to facilitate spectrometer operation, and make the best possible use of the spectrometer.

1.2 Terms

This manual uses both the term *spectrometer* or *ALPHA* when the spectrometer is described.

1.3 Safety

Always observe the instructions described in this manual to ensure user safety and to avoid property damage. Keep this manual for further reference available at any time.

Improper use or failure to follow the safety instructions can result in serious injuries and/or property damage. Any non-observance infringes the intended use (i.e. spectroscopic measurements) of the spectrometer. In this case Bruker Optik GmbH does not assume any liability.

It is the operator's duty to plan and implement all necessary safety measures and to supervise their observance. Moreover, the operator must ensure that the spectrometer is in proper condition and fully functioning. A safe and faultless operation can only be guaranteed if the spectrometer is transported, stored, installed, operated and maintained properly according to the procedures and instructions described in this manual.

Never remove or deactivate any supporting safety systems during spectrometer operation. Objects and/or material not required for operation must not be kept near the spectrometer operating area.

The spectrometer has been developed according to the EN 61010-1:2010 (IEC 61010-1:2010+Cor.:2011) safety regulations for electrical measuring, control and laboratory devices.

1.3.1 Warning labels

When operating the spectrometer you have to observe a number of safety instructions which are highlighted by the appropriate warning label. The warning labels and their meaning are described in the following.

All warning labels on the spectrometer must always be kept legible. Immediately replace worn or damaged labels.

Label	Definition
	General Hazard: This warning symbol indicates general hazard. Observe the safety instructions and follow the precautions described to avoid personal injury and/or property damage.

Table 1.1: Warning labels

Label	Definition
	Hot Surface: This warning symbol refers to components and surfaces which can become very hot during spectrometer operation. Do not touch these components and surfaces. Risk of skin burn! Be careful when operating near hot components and/or surfaces.
	Laser Radiation: This warning symbol indicates the existence of laser radiation. Never look directly into the laser beam, or use any kind of optical instruments to look into the beam as this may cause permanent eye damage.

Table 1.1: Warning labels

1.3.1.1 In case of hazardous sample material

Besides the dangers described above, there can also be hazards caused by the sample material. Depending on the type of hazardous substances used, you have to observe the specific substance-relevant safety instructions. Put on the specific warning label on the corresponding module position. The label must be legible and permanently discernible.

The following list exemplifies types of hazardous sample material:

Label	Definition
	Infectious Material This warning symbol indicates the possible existence of biologically dangerous and infectious material. When working with this kind of material always observe the prevailing laboratory safety regulations and take necessary precautions and disinfection measures (e.g. wearing protective clothing, masks, gloves etc.). Non-observance may cause severe personal injury or even death. For information on how to use, dilute and efficiently apply disinfectants, refer to the Laboratory Biosafety Manual: 1984 by WHO - World Health Organization.
	Radioactive Material This warning symbol indicates the possible existence of radioactivity. When working with radioactive material always observe the safety regulations and take necessary protective measures. Wear protective clothing, e.g. masks and gloves. Non-observance may cause severe personal injury or even death.

Table 1.2: Warning labels in case of hazardous sample material

Label	Definition
	Corrosive Substances This warning symbol indicates the possible existence of corrosive substances. When working with corrosive substances always observe the laboratory safety regulations, and take protective measures (e.g. wear protective masks and gloves). Non-observance may cause severe personal injury or even death.

Table 1.2: Warning labels in case of hazardous sample material

1.3.2 Waste disposal

Dispose all waste produced (chemicals, infectious and radioactively contaminated substances etc.) according to the prevailing laboratory regulations. Detergents and cleaning agents must be disposed according to the special waste regulations.

1.4 General information

1.4.1 Protective earthing

To avoid personal injuries and/or damage caused by electrical power, the supplied spectrometer power cord is equipped with a safety plug. Connect this safety plug only to an earthed power socket. Make sure that the earthed power socket used complies with IEC (International Electrotechnical Commission¹).

1.4.2 Qualified personnel

Initial installation and all maintenance and repair work not described in this manual should only be performed by Bruker service personnel. Make sure that the spectrometer is only operated and maintained (i.e. only maintenance work that is described in this manual) by authorized operating personnel trained in the spectrometer operation and all relevant safety aspects.

All repairs, adjustments and alignments on any spectrometer component must be performed in accordance with the safety regulations and standards applied in the country where the instrument will be installed.

1. International standards organization for electrical and electronic-related technologies.

1.4.3 Intended use

The spectrometer and its components should only be used according to the instructions described in the manual or advised by a Bruker engineer.

In case of accessories or components made by other manufacturers and used in connection with the spectrometer, Bruker Optik GmbH does not assume any liability for safe operation and proper functioning.

1.5 Questions and concerns

If you have questions or concerns about safety, operating the spectrometer, or if you need assistance with software problems or replacement parts, contact Bruker at the numbers listed below:

- Service hotline hardware: +49 (0) 72 43 504-2020
- Service hotline software: +49 (0) 7243 504-2030
- Fax: +49 (0) 72 43 504-2100
- E-mail: service@brukeroptics.com
- Internet: www.brukeroptics.com

2 ALPHA spectrometer

The ALPHA spectrometer is a compact FT-IR spectrometer, with modular design for routine applications in the laboratory.

ALPHA consists of a basic module and an exchangeable sampling module.

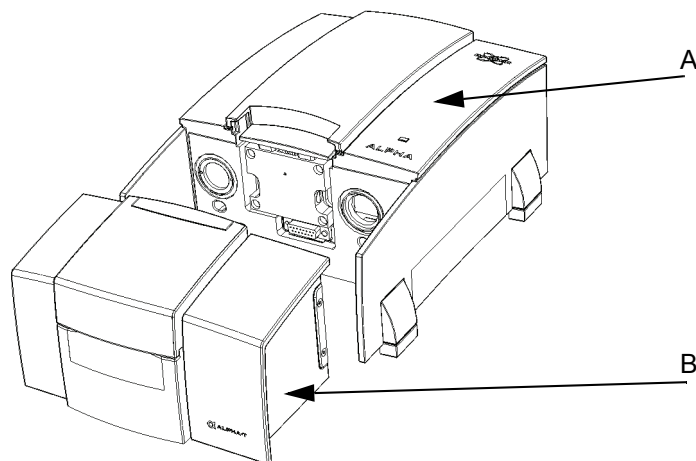


Figure 2.1: Modular spectrometer design

	Definition
A	Basic module
B	Sampling module

Table 2.1: Spectrometer modules

2.1 Basic module

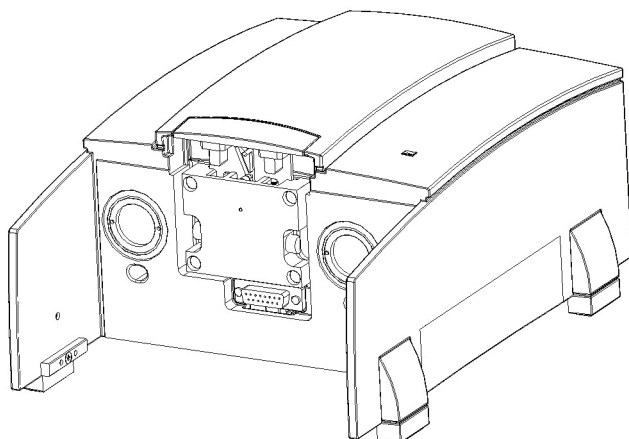


Figure 2.2: ALPHA - Basic module

The basic module is equipped with the following optical components:

Optical component	Specification
IR source	• air cooled • low voltage • lifetime: 44.000 operating hours
Interferometer	• Rocksolid • permanently adjusted
Beamsplitter	• KBr (standard) • ZnSe (optional <i>High Humidity</i> variant)
Window material	• KBr (standard) • ZnSe (optional <i>High Humidity</i> variant)
Detector	DTGS ^a , room temperature

Table 2.2: ALPHA - Optical components

a. DTGS: deuterated triglycine sulphate

2.1.1 Connection ports and LEDs

The following are located on the rear side of the basic module:

- connection ports for power supply and Ethernet
- system diagnostics LEDs

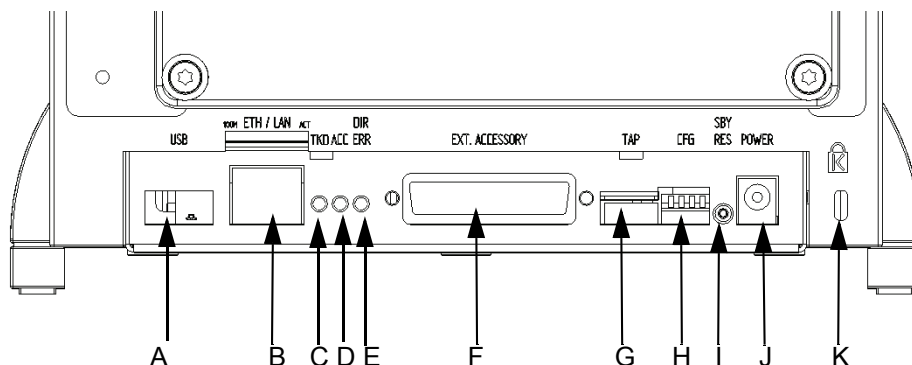


Figure 2.3: Basic module- Connection ports and LEDs

	Type	Definition/Function
A	USB port	5 V power supply, e.g. for WLAN adapter
B	ETH/LAN port	➤ Port to connect the spectrometer to the PC i Use the supplied cross-over data cable, 100Base-T with RJ45 ports. See also the ALPHA installation instructions.
C	TKD LED	➤ If the TKD LED is green, the interferometer mirror is within the data acquisition range. ➤ During data acquisition the light intensity changes to bright green. i The TKD abbreviation means <i>Take Data</i>.
D	ACC LED	➤ If the ACC LED is yellow, the network accesses the optics. ➤ During data transfer the LED will be light yellow. i The ACC abbreviation means <i>Access</i>.

Table 2.3: Basic module - Connection ports and LEDs and their functions

	Type	Definition/Function
E	DIR/ERR LED	➤ If the DIR/ERR LED flashes slightly red in normal operation, the scanner moves. ➤ If the LED is permanently bright red, there must be an interferometer problem, e.g. missing laser signal. i As long as this LED is permanently red no data acquisition is possible. The DIR abbreviation means <i>Direction</i>, ERR means <i>Error</i>.
F	Ext. Accessory port	Port to connect spectrometer to external optional accessories, e.g. detector, remote trigger etc.
G	TAP port	Male connector with 10 socket contacts for service and diagnostics i The TAP abbreviation means <i>Test Access Port</i>.
H	CFG switch	Fourfold DIP switch, the following lever positions are possible: • #1 ON and #2 OFF: DHCP mode - use last DHCP address • #1 OFF and #2 ON: use ON-IP address 10.10.0.1 • #3: stay in boot loader, do not load the application program • #4: reserved i The CFG abbreviation means <i>Configuration</i>.
I	SBY/RES button	➤ Pressing this button about 2 seconds, activates/deactivates the stand-by mode. ➤ Pressing this button more than 6 seconds, resets the spectrometer. i The abbreviation SBY means <i>Standby</i>, RES means <i>Reset</i>.
J	POWER port	Port to connect the spectrometer to power supply i Use the supplied power cord with mains adapter. The POWER port is suited for extra low-voltage connectors. Only use power cords which have been approved by Bruker. See also the ALPHA installation instructions.

Table 2.3: Basic module - Connection ports and LEDs and their functions

	Type	Definition/Function
K	Kensington lock	Optional safety (anti-theft) lock for mobile hardware

Table 2.3: Basic module - Connection ports and LEDs and their functions

2.1.2 Stand-by mode

The spectrometer can be set into stand-by mode by pressing the SBY/RES button. The button is located on the rear spectrometer side (chapter 2.1.1).

i If the spectrometer is in stand-by mode, it is not switched off completely. There is still some power consumption which is, however, significantly lower than in operation mode as the IR source, laser and detector are switched off.

2.1.2.1 Activating

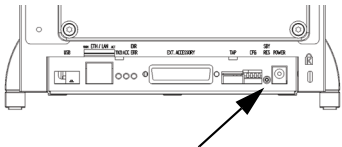
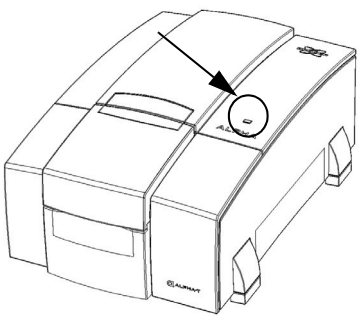
1		Press the SBY/RES button for a short time (about 1 to 2 seconds). ➤ The button is located at the spectrometer rear side.
2		• The spectrometer beeps. • The green status indicator on the spectrometer top side (marking in figure) begins to flash in short intervals. • The status light on the OPUS spectroscopy software interface becomes gray.

Table 2.4: Activating stand-by mode

2.1.2.2 Deactivating

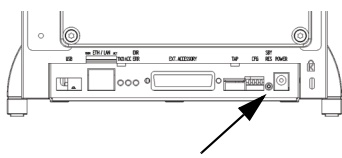
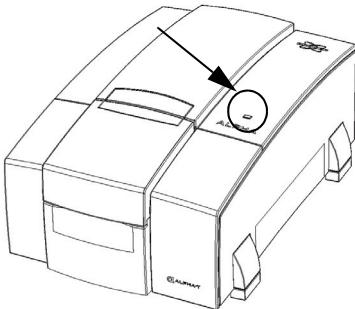
1		Press the SBY/RES button for a short time (about 1 to 2 seconds). ➤ The button is located at the spectrometer rear side.
2		• The spectrometer beeps. ➤ Laser, IR source and detector are switched on again. • The status indicator on the spectrometer top side (marking in figure) and on the software interface are yellow. ➤ Reason: The IR source needs about 7 minutes to reach its operating temperature.
i As soon as the IR source has reached its operating temperature, the status indicator on the spectrometer top side and on the specktroscopy software interface is permanently green.		

Table 2.5: Deactivating stand-by mode

2.2 Exchanging sampling module

The different types of ALPHA sampling modules can be attached and detached from the basic module. The procedure is the same for all types of sampling modules.

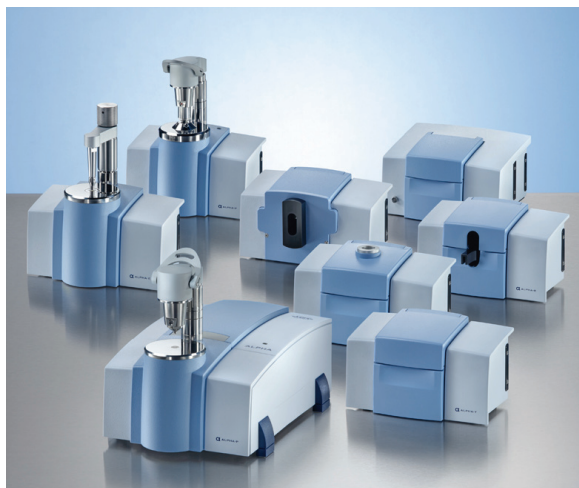


Figure 2.4: Different ALPHA sampling modules

2.2.1 Procedure

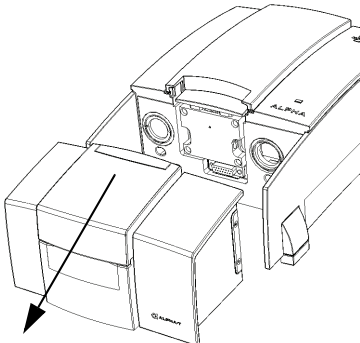
1		Press the lock, located on the basic module, right down. ➤ The sampling module is released from the basic module. The lock stops at half-height.
2		Detach the sampling module from the basic module.

Table 2.6: Exchanging sampling module

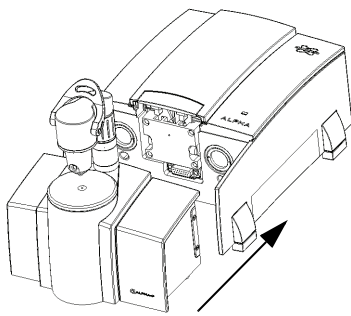
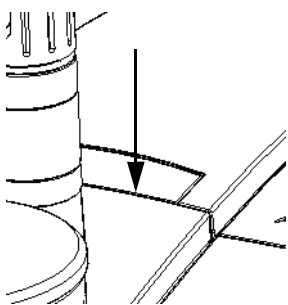
3		Push the new sampling module right towards the basic module.
4		Press the lock on the basic module once again right down. ➤ The sampling module engages into the basic module. The lock snaps right up.

Table 2.6: Exchanging sampling module

2.2.2 After exchanging the sampling module

After exchanging the sampling module, a calibration is performed. The calibration is indicated by a yellow message bubble on the bottom right end of the software interface.

If the calibration has finished, the performance test starts (chapter 7.2.1). A yellow bubble appears on the bottom right of the OPUS graphical interface, and contains an appropriate message.

2.3 Spectrometer configuration

2.3.1 ALPHA for transmission measurements

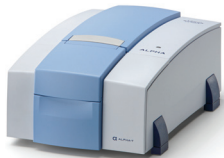
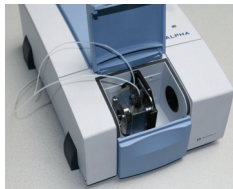
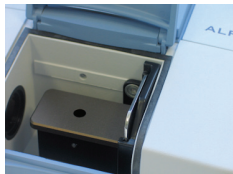
ALPHA		Sample type
ALPHA-T (Universal Sampling Module)		• Solids • Liquids • Gases
ALPHA-T with liquid cells		Liquids
ALPHA-T with gas cells (different types of path lengths possible)		Gases
ALPHA-T with 30° reflection accessory		Solids

Table 2.7: ALPHA for transmission measurements

2.3.2 ALPHA for ATR measurements

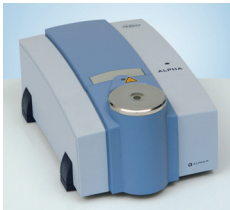
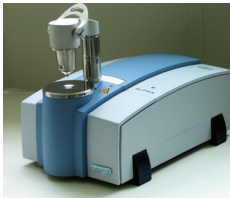
ALPHA		Sample type
ALPHA-E (Eco ATR module)		• Solids • Liquids
ALPHA-P (Platinum ATR module)		• Solids • Liquids
ALPHA-P (Platinum ATR module, high pressure variant)		• Solids • Liquids
ALPHA-P (Platinum ATR module, heatable, without pressure applicator)		Liquids
ALPHA-P (Platinum ATR module, heatable, with pressure applicator)		• Solids • Liquids

Table 2.8: ALPHA for ATR measurements

2.3.3 ALPHA for reflection measurements

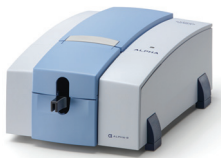
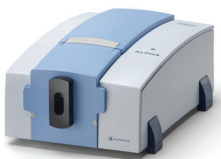
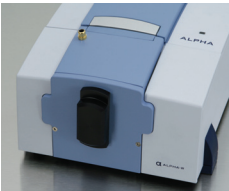
ALPHA		Sample type
ALPHA-R (Drift module, for diffuse reflection)		Solids
ALPHA-R (A241/D, contactless reflection)		Solids
ALPHA-R (A241/DV, contactless, video-assisted reflection)		Solids
ALPHA-R (A240/DU, diffuse and specular reflection from the bottom upwards)		Solids

Table 2.9: ALPHA for reflection measurements

2.3.4 ALPHA for gas analysis

ALPHA		Sample type
ALPHA-G (A139/D, with long-path gas cell)		Gases

Table 2.10: ALPHA for gas analysis

ALPHA		Sample type
ALPHA-G (A139-H1 with long-path gas cell and heating jacket)		Gases

Table 2.10: ALPHA for gas analysis

2.4 Design

Due to its modular design many maintenance tasks, e.g. replacing IR source are easy to perform and can be achieved without any re-alignment of the optics. The consumables and electronic components are located in separate compartments and can therefore be replaced without interfering with the sealed, desiccated optics area. For further details on maintenance refer to chapter 9.

Full support of industry standard communication protocols makes the integration simple. A permanent on-line diagnostics of each spectrometer component facilitates troubleshooting and maintenance.

i Depending on the spectrometer configuration ordered, the spectrometer may not include all options that are described in this manual.

2.5 Applications

- Analysis of liquids and solids
- Analysis of gases
- Contactless and non-destructive analysis

2.6 Spectrometer housing

Under no circumstances is the operator allowed to open the sealed spectrometer housing.

2.7 Optics

The spectrometer incorporates state-of-the-art optics for outstanding sensitivity and stability, which enables

- high quality measurement results
- less down-time
- direct method transfer

The spectrometer uses a permanently aligned ROCKSOLID interferometer which ensures a high energy throughput and low polarization effects.

In the standard configuration the spectrometer is equipped with a room-temperature DTGS¹ detector.

1. DTGS: deuterated triglycine sulphate

2.8 Electronics

The electronics are based on a high speed 24 bit data sampling unit that guarantees experimental results with an outstanding accuracy. Any modern data system (PC workstation, laptop etc.) with the OPUS spectroscopic software installed can be used to control the instrument and perform data processing. The spectrometer is linked to the data system by a standard 100Base-T Ethernet connection, which allows the spectrometer to be integrated into an existing data network.

The spectrometer is completely software controlled. All components can be operated using the OPUS spectroscopic software. Diagnostic routines help to maintain optimum status and maximum instrument performance which is validated by the OPUS Validation Program (OVP) using the internal validation unit (IVU) which is also called aperture or filter wheel.

2.9 Spectroscopy software

The spectrometer is operated by the OPUS spectroscopy software. More detailed information on the OPUS software is described in the Quick User Guide in chapter 7 and in the OPUS/MENTOR manual.

3 ALPHA for transmission measurement

Measurement type:	transmission
Usable sample material:	• solids (films, KBr pellets) • liquids • gases
Optional measuring tools:	• liquid cell • gas cell • 30° reflection accessory

3.1 Overview on ALPHA spectrometer configurations for transmission

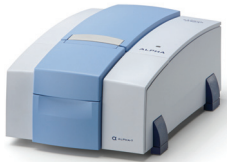
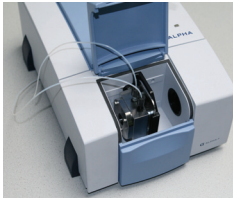
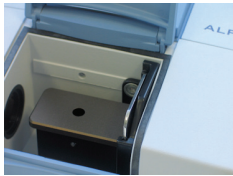
ALPHA spectrometer configurations for transmission	
	
ALPHA-T universal sampling module	ALPHA-T with liquid cells
	
ALPHA-T with gas cells (different types of path lengths possible)	ALPHA-T with 30° reflection accessory

Table 3.1: ALPHA spectrometer configurations for transmission

3.2 ALPHA-T universal sampling module

Measurement type:	transmission
Usable sample material:	• solids (films, KBr pellets) • liquids • gases

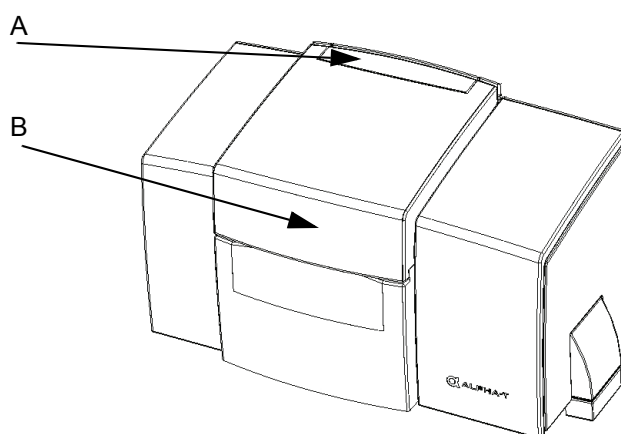


Figure 3.1: ALPHA-T universal sampling module

	Definition
A	Lock/release
B	Sample compartment lid

Table 3.2: ALPHA-T sampling module - Components

3.2.1 Specifications

Spectral range:	• 375 – 7,500 cm^{-1}, with standard KBr beamsplitter • 500 – 6,000 cm^{-1}, optionally with high humidity ZnSe optics
Spectral resolution:	• better than 2 cm^{-1} • optionally better than 0.8 cm^{-1}
Wavenumber accuracy:	better than 0.05 cm^{-1} @ 1,576 cm^{-1}

3.2.2 Optical path

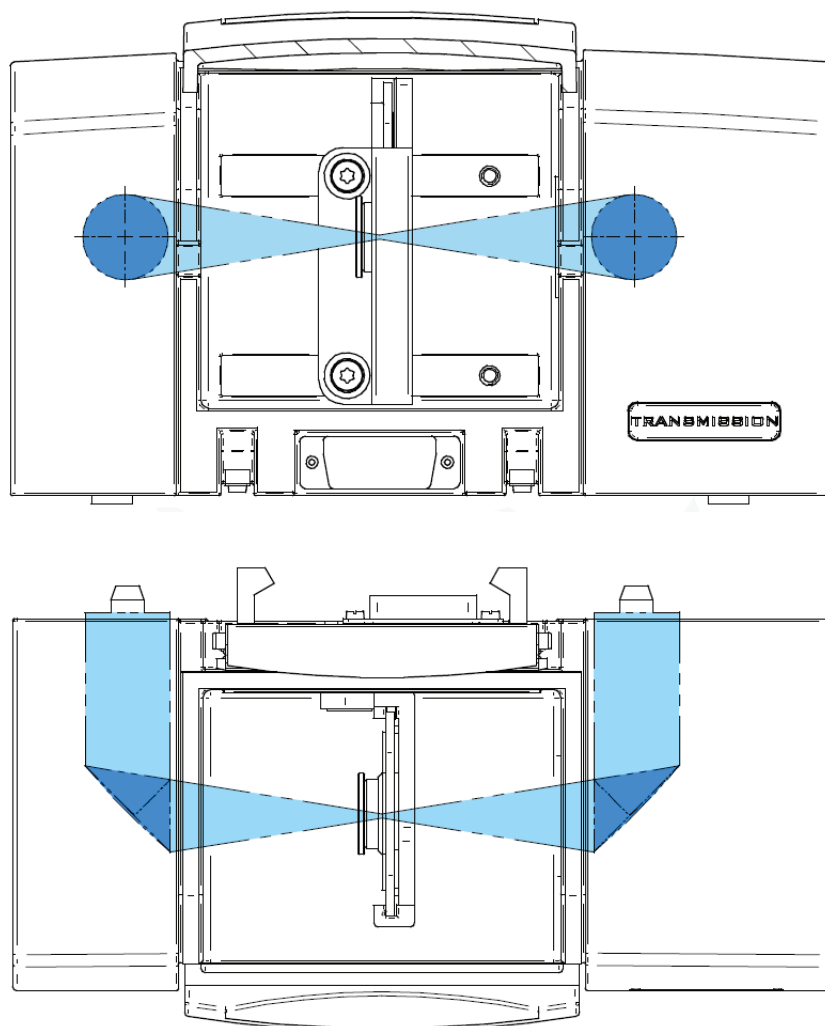


Figure 3.2: ALPHA-T - Optical path

3.2.3 Opening sample compartment

Opening the sample compartment is identical with all ALPHA-T spectrometer configurations.

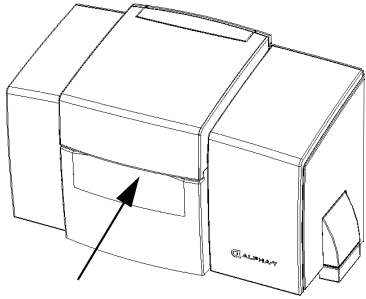
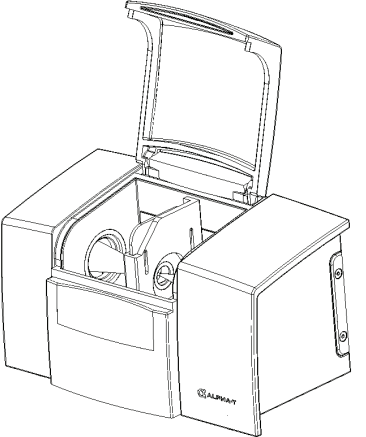
1		Reach under the bottom edge of the sample compartment lid.
2		Lift up the sample compartment lid.

Table 3.3: ALPHA-T sampling module - Opening sample compartment

3.2.4 Starting measurement

3.2.4.1 Loading measurement experiment

1. Measurement experiment file is loaded.
 - If AAR accessory recognition is activated (which is the case by default), the OPUS spectroscopy software recognizes the sampling module used. Thus, the experiment file¹ (TRANS.XPM) assigned to the sampling module is automatically loaded in OPUS. If the experiment file is not loaded automatically, perform the steps 2 et seq.
2. Start OPUS. On the *Measure* menu, select the *Setup Measurement Experiment* command.
 - The *Setup Measurement Experiment* dialog opens.
3. Click the *Basic* tab.
4. Click the *Load* button and select the *TRANS.XPM* experiment file from the dialog that opens.
5. Click the *Accept & Exit* button.

3.2.4.2 Starting background measurement

1. The spectrometer status indicator and the OPUS status light must be green.
2. Make sure that not any sample is available in the sample compartment.
3. Turn down the sample compartment lid.
4. On the OPUS *Measure* menu select the *Measurement* command.
 - The *Measure* dialog opens.
5. Click the *Start Background Measurement* button.
6. Wait until the background measurement has finished.

3.2.4.3 Starting sample measurement

1. Lift up the sample compartment lid.
2. Insert the sample.
3. Turn down the sample compartment lid.
4. On the OPUS *Measure* menu, select the *Setup Measurement Parameters* command.
5. On the OPUS *Measure* menu select the *Measurement* command.
 - The *Measure* dialog opens.
6. Click the *Start sample Measurement* button.
7. Wait until the sample measurement has finished.

1. The universal ALPHA-T sampling module is supplied with the *TRANS.XPM* experiment file. This experiment file contains the default parameters set by Bruker, to be used to perform transmission measurement.

3.2.5 Potential operating errors

Possible problems	Troubleshooting
Instrument/performance test not passed	• Check whether there is a sample or any other kind of object inside the sample compartment. • Remove sample or object from the sample compartment. • Start instrument/performance test again.

Table 3.4: Potential operating errors

3.3 Measuring accessories

Depending on the type of sample used (solid, liquid or gas) different types of measuring accessories are available to position the sample into the sample compartment of the sampling module:

Included in delivery content:	2 x 3" sample holder for KBr pellets (13 mm)
Optional measuring accessory:	• foil holder • liquid cell • gas cell • 30° reflection accessory

Table 3.5: ALPHA-T sampling module - Measuring accessory

3.3.1 Sample holder (standard)

By default, the sample holder is mounted in the middle of the sample compartment. On the left side of the sample holder there is a fixture to insert the sample.

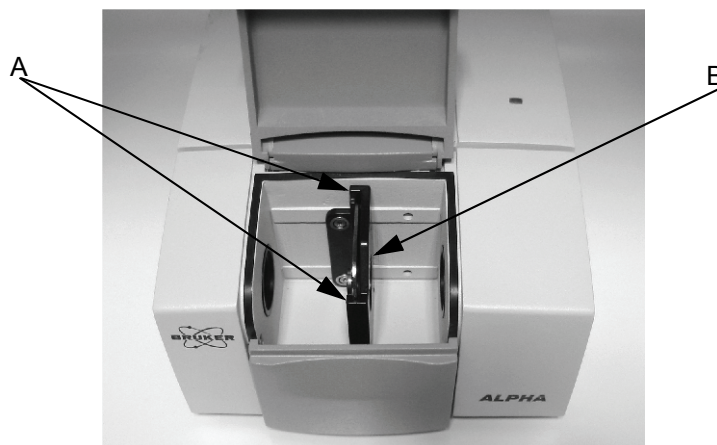


Figure 3.3: ALPHA-T sampling module with sample holder

	Definition
A	Fixture
B	Sample holder

Table 3.6: Sample holder - Components

3.3.2 Foil holder (option)

To perform spectroscopic analysis of foils you can fix the foils in a foil holder. To measure the sample is inserted into the foil holder. The foil holder is mounted in the sample compartment, by default.



Figure 3.4: Foil holder for ALPHA-T

3.3.3 Liquid cell (option)

To perform spectroscopic analysis of liquid samples, liquid cells are inserted into the sample holder. The sample holder is mounted in the sample compartment, by default.

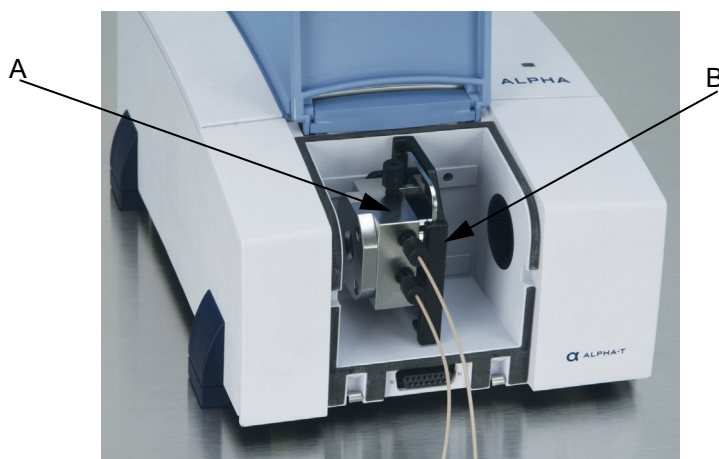


Figure 3.5: ALPHA-T sampling module with liquid cell

	Definition
A	Liquid cell
B	Sample holder

Table 3.7: Liquid cell - Components

When analyzing the sample in a liquid cell observe the following:

- Avoid any air bubbles when filling in the sample into the liquid cell.
- Thoroughly empty and clean the liquid cell after each measurement.
- Clean the liquid cell using an appropriate solvent.

i The solvent to be used for cleaning depends on the window material of the cell. Do not use water for cleaning in case of KBr windows. Do not use acidic solvents for cleaning in case of ZnSe windows.

3.3.4 Gas cell (option)

To perform spectroscopic analysis of gaseous samples the gas cell is inserted into the sample compartment together with a special gas cell holder.



Figure 3.6: Gas cell holder and gas cell (example)

	Definition
A	Gas cell holder
B	Gas cell

Table 3.8: Gas cell holder and gas cell for ALPHA-T sampling module

3.3.4.1 Displacing mounted sample holder

To insert the gas cell into the ALPHA-T sample compartment you may first have to displace the sample holder in the sample compartment.

Procedure

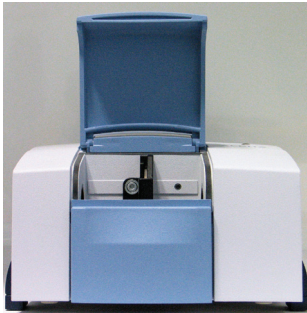
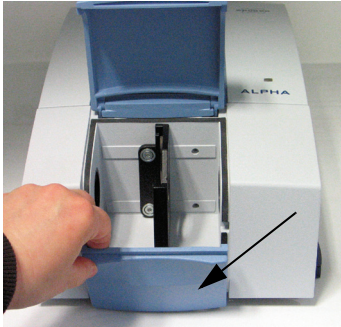
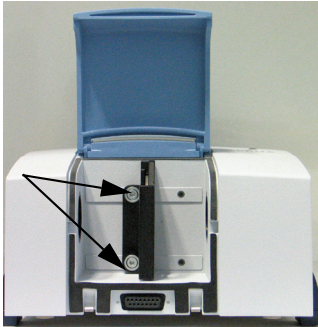
1		Lift up the sample compartment lid.
2		Remove the blue casing at the front side of the sample compartment.
3		Use the TX20 screw driver supplied to loosen the two TORX screws which fix the sample holder.

Table 3.9: Displacing sample holder when using ALPHA-T with gas cell

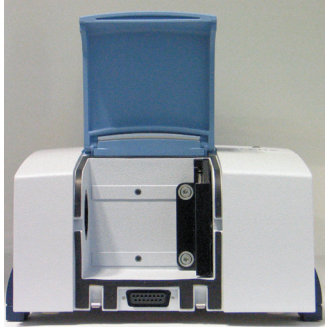
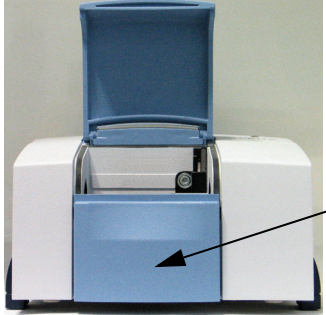
4		Mount the sample holder to the most right position.
5		Attach the blue casing onto the front side of the sample compartment.

Table 3.9: Displacing sample holder when using ALPHA-T with gas cell

3.3.4.2 Inserting gas cell into the sample compartment

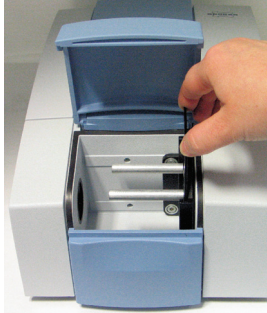
1		Insert the gas cell holder into the fixture of the sample holder.
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Table 3.10: ALPHA-T sampling module - Inserting gas cell into the sample compartment

2		Position the gas cell onto the gas cell holder as shown in the left figure.
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Table 3.10: ALPHA-T sampling module - Inserting gas cell into the sample compartment

3.4 ALPHA-T 30° reflection accessory

To perform fast surface spectroscopic analysis of solid samples by specular reflection, the 30° reflection accessory is inserted into the sample compartment.

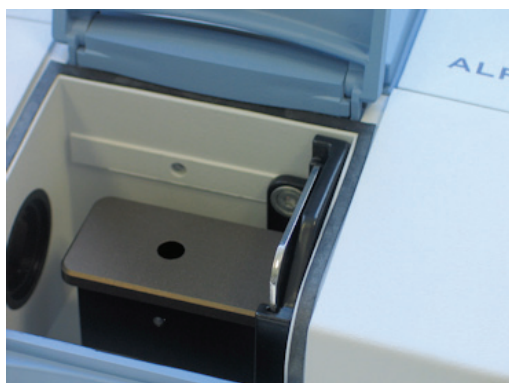


Figure 3.7: ALPHA-T 30° reflection accessory

3.4.1 Specifications

Spectral range:	375 to 7,500 cm^{-1}
Angle of incidence:	30°
Sample spot:	8 mm Ø
Additional components:	reference mirror (gold)
Special features:	horizontal face-down sample position, to fix into sample holder

3.4.2 Displacing mounted sample holder

To insert the 30° reflection accessory into the ALPHA-T sample compartment you first have to displace the sample holder located in the sample compartment, into the outer right position.

Procedure

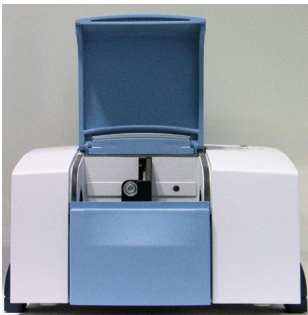
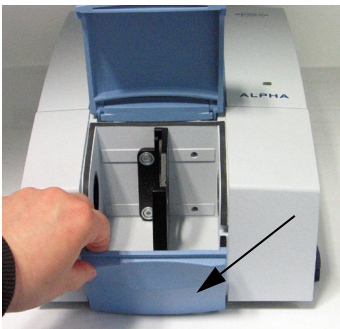
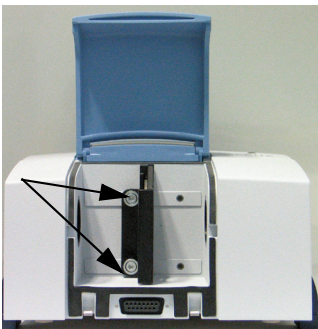
1		Lift up the sample compartment lid.
2		Remove the blue casing at the front side of the sample compartment.
3		Use the TX20 screw driver supplied to loosen the two TORX screws which fix the sample holder.

Table 3.11: Displacing sample holder when using ALPHA-T with 30° reflection accessory

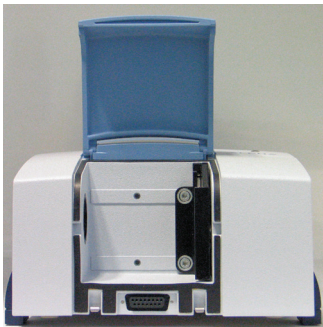
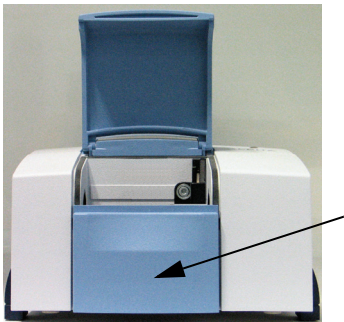
4		Mount the sample holder to the most right position.
5		Attach the blue casing onto the front side of the sample compartment.

Table 3.11: Displacing sample holder when using ALPHA-T with 30° reflection accessory

3.4.3 Inserting 30° reflection accessory into the sample compartment

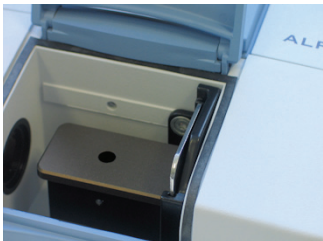
1		Insert the 30° accessory reflection into the fixture of the sample holder.
2		Position the 30° reflection accessory as shown in the left figure.

Table 3.12: ALPHA -T sampling module - Inserting 30° reflection accessory into the sample compartment

3.4.4 Performing measurement

Measurement is performed by the OPUS spectroscopy software. Before starting a sample measurement you first have to perform a background measurement. Observe the steps described in chapter 3.4.4.1 to 3.4.4.3.

3.4.4.1 Loading measurement experiment

1. Measurement experiment file is loaded.
 - *If AAR accessory recognition is activated (which is the case by default), the OPUS spectroscopy software recognizes the sampling module used. Thus, the experiment file¹ (TRANS.XPM) assigned to the sampling module is automatically loaded in OPUS. If the experiment file is not loaded automatically, perform the following steps.*
2. Start OPUS. On the *Measure* menu, select the *Setup Measurement Experiment* command.
 - *The Setup Measurement Experiment dialog opens.*
3. Click the *Basic* tab.
4. Click the *Load* button and select the *TRANS.XPM* experiment file from the dialog that opens.
5. Click the *Accept & Exit* button.

1. The universal ALPHA-T sampling module is supplied with the *TRANS.XPM* experiment file. This experiment file contains the default parameters set by Bruker, to be used to perform transmission measurement.

3.4.4.2 Starting background measurement

1. Put the reference mirror onto the reflection unit and push it completely to the right.

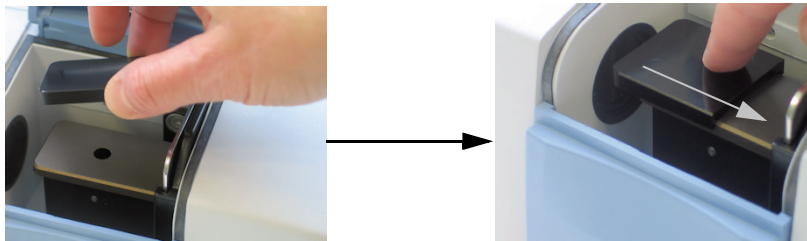


Figure 3.8: Putting reflection mirror onto reflection unit

2. Turn down the sample compartment lid.
 - *The spectrometer status indicator must be green (chapter 10.2.1).*
3. Start the OPUS spectroscopy software and wait until the performance test has finished.
 - *The OPUS status light must be green after the performance test has finished.*
4. On the OPUS Measure menu select the *Measure* command.
 - *The Measure dialog opens.*
5. Click the *Start Background Measurement* button.
6. Wait until the background measurement has finished.

3.4.4.3 Starting sample measurement

1. Lift up the sample compartment lid.
2. Remove the reference mirror.
3. Position the sample exactly onto the reflection unit.
4. Turn down the sample compartment lid.
 - *The spectrometer status indicator must be green (chapter 10.2.1).*
5. Click the *Start Sample Measurement* button.
6. Wait until the sample measurement has finished.

4 ALPHA for ATR measurement

Measurement type:	attenuated total reflection
Usable sample material:	• solids (paste, powder, gel, granular material, work pieces) • liquids • polymer film • surface coating
Available crystal types:	• diamond • germanium (Ge) • zinc selenide (ZnSe)

4.1 Overview on ALPHA spectrometer configurations for ATR

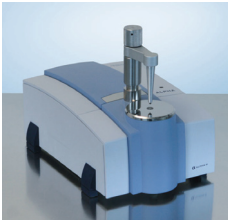
ALPHA spectrometer configurations for ATR		
		
ALPHA-E (Eco ATR module)	ALPHA-P (Platinum ATR module)	ALPHA-P (Platinum ATR module, high pressure variant)

Table 4.1: ALPHA spectrometer configurations for ATR

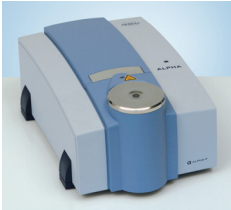
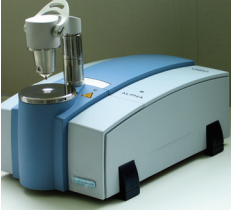
ALPHA spectrometer configurations for ATR		
		
ALPHA-P (Platinum ATR module, heat-able, without pressure applicator)	ALPHA-P (Platinum ATR module, heat-able, with pressure applicator)	

Table 4.1: ALPHA spectrometer configurations for ATR

4.2 ALPHA-E sampling module

Measurement type: attenuated total reflection

Usable sample material:

- solids (paste, powder)
- liquids
- foils

Available crystal types:

- germanium (Ge)
- zinc selenide (ZnSe)

Actual sampling surface on ATR crystal: 5 mm Ø

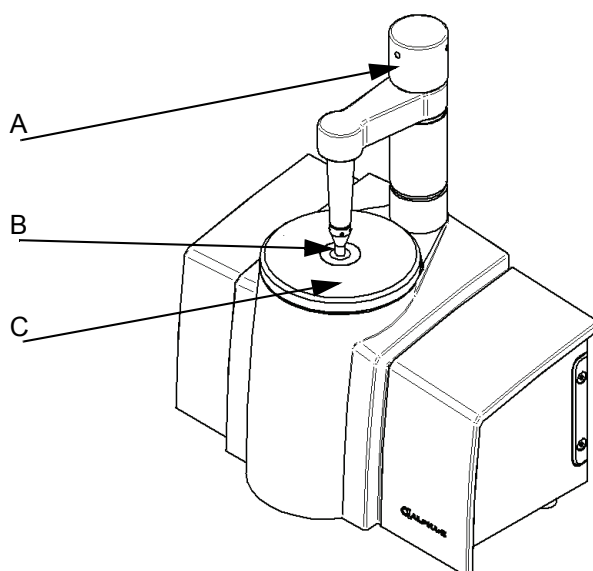


Figure 4.1: ALPHA-E sampling module

	Definition
A	Knob to press and release the anvil
B	Anvil
C	Crystal plate with integrated ATR crystal

Table 4.2: ALPHA-E sampling module - Components

4.2.1 Optical path

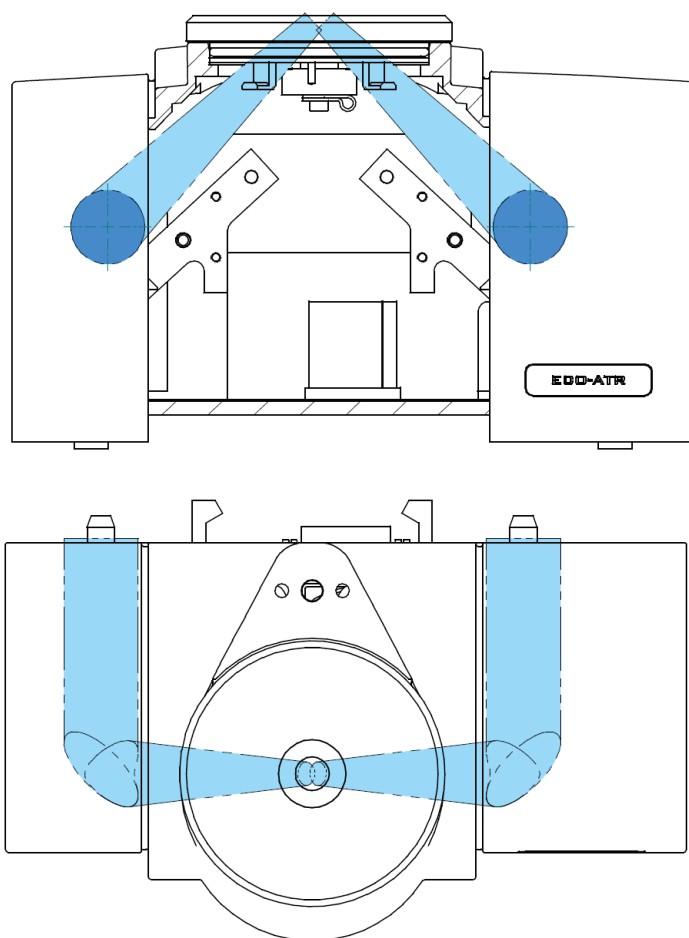


Figure 4.2: ALPHA-E - Optical path

4.3 ALPHA-P sampling module

Measurement type:	attenuated total reflection
Usable sample material:	• solids (paste, powder) • liquids
Available crystal types:	• diamond • germanium (Ge)
Actual sampling surface on ATR crystal:	• diamond: 2 x 2 mm • germanium: 5 mm Ø
Anvil tip types:	• Standard tip type for powder, foils, work pieces • Special tip type for granular material

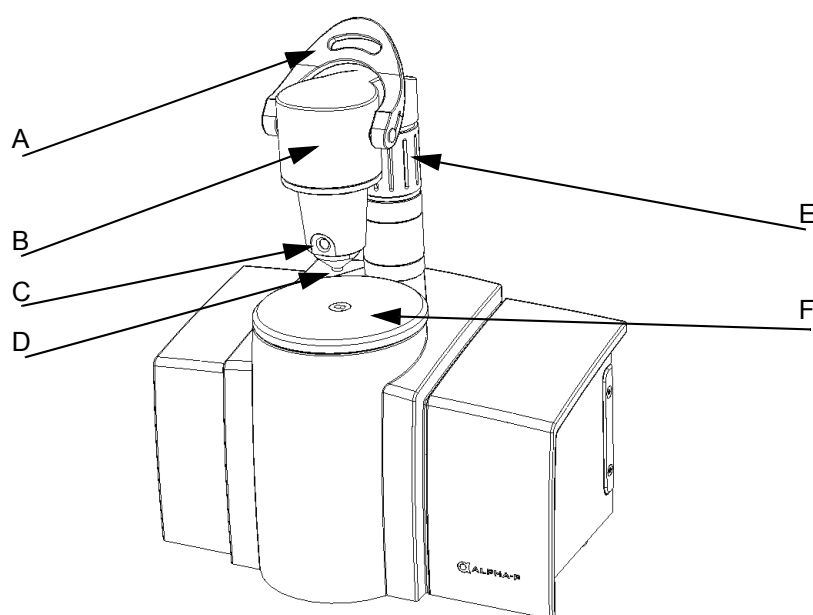


Figure 4.3: ALPHA-P sampling module

	Definition
A	Lever to move the anvil up and down
B	Pressure arm
C	Pressure control spot (red spot)

Table 4.3: ALPHA-P sampling module - Components

	Definition
D	Anvil
E	Knob to adjust height adjustment of pressure arm
F	Crystal plate with integrated ATR crystal

Table 4.3: ALPHA-P sampling module - Components

4.3.1 Optical path

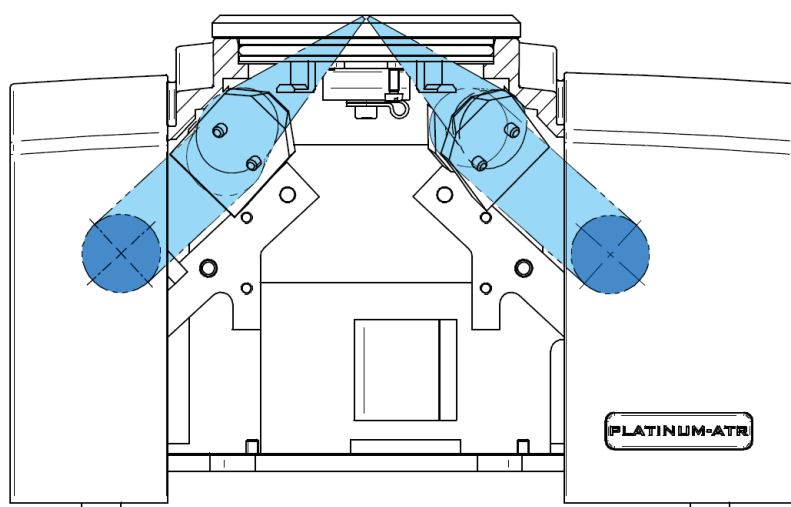


Figure 4.4: ALPHA-P - Optical path

4.3.2 High-pressure variant

Measurement type:	attenuated total reflection
Usable sample material:	• solids (paste, powder) • liquids
Available crystal types:	• diamond
Actual sampling surface on ATR crystal:	2 x 2 mm
Anvil tip types:	• Standard tip type for powder, foils, work pieces • Special tip type for granular material

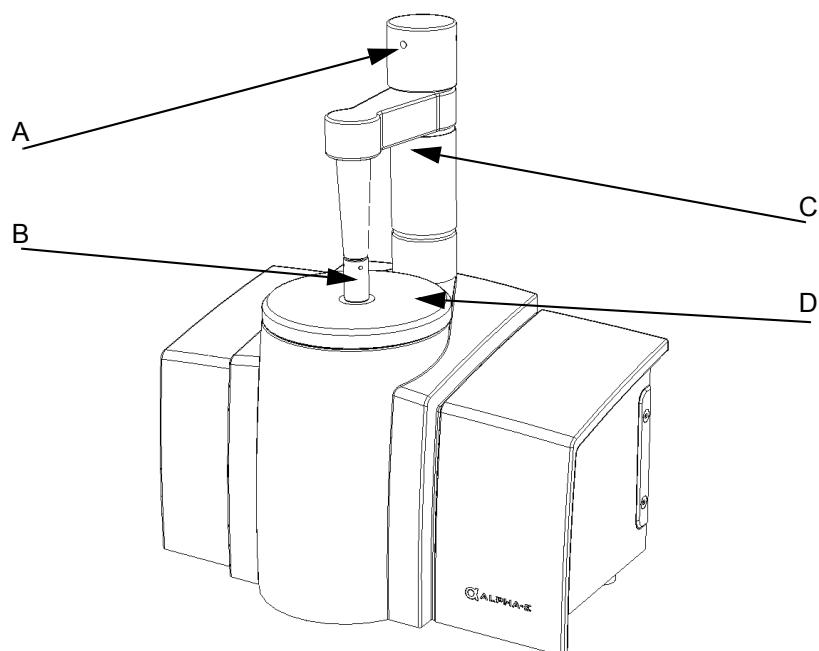


Figure 4.5: ALPHA-P sampling module - High-pressure variant

	Definition
A	Knob to press and release the anvil
B	Anvil
C	Pressure arm
D	Crystal plate with integrated ATR crystal

Table 4.4: ALPHA-P sampling module - Components of high-pressure variant

4.3.3 With heatable ATR unit

Measurement type:	attenuated total reflection up to a temperature of 120°C
Usable sample material ^a :	• solids (paste, powder) • liquids
Available crystal types:	diamond
Actual sampling surface on ATR crystal:	2 x 2 mm
Maximum sample temperature:	120°C

a. In case of ALPHA-P with pressure applicator only.

The desired temperature value is set by using the OPUS spectroscopy software. The temperature controller integrated into the ALPHA-P sampling module heats the sample position up to the temperature set in the software.

The ALPHA-P sampling module with heatable ATR unit is available in different variants:

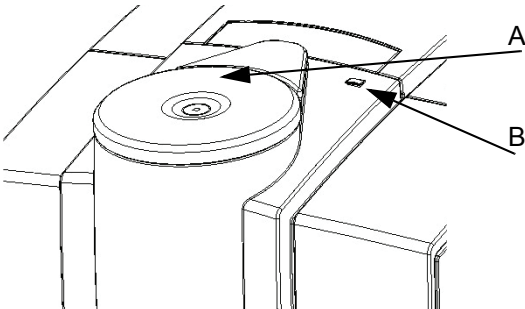
Variant	Illustration
Without pressure applicator:	 • A: crystal plate with integrated ATR crystal • B: temperature LED

Table 4.5: ALPHA-P sampling module with heatable ATR unit - Variants

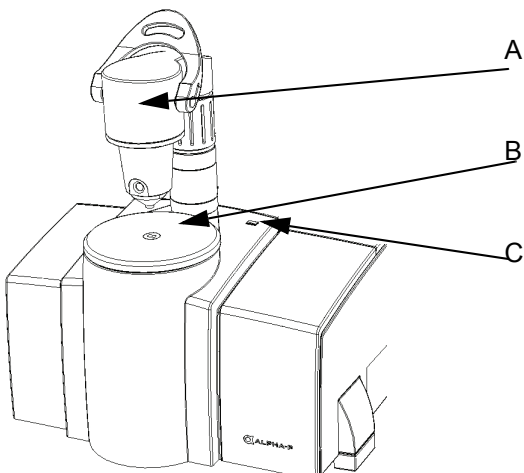
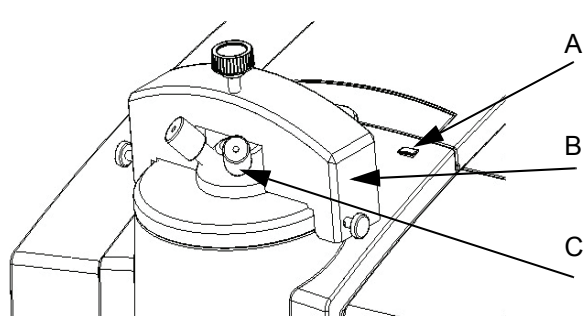
Variant	Illustration
With pressure applicator:	 • A: pressure arm • B: crystal plate with integrated ATR crystal • C: temperature LED
With flow-through cell:	 • A: temperature LED • B: clamping bracket • C: flow-through cell

Table 4.5: ALPHA-P sampling module with heatable ATR unit - Variants

4.3.3.1 Temperature controller status indicated by temperature LED

The temperature LED indicates the temperature controller status as follows:

Temperature LED	Definition
Flashing quickly (green)	Temperature controller is heating, but the desired final temperature has not yet been reached.
Flashing slowly (green)	Temperature controller does not heat as the current temperature at the sample position is higher than the temperature defined in the OPUS spectroscopy software.
Permanently green	The desired final temperature is reached and OPUS starts the measurement. i As soon as the desired final temperature is reached, the temperature is kept at a constant level of $\pm 0.5^{\circ}\text{C}$.
Off	The temperature controller is switched off.

Table 4.6: Temperature controller status indicated by temperature LED

4.3.3.2 Software requirements in case of OPUS version <7

When working with an OPUS spectroscopy software version <7, and you have purchased the sampling module at a later date than the software, the following requirements must be met to be able to use the *heatability* feature:

- Firmware version: 1.026
- OVP database version 6.5.4

To implement these requirements, you have to update the spectrometer firmware (appendix C) and the OVP database:

1. Insert the USB stick, which contains the OVP database, into the USB port of the PC.
2. Open the Windows explorer and browse to the directory of the USB stick.
3. Double click the *OVP_Database_Install.exe* file to start the installation program.
 - *The installation program first checks for previously installed components. This automatic search process may take a few seconds.*
4. Follow the on-screen instructions.

4.3.3.3 Installing flow-through cell

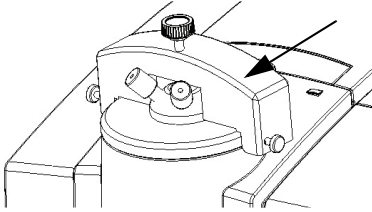
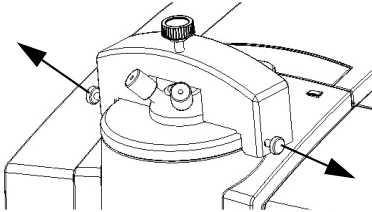
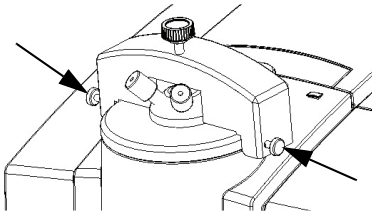
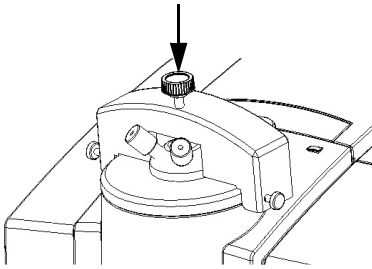
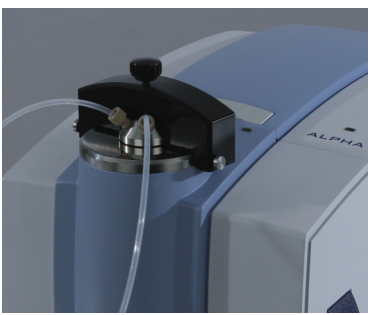
1	Place the flow-through cell onto the ATR crystal.	
2		Position the retaining clip above the flow-through cell.
3		Pull out the two locking pins. ☞ Orientate the retaining clip such that the two locking pins can exactly be inserted into the two bore holes.
4		Push the locking pins into the corresponding bore hole to fix the retaining clip.
5		Rotate the fastening screw downwards.
6		Direct the drain hose into a bucket for collecting the sample liquid. i Alternatively, you can connect the drain and filling hoses to a pump to ensure a continuous liquid flow-through.

Table 4.7: Installing flow-through cell

4.3.3.4 De-installing flow-through cell

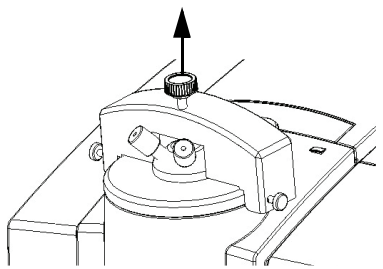
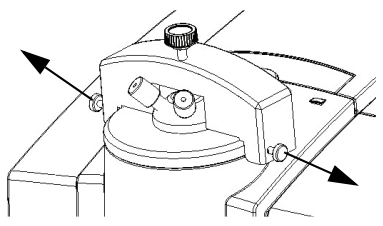
1		Rotate fastening screw upwards.
2		Pull out the two locking pins.
3	Lift up the retaining clip.	
4	Remove the flow-through cell from the ATR crystal.	

Table 4.8: De-installing flow-through cell

4.4 Crystal material

Depending on the sampling module used, the ATR crystal plate is available with the following types of crystal material:

- diamond
- germanium (Ge)
- zinc selenide (ZnSe)

Due to the accessory recognition (AAR), which is activated by default, the crystal material is automatically recognized with ALPHA. The color of the crystal indicates which type of crystal material is used for the crystal plate.

Crystal material	Crystal color
Diamond	Transparent
Germanium	Silver gray
Zinc selenide	Yellow

Table 4.9: Crystal colors

The specifications of the crystal material are described in appendix A.5.

4.5 Replacing ATR crystal plate

i As it is not easy to rotate the ATR crystal plate manually, it is recommended to use the pin supplied. The leverage effect of the pin facilitates rotating the crystal plate.

CAUTION



Improper use of the ZnSe crystal material, broken crystal

Health damage, risk of poisoning

- Carefully handle broken crystal fragments.
- Avoid material abrasion or dust.
- Do not inhale nor ingest material abrasion or dust.

4.5.1 Procedure

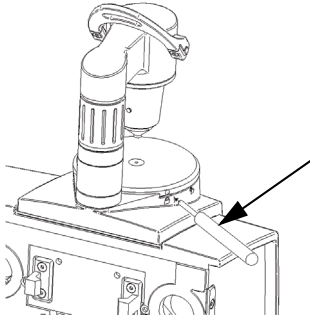
1		Insert the pin supplied into the borehole of the crystal plate.
2	Use the pin and rotate the crystal plate counterclockwise up to the stop (about 20°).	
3	Remove the crystal plate.	
4	Position the new crystal plate onto the ATR sampling module and insert the pin into the borehole of the crystal plate.	
5	• Use the pin and rotate the crystal plate counterclockwise up to the stop (about 20°). • Pull out the pin from the borehole of the crystal plate.	

Table 4.10: Replacing ATR crystal plate

4.6 Working with pressure applicator

How to work with the pressure applicator depends on the sampling module used.

4.6.1 With the ALPHA-E sampling module

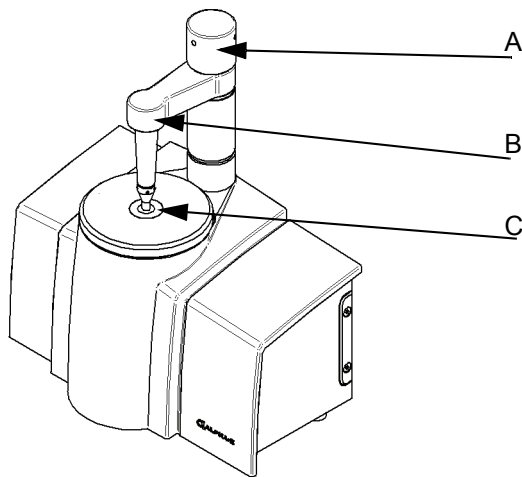


Figure 4.6: Pressure applicator with the ALPHA-E sampling module

	Definition	Function
A	Knob	Moving pressure arm (B) upwards: ☞ Rotate the knob counterclockwise Moving pressure arm (B) downwards: ☞ Rotate the knob clockwise
B	Pressure arm	• Rotatable by 360° • Can be swivelled out of measuring position i If you swivel the pressure arm into the measuring position again, the pressure arm snaps into place exactly above the ATR crystal.

Table 4.11: ALPHA-E sampling module - Components of pressure applicator

	Definition	Function
C	Anvil	Pressing sample against the ATR crystal - Increasing contact pressure: - rotate knob (A) further downwards - Decreasing contact pressure: - rotate knob (A) further upwards i The inner part of the pressure arm (B) is equipped with a slip-clutch mechanism which is activated as soon as the optimal contact pressure between sample and anvil has been applied. Further rotating the knob, which is only possible by substantial force, would not have any effect on the contact pressure already applied.

Table 4.11: ALPHA-E sampling module - Components of pressure applicator

4.6.2 With the ALPHA-P sampling module

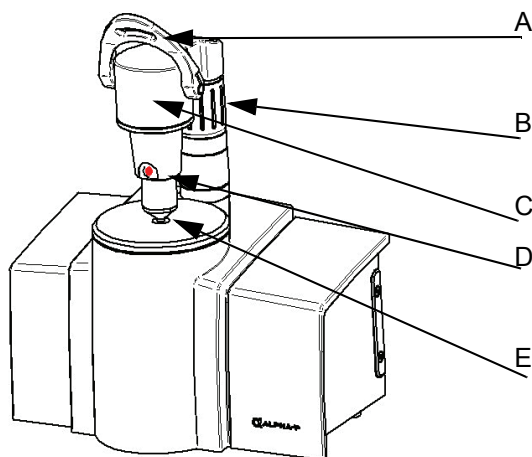


Figure 4.7: Pressure applicator with ALPHA-P sampling module

	Definition	Function
A	Lever	• Moving anvil (E) upwards: ☞ turn up the lever • Moving anvil (E) downwards: ☞ turn down the lever
B	Knob	Adjusting height of pressure arm (C) • Moving pressure arm (C) upwards: ☞ Rotate knob counterclockwise • Moving pressure arm (C) downwards: ☞ Rotate knob clockwise
C	Pressure arm	• Rotatable by 360° • Can be swivelled out of measuring position i If you swivel the pressure arm into the measuring position again, the pressure arm snaps into place exactly above the ATR crystal.
D	Red spot	Indicating contact pressure i The red spot must be exactly in the middle of the round recess on the front of the pressure arm to ensure optimal contact pressure (see figure 4.7).
E	Anvil	Pressing sample against the ATR crystal

Table 4.12: ALPHA-P sampling module - Pressure applicator

4.6.2.1 Adjusting height of pressure arm

To adjust the height of the complete pressure arm (C in figure 4.7) is required if:

Problem	Setting
• Lever (A in figure 4.7) turned to the outermost front position • Anvil (E) has not yet any contact with the sample • Red spot below round recess	Move pressure arm (C) downwards: ➡ rotate knob (B) clockwise ➡ rotate knob (B) until the red spot (D) is exactly in the middle of the round recess
• Lever (A in figure 4.7) cannot be turned to the outermost front position as the sample is too thick • Red spot above round recess	Move pressure arm (C) upwards: ➡ rotate knob (B) counterclockwise ➡ rotate knob (B) until the red spot (D) is exactly in the middle of the round recess

Table 4.13: When must the height of the pressure arm be adjusted?

4.7 Cleanness test

Before you start measuring, a cleanness test can be performed. This kind of test is a reference measurement of the ATR crystal. Based on the maximum contamination limit allowed, the test verifies whether the crystal is clean. The tested wavenumber range is between 2800 and 3030 cm^{-1} (absorption).

The cleanness test is performed by the OPUS spectroscopy software. Details on the cleanness test are described in the OPUS/MENTOR manual.

4.8 Positioning sample on the ATR crystal

Before positioning the sample on the ATR crystal make sure that the sample plate, ATR crystal and the anvil are clean.

- i** If the sample cannot be placed onto the ATR crystal as the distance between the ATR crystal and anvil is too small, you first have to move the pressure arm upwards.

4.8.1 Procedure in case of ALPHA-E and ALPHA-P (non-heatable)

1	Position sample in the middle of the sample plate. ➤ <i>The ATR crystal must completely be covered by the sample material. A small amount of sample material is sufficient.</i>  CAUTION Improper use of flammable and/or explosive substances: Property damage and personal injury ➤ Always wear safety goggles. ➤ Only use small amounts of sample material. ➤ Do not inhale the vapor of volatile substances. ➤ Observe the safety instructions.
2	Press anvil against the sample. ➤ <i>Make sure that the optimum contact pressure is used to press the sample against the ATR crystal.</i>
3	Measure sample (chapter 4.9).
4	• Move pressure arm upwards. • Remove sample. i Sample material should not be left in contact with the crystal for an extended period of time, as chemically reactive sample material may degrade the crystal quality and discolor the metal plate. Therefore, once the measurement has finished, remove the sample from the crystal.
5	Thoroughly clean the sample plate, ATR crystal and anvil, see chapter 4.10.

Table 4.14: Positioning sample on ATR crystal in case of ALPHA-E and ALPHA-P (non-heatable)

4.8.2 Procedure in case of ALPHA-P (heatable)

Observe the following safety instructions when heating sample material:

Hot liquids directly on the ATR crystal:	When putting hot liquids onto the ATR crystal without using a flow-through cell, be careful as some hot liquids may splash around. ☞ Always wear safety goggles. ☞ Do not heat up a liquid above its boiling temperature.
When using flow-through cells:	During measurement, the flow-through cell and the sample can be heated up to 120°C. ☞ Avoid any skin contact with hot objects and surfaces. Improper handling can cause burns!
When heating flammable liquids:	Heating up flammable liquids involves the risk of deflagration, explosion and fire. ☞ Heat only liquids with a burning point above 200°C. ☞ Consider that flammability is increased with heated liquids. ☞ Keep heated liquids away from naked flames.
When heating corrosive liquids or liquids that are hazardous to health:	Heating corrosive liquids or liquids that are hazardous to health can produce toxic or corrosive gases and vapors. ☞ Extract toxic or corrosive gases and vapors.

Table 4.15: Safety instructions when heating sample material

In case of flow-through cell

1	Fill the syringe with sample liquid.
2	Insert the syringe into the injection port at the flow-through cap or filling hose.
3	• Direct the drain hose into a bucket for collecting the sample liquid. • Inject the sample liquid.
4	Set the temperature (chapter 4.9.2.1).
5	Measure sample (chapter 4.9.2).
6	Clean flow-through cell (chapter 4.11).

Table 4.16: Positioning sample on ALPHA-P equipped with flow-through cell

In case of pressure applicator

1	Position sample in the middle of the sample plate. ➤ <i>The ATR crystal must completely be covered by the sample material. A small amount of sample material is sufficient.</i>  CAUTION Improper use of heated corrosive substances and/or substances that are hazardous to health: Property damage and personal injury ➤ Always wear safety goggles. ➤ Extract corrosive or hazardous gases or vapors. ➤ Observe the safety instructions.
2	Press anvil against the sample. ➤ <i>Make sure that the optimum contact pressure is used to press the sample against the ATR crystal.</i>
3	Measure sample (chapter 4.9).
4	• Move pressure arm upwards. • Remove sample. i Sample material should not be left in contact with the crystal for an extended period of time, as chemically reactive sample material may degrade the crystal quality and discolor the metal plate. Therefore, once the measurement has finished, remove the sample from the crystal.
5	Thoroughly clean the sample plate, ATR crystal and anvil, see chapter 4.10.

Table 4.17: Positioning sample on ALPHA-P equipped with pressure applicator

4.9 Performing measurement

Measurement is performed by the OPUS spectroscopy software. Before starting a sample measurement you first have to perform a background measurement. Observe the steps described in the following chapters.

4.9.1 Performing non-temperature controlled measurement

4.9.1.1 Loading measurement experiment

1. Measurement experiment file is loaded.
 - *If AAR accessory recognition is activated (which is the case by default), the OPUS spectroscopy software recognizes the sampling module and ATR crystal type used. Thus, the experiment file¹ is automatically loaded in OPUS. If the experiment file is not loaded automatically, perform the following steps.*
2. Start OPUS. On the *Measure* menu, select the *Setup Measurement Experiment* command.
 - *The Setup Measurement Experiment dialog opens.*
3. Click the *Basic* tab.
4. Click the *Load* button. Based on the ATR crystal material used, select the respective experiment file from the dialog that opens.
5. Click the *Accept & Exit* button.

4.9.1.2 Starting background measurement

1. Flap the pressure arm upwards.
2. If necessary, start OPUS and wait until the performance test has finished.
 - *After the performance test has finished, the spectrometer status indicator and the OPUS status light must be green.*
3. On the OPUS *Measure* menu select the *Measurement* command.
 - *The Measure dialog opens.*
4. Click the *Start Background Measurement* button.
5. Wait until the background measurement has finished.

1. Depending on the ATR crystal type used, the ATR sampling module is supplied with the following experiment file: ATR_Di.XPM, ATR_Ge.XPM, ATR_ZnSE.XPM. This experiment file contains the default parameters set by Bruker, to be used to perform ATR measurement.

4.9.1.3 Starting sample measurement

1. Flap the pressure arm upwards.
2. Position sample onto the ATR crystal.
3. Flap the pressure arm downwards.
 - *The optimal contact pressure is given if the red spot is exactly in the middle of the round recess on the front of the pressure arm (chapter 4.6.2). The spectrometer status indicator and the OPUS status light must be green.*
4. On the *Measure* menu, select the *Measurement* command.
 - *The Measurement dialog opens.*
5. Click the *Start Sample Measurement* button.
6. Wait until the sample measurement has finished.

4.9.2 Performing temperature-controlled measurement (only with heatable ALPHA-P)

4.9.2.1 Setting temperature

Minimum temperature:	5°C above the current temperature ^a
Maximum temperature:	120°C ^b
Temperature controller off:	when entering a temperature value of -300°C
Temperature controller on:	when entering a temperature value between 30 and 120°C

Table 4.18: Temperature features

- a. The minimum temperature value specified must be at least 5°C above the current temperature. Otherwise, the spectrometer does not start heating, and the *Instrument ready* status cannot be achieved, i.e. measurement cannot be performed.
- b. The spectrometer is not equipped with a cooling device. The heatable sampling module allows temperature-controlled measurements up to a maximum temperature of 120°C. When entering a higher temperature value a warning message is displayed. Confirm the message and enter a temperature value of <120°C.

i The temperature accuracy of the ATR crystal is $\pm 1^\circ\text{C}$. In case of a strongly heat-dissipating sample, wait a few minutes before starting the measurement to be able to ensure the stated temperature accuracy.

Procedure

1. On the *Measure* menu, select the *Setup Measurement Parameters* command.
2. Click the *Optic* tab.
3. Enter the desired sample temperature.
 - *The temperature must be between 5°C above the current temperature and 120°C.*
4. Click the *Accept & Exit* command.

4.9.2.2 Loading measurement experiment

1. The desired temperature must be set in OPUS (chapter 4.9.2.1).
2. Measurement experiment file is loaded.
 - *If AAR accessory recognition is activated (which is the case by default), the OPUS spectroscopy software recognizes the sampling module and ATR crystal type used. Thus, the experiment file¹ (ATR_Di.XPM) is automatically loaded in OPUS. If the experiment file is not loaded automatically, perform the following steps.*
3. Start OPUS. On the *Measure* menu, select the *Setup Measurement Experiment* command.
 - *The Setup Measurement Experiment dialog opens.*
4. Click the *Basic* tab.
5. Click the *Load* button and select the *ATR_Di.XPM* experiment file from the dialog that opens.
6. Click the *Accept & Exit* button.

1. Depending on the ATR crystal type used, the ATR sampling module is supplied with the following experiment file: ATR_Di.XPM, ATR_Ge.XPM, ATR_ZnSE.XPM. This experiment file contains the default parameters set by Bruker, to be used to perform ATR measurement.

4.9.2.3 Performing background measurement



If you use ALPHA-P equipped with pressure applicator, start with step 3.

1. Flap the pressure arm upwards.
2. If necessary, start the OPUS spectroscopy software and wait until the performance test has finished.
 - *The OPUS status light must be green after the performance test has finished (chapter 10.2.1).*
3. On the *Measure* menu, click the *Measurement* command.
 - *The Measurement dialog opens.*
4. Click the *Start Background Measurement* button.
 - *The spectrometer starts heating. This is indicated by a quickly flashing LED (green). As soon as the desired temperature is reached (the LED is permanently green), OPUS automatically starts background measurement.*
5. Wait until the background measurement has finished.

4.9.2.4 Measuring liquid sample

1. Flap the pressure arm (if available) upwards and put the liquid sample onto the ATR crystal. If a flow-through cap is used, inject the liquid sample by a syringe.
 - *Do not remove the syringe from the injection port during measurement, to avoid any dry run of the flow-through cap.*
2. Wait until the sample is heated.
3. On the *Measure* menu, select the *Measurement* command.
 - *The Measurement dialog opens.*
4. Click the *Start Sample Measurement* button.
5. Wait until the sample measurement has finished.

4.10 Cleaning ATR sampling module

After each sample measurement performed, you thoroughly have to clean the sample plate, ATR crystal and anvil. Remaining sample material on any of these elements may cause falsified measurement results.

4.10.1 Cleaning solvents

Never use caustic, abrasive substances nor strong acids. These substances can damage the sample plate and the ATR unit.

Possible cleaning solvents:

- alcohol
- acetone
- ethanol
- isopropanol

i When using solvents read the materials safety data sheets supplied with the solvents and observe any recommended handling procedure.

The solvent used for cleaning depends on the sample type that has been analyzed. In any case, it is best to try to clean the sample plate by the mildest solvent possible.

4.10.2 Procedure

If you use a heatable ALPHA-P sampling module, wait until the sample plate has cooled down sufficiently.

1	Flap the pressure arm upwards and move it out of the measuring position.
2	Clean the ATR crystal by a lint-free cloth or Q-tip. Apply only low pressure. i Never use filled paper tissue to rub on the surface. Due to the abrasive effect of the paper tissue you may scratch the crystal surface, especially in case of crystals made of germanium (Ge) and zinc selenide (ZnSe).
3	Clean the anvil.
4	Perform a cleanness test (chapter 4.7) to check whether the ATR crystal is free of sample residues.  If the cleanness test fails, repeat the cleaning procedure.

Table 4.19: Cleaning ATR sampling module

4.11 Cleaning flow-through cell

1	Purge the flow-through cell thoroughly with an appropriate cleaning solvent and/or water. i It is recommended to purge the flow-through cell by using air.
2	De-install the flow-through cell (chapter 4.3.3.4).
3	Carefully clean the sampling surface and the ATR crystal surface.
4	Perform a cleanness test (chapter 4.7) to check whether the ATR crystal is free of sample residues.  If the cleanness test fails, repeat the cleaning procedure.

Table 4.20: Cleaning ATR sampling module

5 ALPHA for reflection measurement

Measurement type:	• specular reflection^a • diffuse reflection^b
-------------------	---

Usable sample material: sample surfaces

- a. When IR light falls on a surface, two types of reflection occur: specular reflection (i.e. the light reflects directly off the surface) and diffuse reflection (the IR light penetrates the sample surface is reflected in all directions). The amount of reflection type occurring on the sample depends on the roughness of the sample surface.
- b. The optics of a diffuse reflection accessory is designed in such a way that the detection of diffusely reflected light is optimized and the detection of specularly reflected light is minimized. DRIFTS (diffuse reflectance infrared fourier transform spectroscopy) is an analyzing technique in FT-IR spectroscopy that makes use of the phenomenon of the diffuse reflection.

5.1 Overview on ALPHA spectrometer configurations for reflection

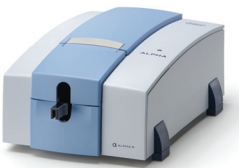
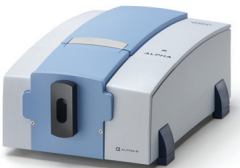
ALPHA spectrometer configurations for reflection	
	
ALPHA-R (A528/D drift module, diffuse reflection)	ALPHA-R (A241/D module, contactless reflection)

Table 5.1: Overview on ALPHA spectrometer configurations for reflection

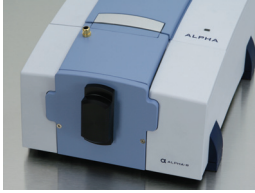
ALPHA spectrometer configurations for reflection	
	
ALPHA-R (A241/DV, contactless, video-based reflection)	ALPHA-R (A240/DU, upward looking reflection)

Table 5.1: Overview on ALPHA spectrometer configurations for reflection

5.2 ALPHA-R sampling module (A528/D)

Measurement type:	diffuse reflection
Usable sample material:	• solids in a powdery or crystalline state • cloths, fibers and coatings that can be abraded • powders, paper, polymers, solid samples having a rough surface • gemstones
Advantages of this kind of measuring technique:	• no sample preparation required in case of powders and crystals • suited for strongly scattering and absorbing samples • high signal intensity^a
Toolkit:	included in delivery content, chapter 5.2.4

a. This measurement technique delivers comprehensive and extensive spectral information. As the scattered and diffusely reflected light of a large number of particles is detected, sample components with a weak signal intensity can be detected as well.

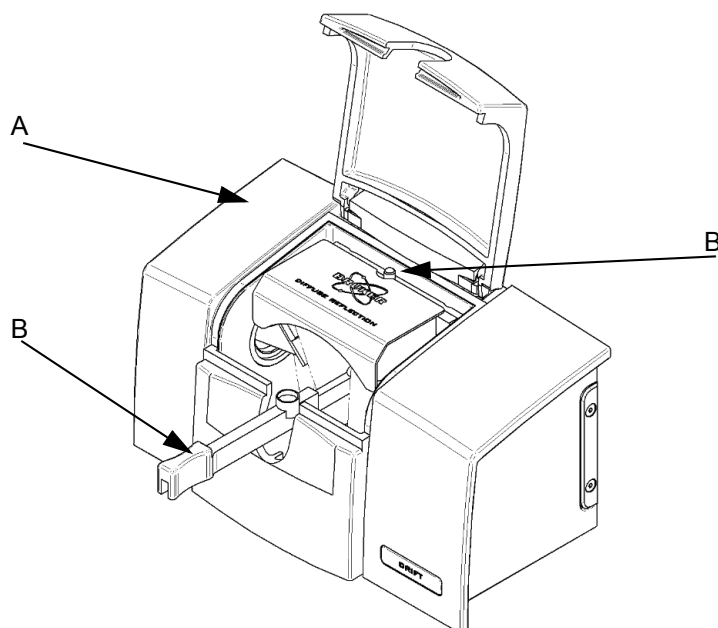


Figure 5.1: ALPHA-R sampling module (A528/D)

	Definition
A	Reflection unit
B	Tool for sample material
C	Height adjustment screw

Table 5.2: ALPHA-R (A528/D) sampling module - Components

5.2.1 Specifications

Spectral range:	375 - 7,500 cm ⁻¹
Mirror material:	Gold
Sampling spot:	Ø 2 mm

5.2.2 Optical path

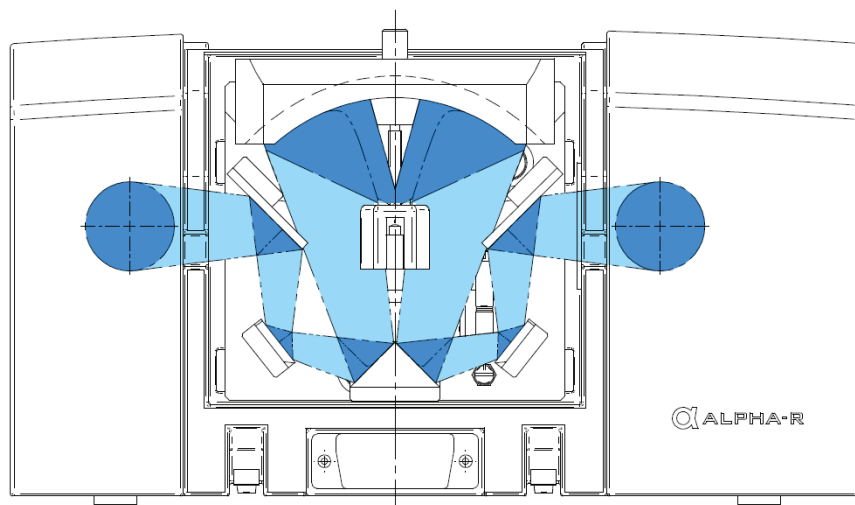


Figure 5.2: ALPHA-R (A528/D) - Optical path

5.2.3 Software requirements in case of OPUS version <7

When working with an OPUS version <7 and you have purchased the sampling module at a later date than the software, the following requirements must be met:

- OVP database version 6.5.6

The OVP database must be updated (chapter 5.2.3.1).

5.2.3.1 Updating OVP data base

1. Insert the USB stick, which contains the OVP database, into the USB port of the PC.
2. Open the Windows explorer and browse to the directory of the USB stick.
3. Double click the *OVP_Database_Install.exe* file to start the installation program.
 - *The installation program first checks for previously installed components. This automatic search process may take a few seconds.*
4. Follow the on-screen instructions.

5.2.4 Tool kit

Apart from spatula and brush, the tool kit for the ALPHA-R (A528/D) sampling module additionally contains the following components:

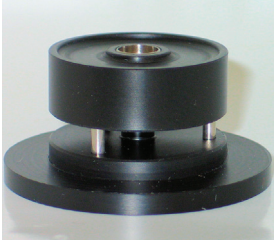
Components	
Cup for sample material	
Cup for reference mirror	
Reference mirror	
Auxiliary device for filling the sample cup	

Table 5.3: ALPHA-R (A528/D) tool kit - Components

Depending on the type of measurement performed, the two removable cups are placed onto the defined position located on a slider. As the slider is mounted on a rail, the slider can be moved forward and backward.

To correctly position the reference mirror or sample cup onto the measuring position, always move the slider forward UP TO the mechanical stop.

5.2.5 Filling sample cup by using the auxiliary device

The auxiliary device supplied for filling the sample cup consists of a base plate and a top piece (figure 5.3).

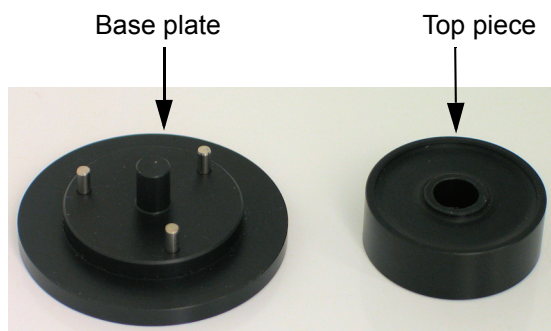


Figure 5.3: Auxiliary device for filling the sample cup

The auxiliary device allows a correct filling, i.e. the powder must be flush with the cup edge.

5.2.5.1 Procedure

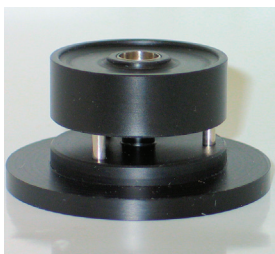
1		Put the top piece onto the base plate such that the three pins at the base plate do not snap in the holes at the bottom of the top piece.
2		Insert the sample cup into the intended hole of the top piece.

Table 5.4: Filling sample cup

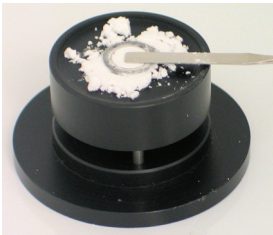
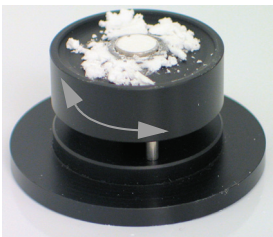
3		• Fill sufficient powder into the sample cup. • Use the supplied spatula and smooth the powder^a. i The powder filling must be flush with the cup edge. Sweep the excess powder in the intended collecting groove.
4		• Rotate the top piece to the left or right until the pins at the base plate snap into the holes at the bottom of the top piece. • Put the top piece onto the base plate.
5		Remove the filled sample cup from the top piece.

Table 5.4: Filling sample cup

a. In terms of powders, there is only sufficient diffuse reflection of infrared radiation by the sample surface if the powder has been smoothed before.

5.2.6 Performing measurement

When performing a measurement observe the order of the chapters described in the following.

5.2.6.1 Checking signal intensity, starting instrument/performance test

If AAR accessory recognition is activated (which is the case by default), the instrument/performance test starts automatically. In case of the ALPHA-R (A528/D) sampling module, however, you first have to check the signal intensity. Otherwise, the tests will fail.

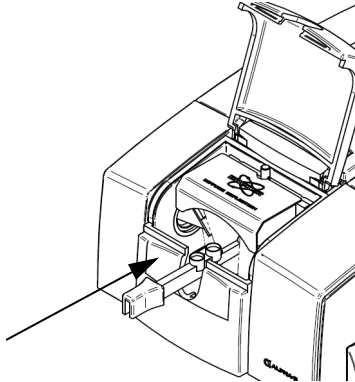
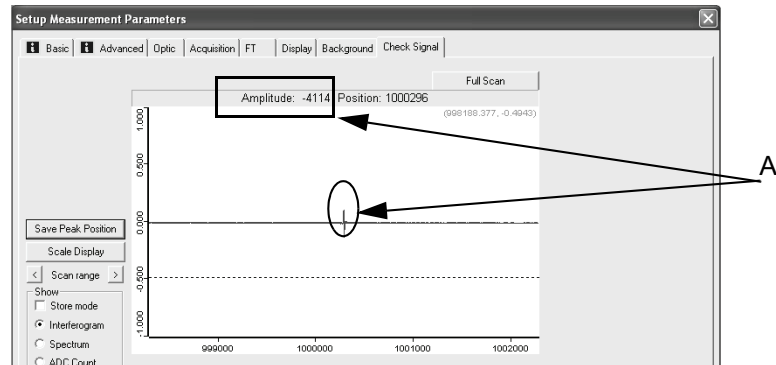
1	Start OPUS spectroscopy software. ➤ <i>On the OPUS interface a message window opens and prompts you to put the reference mirror onto the measuring position.</i> ☞ Lift up the sample compartment lid. ☞ Move the slider with the reference mirror cup backwards into the measuring position, until the mechanical stop. 
2	Dialog Please place the reference at the measurement position Click on "Continue" to proceed with the Test Continue Click the <i>Continue</i> button to start the instrument/performance test.

Table 5.5: Checking signal intensity, starting instrument/performance test

3

If the instrument/performance test failed:

- On the OPUS *Measure* menu, select the *Setup Measurement Parameters* command.
- Click the *Check Signal* tab.



- Check signal intensity (A).
 - Rotate the height adjustment screw (figure 5.1) until there is sufficient signal on the *Check Signal* tab.
 - Close the dialog.

Table 5.5: Checking signal intensity, starting instrument/performance test

5.2.6.2 Loading measurement experiment

1. Measurement experiment file is loaded.
 - If AAR accessory recognition is activated (which is the case by default), the OPUS spectroscopy software recognizes the sampling module used. Thus, the experiment file¹ (DRIFT.XPM) assigned to the sampling module is automatically loaded in OPUS. If the experiment file is not loaded automatically, perform step 2 et seq.
2. Start OPUS. On the *Measure* menu, select the *Setup Measurement Experiment* command.
 - The *Setup Measurement Experiment* dialog opens.
3. Click the *Basic* tab.
4. Click the *Load* button and select the *DRFIT.XPM* experiment file from the dialog that opens.
5. Click the *Accept & Exit* button.

5.2.6.3 Starting background measurement

1. The signal intensity is checked (chapter 5.2.6.1). The spectrometer status indicator and the OPUS status light must be green.
2. On the OPUS *Measure* menu select the *Measurement* command.
 - The *Measure* dialog opens.
3. Click the *Start Background Measurement* button.
4. Wait until the background measurement has finished.

5.2.6.4 Starting sample measurement

1. Move the slider with the filled sample cup forward into the measuring position, until the mechanical stop.
2. On the OPUS *Measure* menu, select the *Setup Measurement Parameters* command.
3. Click the *Check Signal* tab.
4. Position the height adjustment screw (figure 5.1) such that the signal intensity, which can be achieved as maximum, is displayed. The maximum signal intensity depends very much on the sample properties (grain size, color).
5. Close the dialog.
6. On the OPUS *Measure* menu select the *Measurement* command.
 - The *Measure* dialog opens.
7. Click the *Start sample Measurement* button.
8. Wait until the sample measurement has finished.

1. The ALPHA-R sampling module is supplied with the *DRIFT.XPM* experiment file. This experiment file contains the default parameters set by Bruker, to be used to perform diffuse reflection measurement.

5.2.7 Potential operating errors

Problem	Troubleshooting
No or insufficient signal intensity, high spectral noise	Check signal intensity (chapter 5.2.6.1).
Reference mirror or the sample are not correctly placed onto the measuring position	Insert the slider forward or backward until it stops.
Sample cup not properly filled with powder	Fill in the sample cup properly (chapter 5.2.5).

Table 5.6: Potential operating errors

5.3 ALPHA-R (A241/D, A241/DV) sampling module

Measurement type:

- contactless, non-destructive reflection (A241/D)
- contactless, non-destructive reflection, visualization of exact measuring position (A241/D)

Usable sample material:

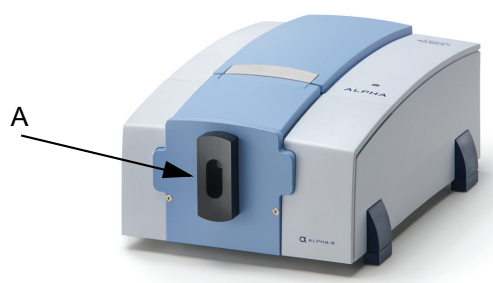
- large sample surfaces
- mural paintings
- coatings on metals

Advantages of this kind of measuring technique:

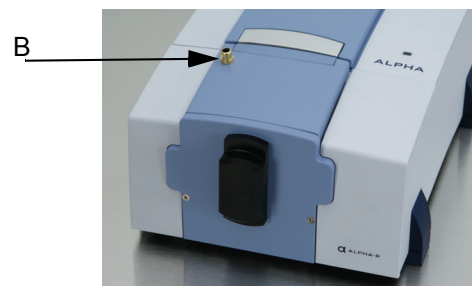
- non-destructive sampling
- no sample preparation required

Application field:

art conservation and restoration



A241/D



A241/DV

Figure 5.4: ALPHA-R sampling module (A241/D, A241/DV)

	Definition
A	Reflection unit
B	Port for video camera cable

Table 5.7: ALPHA-R (A241/D, A241/DV) sampling module - Components

The ALPHA-R (A241/D, A241/DV) sampling module can be mounted on a tripod. Different types of tripods are available with Bruker.



Figure 5.5: ALPHA-R (A241/DV) sampling module mounted on a tripod

5.3.1 Specifications

Spectral range:	375 - 7,500 cm^{-1}
Mirror material:	gold
Sampling spot:	Ø 5 mm (optionally: 3 mm)
Sampling spot distance from spectrometer front:	15 mm

5.3.2 Optical path

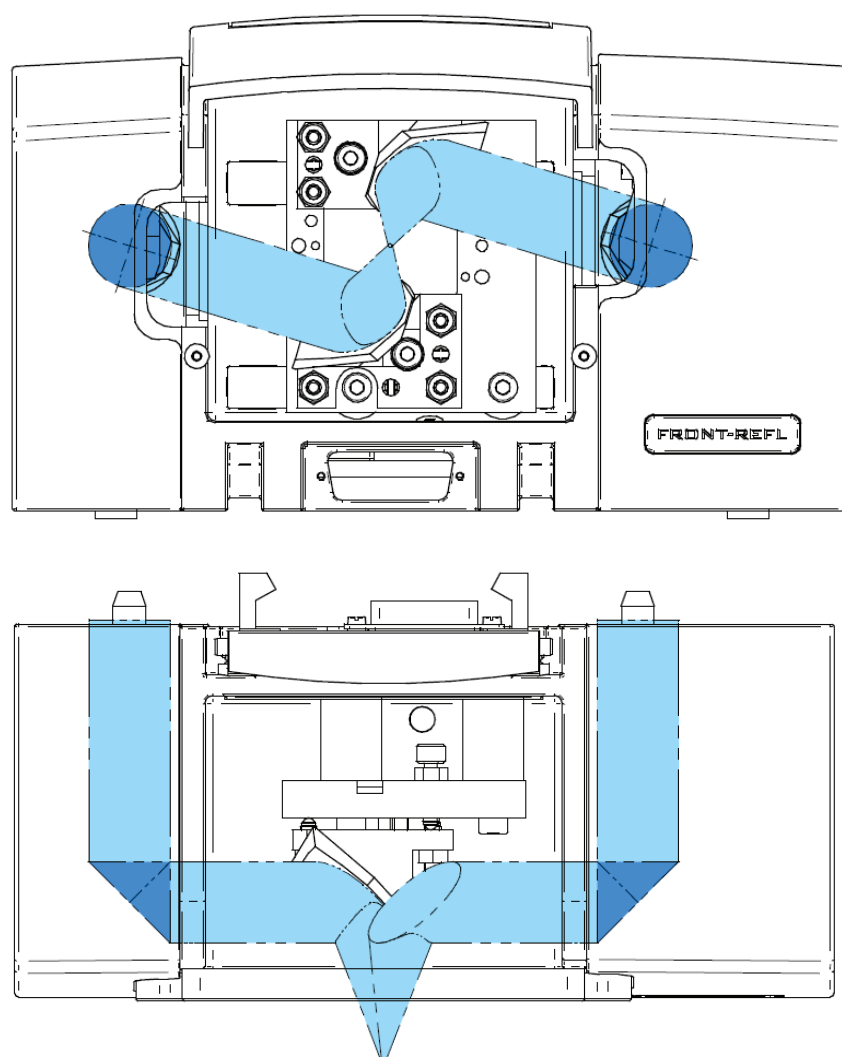


Figure 5.6: ALPHA-R (A241/D) - Optical path

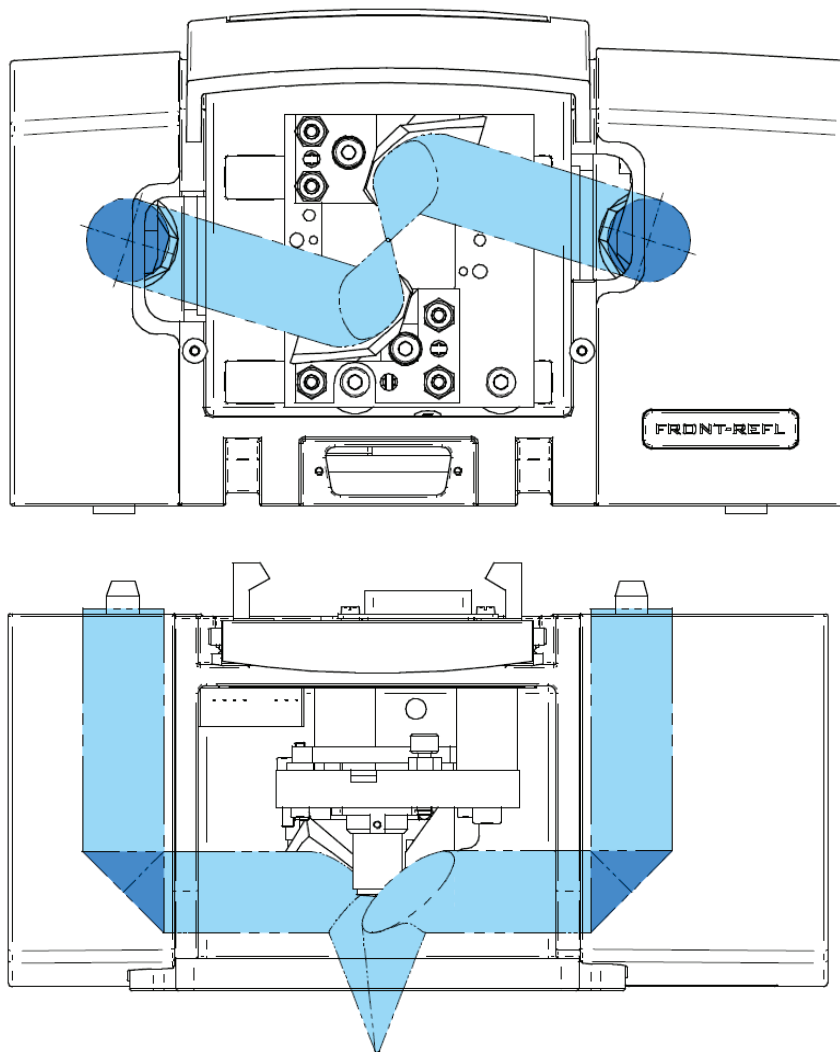


Figure 5.7: ALPHA-R (A241/DV) - Optical path

5.3.3 Software requirements in case of OPUS version <7

When working with an OPUS version <7 and you have purchased the sampling module at a later date than the software, the following requirements must be met:

- OVP database version 6.5.6

The OVP database must be updated (chapter 5.3.3.1).

5.3.3.1 Updating OVP data base

1. Insert the USB stick, which contains the OVP database, into the USB port of the PC.
2. Open the Windows explorer and browse to the directory of the USB stick.
3. Double click the *OVP_Database_Install.exe* file to start the installation program.
 - *The installation program first checks for previously installed components. This automatic search process may take a few seconds.*

Follow the on-screen instructions.

5.3.4 Integrated video camera (only with ALPHA-R A241/DV)

The ALPHA-R A241/DV sampling module contains an integrated video camera for the visualization and documentation of the exact measuring position.

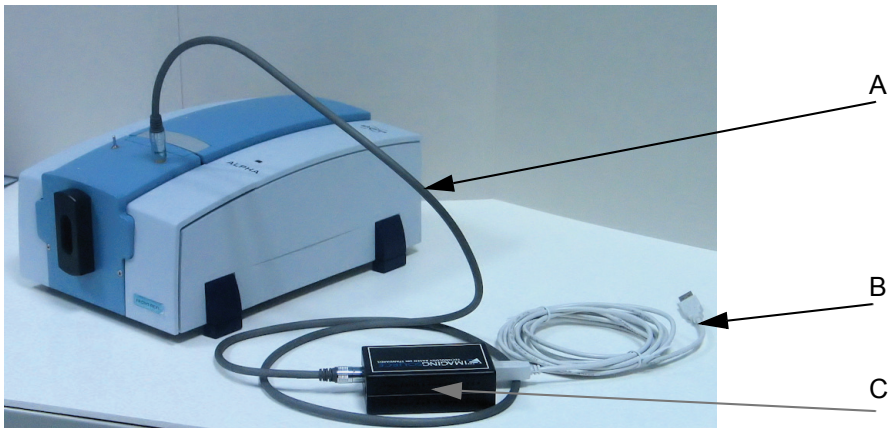


Table 5.8: ALPHA-R (A241/DV)

	Definition
A	Video cable
B	USB cable
C	Video-to-USB converter

Table 5.9: ALPHA-R (A241/DV) sampling module - Components

5.3.4.1 Installing video-to-USB converter

1		Insert one end of the video cable into the video port, located on the spectrometer.
---	---	---

Table 5.10: Installing video-to-USB converter

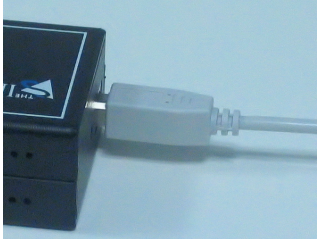
2		Insert the other end of the video cable into the video port, located on the video-to-USB converter.
3		• Insert one end of the USB cable into the USB port, located on the video-to-USB converter. • Insert the other end of the USB cable into the USB port, located on the PC.

Table 5.10: Installing video-to-USB converter

5.3.4.2 Functions of camera control software

The ALPHA-R (A241/DV) sampling module uses the *IC-Capture* camera control software. The camera control software provides the following features:

- setting device parameters
- displaying live image
- acquiring single images and storing them in BMP or JPG file format
- acquiring sequences of single images and incrementing them in BMP or JPG file format
- acquiring image data streams and storing them in AVI file format

5.3.4.3 Installing driver for camera control software

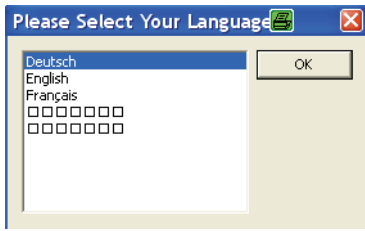
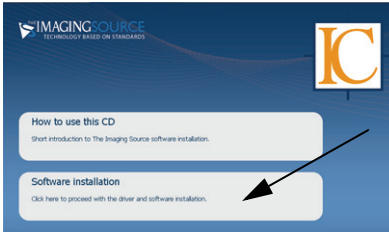
1	• Insert the <i>IC-Capture</i> CD into the CD drive of your PC. • Open the Windows Explorer and double click the <i>Setup.exe</i> file from the main path of the CD. ➤ <i>The installation starts.</i>	
2		Select the program language.
3		Select <i>Software Installation</i> .
4		Select <i>Driver</i> .
5		Select <i>Video Converter</i> .

Table 5.11: Installing driver for camera control software

6		Select <i>USB Converter</i> .
7	Continue installation. - If a message pops up requesting a <i>logo test</i>, click the <i>Next</i> button. - If the driver installation is completed, click the <i>Finish</i> button.	

Table 5.11: Installing driver for camera control software

5.3.4.4 Installing camera control software

1	On the Windows Explorer, double click the <i>Setup.exe</i> file from the main path of the CD. ➤ <i>The installation starts.</i>	
2		Select <i>Software Installation</i> .
3		Select <i>Software for end users</i> .

Table 5.12: Installing camera control software

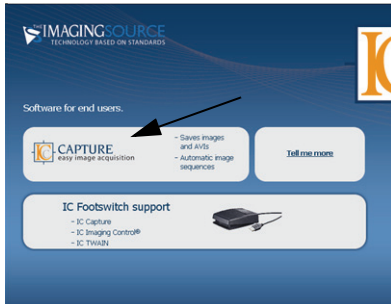
4		Select <i>Capture</i> .
5	Continue installation. ➡ Click the <i>Next</i> button. ➡ If the software installation is completed, click the <i>Finish</i> button.	

Table 5.12: Installing camera control software

5.3.4.5 Starting and configuring camera control software

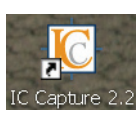
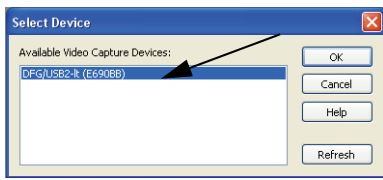
1		Double click the <i>IC-Capture</i> software shortcut, located on the desktop.
2		Select the camera type connected and click OK to confirm.

Table 5.13: Starting and configuring camera control software

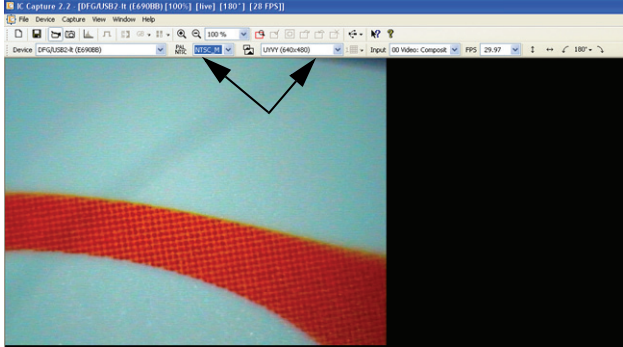
3	When the software starts, the video camera shows the live image of the sample. If the live image is distorted, the settings made in the <i>PAL/NTSC_M</i> menu must be changed.  ☞ Check the correct orientation of the live image by means of a printed object. ☞ If the view is rotated, select the <i>Rotation</i> command on the <i>Device</i> menu.
4	To change between live image and snapshots taken, use the icon  or  in the toolbar.  i Further details on the camera control software are described in the <i>Help</i> menu.

Table 5.13: Starting and configuring camera control software

5.3.5 Performing measurement

When performing a measurement observe the order of the chapters described in the following.

5.3.5.1 Mounting reference cap, starting instrument/performance test

If AAR accessory recognition is activated (which is the case by default), the instrument/performance test starts automatically. In case of the ALPHA-R (A241/D, A241/DV) sampling module, however, you first have to mount the supplied reference cap onto the front side of the sampling module. Otherwise, the tests will fail.

Reference cap with built-in gold coated mirror:	
Spacer:	
Sample cap with sample spot diameter of 3 mm:	
Sample cap with sample spot diameter of 5 mm:	

Table 5.14: Measuring tools supplied

Procedure

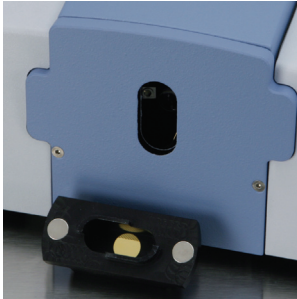
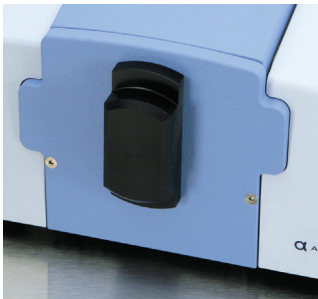
1		Mount the reference cap onto the front side of the sampling module. ➤ <i>The reference cap is kept in position by magnets.</i>
2		• Start OPUS spectroscopy software. ➤ <i>The instrument/performance test starts.</i> • Wait until the test has finished. ➤ <i>The spectrometer status indicator and the OPUS status light must be green.</i>

Table 5.15: Mounting reference cap

5.3.5.2 Loading measurement experiment

- Measurement experiment file is loaded.
 - *If AAR accessory recognition is activated (which is the case by default), the OPUS spectroscopy software recognizes the sampling module used. Thus, the experiment file¹ (DRIFT.XPM) is automatically loaded in OPUS. If the experiment file is not loaded automatically, perform the steps 2 et seq.*
- Start OPUS. On the OPUS Measure menu, select the *Setup Measurement Experiment* command.
 - *The Setup Measurement Experiment dialog opens.*
- Click the *Basic* tab.
- Click the *Load* button and select the *DRIFT.XPM* experiment file from the dialog that opens.
- Click the *Accept & Exit* button.

1. The ALPHA-R sampling module is supplied with the *DRIFT.XMP* experiment file. This experiment file contains the default parameters set by Bruker, to be used to perform reflection measurement.

5.3.5.3 Starting background measurement

1. Keep the reference cap mounted. Make sure that the spectrometer status indicator and OPUS status light are green.
2. On the OPUS *Measure* menu select the *Measurement* command.
 - *The Measure dialog opens.*
3. Click the *Start Background Measurement* button.
4. Wait until the background measurement has finished.

5.3.5.4 Starting sample measurement

1. Remove the reference cap and fix the spacer or sample cap with the particular sample spot diameter onto the front side of the sampling module.
 - *The spacer or holder is kept in position by magnets.*



Figure 5.8: Spacer fixed on the front side of the sampling module

2. Position the sample directly in front of the spacer or sample cap.
 - *If the sample must not be touched, you can measure the sample without using the spacer. Position the sample in front of the sampling module. Keep a distance of about 15 mm from the reflection unit.*
3. Illuminate sample (**optionally**): turn the switch, located on top of the spectrometer housing, to the left.

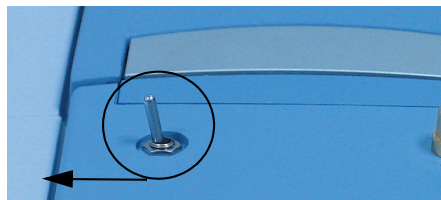


Figure 5.9: Switch to illuminate sample

4. On the *OPUS Measure* menu, select the *Setup Measurement Parameters* command.
5. On the *OPUS Measure* menu select the *Measurement* command.
➤ *The Measure dialog opens.*
6. Click the *Start sample Measurement* button.
7. Wait until the sample measurement has finished.

5.3.6 Potential operating errors

Problem	Troubleshooting
Instrument/performance test failed	Check whether reference cap is mounted (chapter 5.3.5.1).
No sample spectrum or bad spectrum quality	• Sample not properly positioned when measuring without spacer • Keep a distance of at least 15 mm from the reflection unit. • If required, optimize signal: ☞ On the <i>OPUS Measure</i> menu, select the <i>Advanced Measurement</i> command. ☞ Click the <i>Check Signal</i> tab. • Sample consistency does not allow reflection measurement (e.g. in case of very dark, coarse or porous sample material), other type of measuring required (e.g. ATR)

Table 5.16: Potential operating errors

5.4 ALPHA-R (A240/DU) sampling module

Measurement type:	• specular reflection from bottom up • diffuse reflection from bottom up
Usable sample material:	• reflective materials • textiles
Application field:	art conservation and restoration
Two sample holders included in delivery content:	for solid samples with a diameter of 3 mm or 5 mm

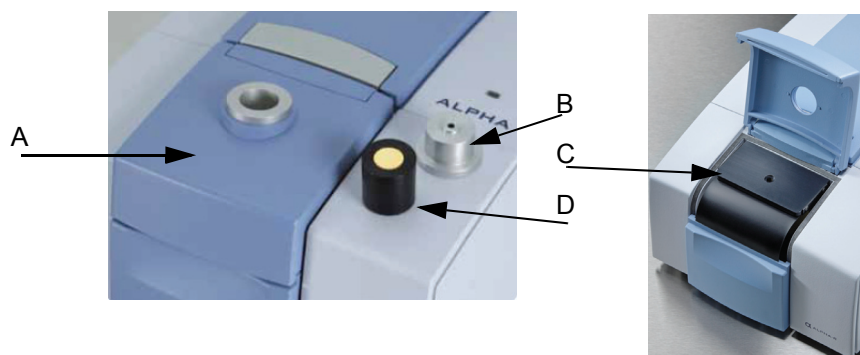


Figure 5.10: ALPHA-R sampling module (A240/DU)

	Definition
A	Reflection unit, closed with sample holder
B	Sample holder
C	Reflection unit, open with fixed sampling plate
D	Gold mirror for reference measurements

Table 5.17: ALPHA-R (A240/DU) sampling module - Components

5.4.1 Specifications

Spectral range:	375 - 7,500 cm^{-1} (with KBr windows)
Sampling spot (depending on sample holder):	• 5 mm • 3 mm
Sampling holder:	• holder for flat samples • sample cup for granules (>3 mm, >5 mm)

5.4.2 Optical path

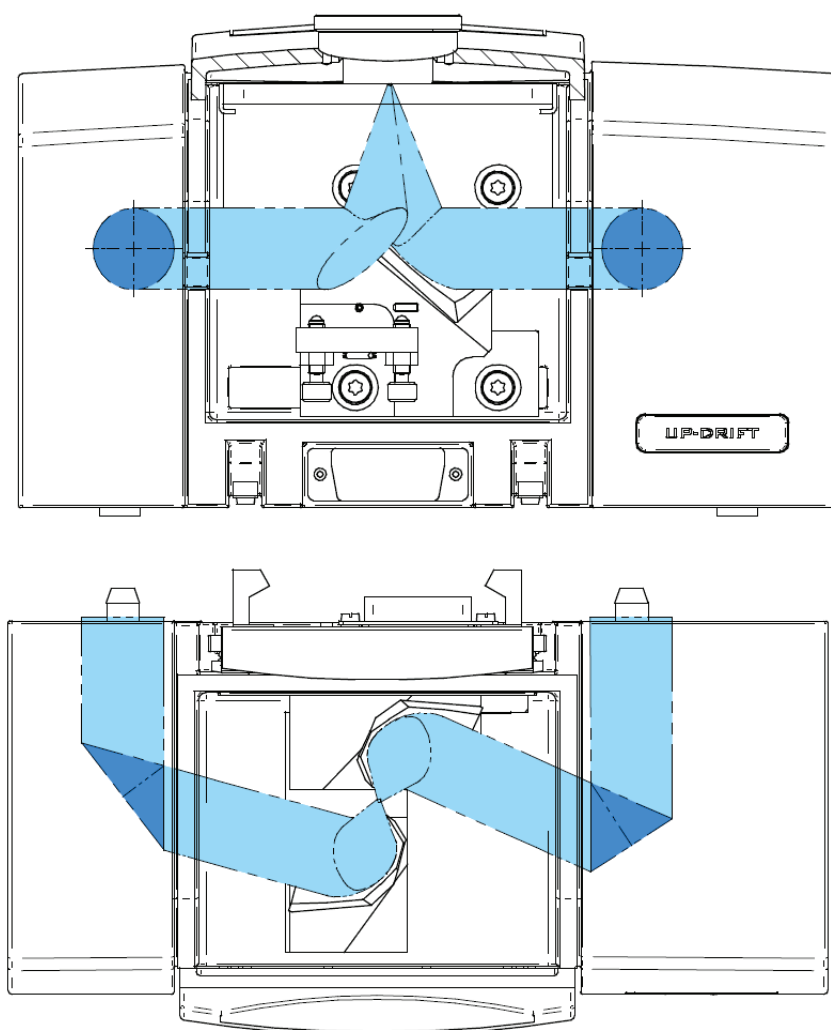


Figure 5.11: ALPHA-R (A240/DU) - Optical path

5.4.3 Software requirements in case of OPUS version <7

When working with an OPUS version <7 and you have purchased the sampling module at a later date than the software, the following requirements must be met:

- OVP database version 6.5.6

The OVP database must be updated (chapter 5.4.3.1).

5.4.3.1 Updating OVP data base

1. Insert the USB stick, which contains the OVP database, into the USB port of the PC.
2. Open the Windows explorer and browse to the directory of the USB stick.
3. Double click the *OVP_Database_Install.exe* file to start the installation program.
 - *The installation program first checks for previously installed components. This automatic search process may take a few seconds.*

Follow the on-screen instructions.

5.4.4 Performing measurement

When performing a measurement observe the order of the chapters described in the following.

5.4.4.1 Inserting reference mirror, starting instrument/performance test

If AAR accessory recognition is activated (which is the case by default), the instrument/performance test starts automatically. In case of the ALPHA-R (A240/DU) sampling module, however, you first have to insert the supplied reference mirror into the measuring position. Otherwise, the tests will fail.

1		Insert reference mirror into the measuring position, with the mirror surface facing downward.
2	• Start OPUS spectroscopy software. ➤ <i>The instrument/performance test starts.</i> • Wait until the test has finished. ➤ <i>The spectrometer status indicator and the OPUS status light must be green.</i>	

Table 5.18: Inserting reference mirror

5.4.4.2 Loading measurement experiment

1. Measurement experiment file is loaded.
 - If AAR accessory recognition is activated (which is the case by default), *the OPUS spectroscopy software recognizes the sampling module used. Thus, the experiment file¹ (DRIFT.XPM) assigned to the sampling module is automatically loaded in OPUS. If the experiment file is not loaded automatically, perform step 2 et seq.*
2. Start OPUS. On the OPUS *Measure* menu, select the *Setup Measurement Experiment* command.
 - *The Setup Measurement Experiment dialog opens.*
3. Click the *Basic* tab.
4. Click the *Load* button and select the *DRIFT.XPM* experiment file from the dialog that opens.
5. Click the *Accept & Exit* button.

5.4.4.3 Starting background measurement

1. Keep the reference mirror inserted into the measuring position. Make sure that the spectrometer status indicator and the OPUS status light must be green.
2. On the OPUS *Measure* menu select the *Measurement* command.
 - *The Measure dialog opens.*
3. Click the *Start Background Measurement* button.
4. Wait until the background measurement has finished.

1. The ALPHA-R sampling module is supplied with the *DRIFT.XPM* experiment file. This experiment file contains the default parameters set by Bruker, to be used for reflection measurement.

5.4.4.4 Starting sample measurement

1. Position sample:

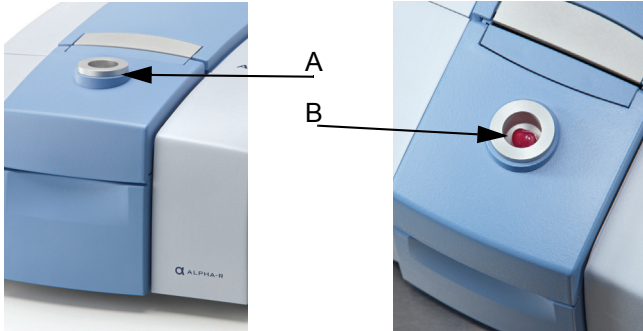
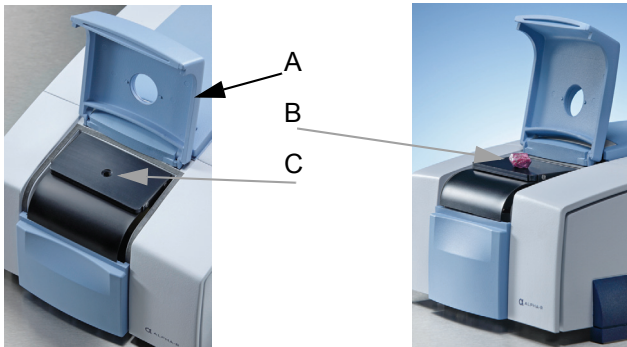
Small samples^a: ➤ Insert sample holder (A) into measuring position. ➤ Fill sample (B) into sample holder.	 The left photograph shows a small, circular sample holder (A) being inserted into the measuring position of the ALPHA-5. The right photograph shows the same holder (A) filled with a red sample (B).
Larger samples: ➤ Lift up the sample compartment lid (A). ➤ Place the sample (B) directly onto the measuring position (C) of the sample stage. The measuring position must be completely covered by the sample. ➤ Keep the lid open during measurement.	 The left photograph shows the sample compartment lid (A) being lifted. The right photograph shows a larger sample (B) placed directly onto the measuring position (C) of the sample stage.

Table 5.19: Positioning sample

a. Depending on the sample holder used, the sample must have a diameter of at least 3 or 5 mm.

- On the OPUS *Measure* menu select the *Measurement* command.
➤ *The Measure dialog opens.*
- Click the *Start Sample Measurement* button.
- Wait until the sample measurement has finished.

5.4.5 Potential operating errors

Possible cause	Troubleshooting
Instrument/performance test failed	Check whether the reference is correctly inserted, with the mirror surface facing downwards (chapter 5.4.4.1).
No sample spectrum or bad spectrum quality	Sample consistency does not allow reflection measurement (e.g. in case of very dark sample material), other type of measuring required (e.g. ATR)

Table 5.20: Potential operating errors

6 ALPHA for gas analysis

Measurement type:	gas analysis
Usable sample material:	gaseous samples
Application field:	industrial research or process applications

6.1 Overview on ALPHA spectrometer configurations for gas analysis

ALPHA spectrometer configurations for gas analysis	
	
ALPHA-G (A139, with long-path gas cell)	ALPHA-G with A139-H1 option (heating jacket)

Table 6.1: Overview on ALPHA spectrometer configurations for gas analysis

6.2 ALPHA-G sampling module (A139/D)

Measurement type:	gas analysis
Usable sample material:	gases
Accessory set:	heating jacket for gas cell with a path length of 4.8 m

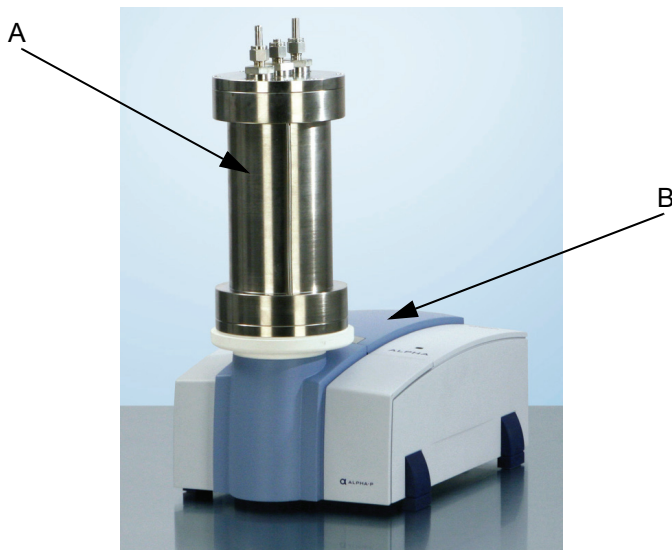


Figure 6.1: ALPHA-G (A139/D) sampling module

	Definition
A	Long-path gas cell
B	Basic module

Table 6.2: ALPHA-G (A139/D) sampling module - Components

6.2.1 Specifications

Path length:	4.8 m
Volume:	approx. 0.5 l
Gas cell hardware:	aluminium electroless nickel
Temperature range:	from ambient to 200°C

Pressure range:	from vacuum to 3.5 bar
Power consumption:	300 W
Maximum gas flow rate:	50 l/min
Protection class:	I, complies with IEC 61140

6.2.2 Optical path

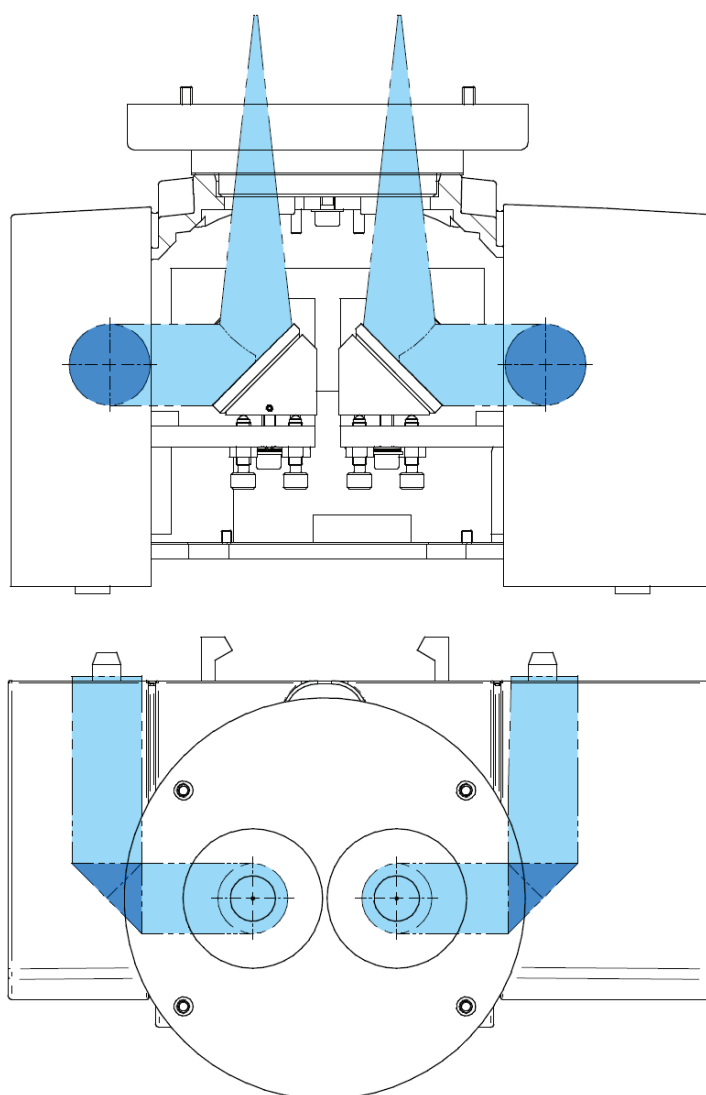


Figure 6.2: ALPHA-G (A139/D) sampling module - Optical path

6.2.3 Software requirements in case of OPUS version <7

When working with an OPUS version <7 and you have purchased the sampling module at a later date than the software, the following requirements must be met:

- OVP database version 6.5.6

The OVP database must be updated (chapter 6.2.3.1).

6.2.3.1 Updating OVP data base

1. Insert the USB stick, which contains the OVP database, into the USB port of the PC.
2. Open the Windows explorer and browse to the directory of the USB stick.
3. Double click the *OVP_Database_Install.exe* file to start the installation program.
 - *The installation program first checks for previously installed components. This automatic search process may take a few seconds.*
4. Follow the on-screen instructions.

6.2.4 Installing gas connections

The gas cell is equipped with a gas inlet and outlet. The gas inlet and outlet are 1/4" Swagelok connections.

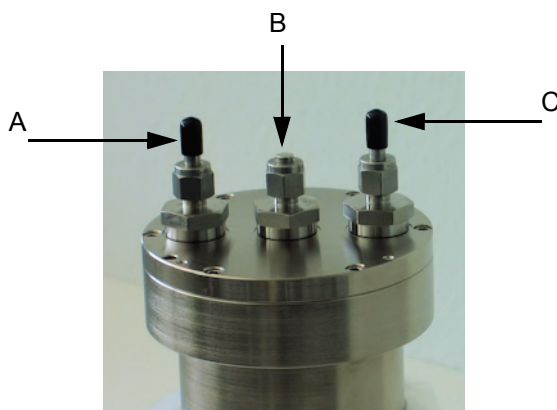


Figure 6.3: ALPHA-G (A139/D) sampling module - Gas cell ports

	Definition
A	Gas outlet
B	Blind plug
C	Gas inlet

Table 6.3: ALPHA-G (A139/D) sampling module - Components

To connect the gas supply lines you first have to remove the black protective caps.

6.2.4.1 Safety instructions when operating the gas cell

Before being able to operate the gas cell you have to observe the following safety instructions:



CAUTION

Improper sampling techniques

Personal injury or property damage.

➤ Never use oxidizing, corrosive or toxic samples in connection with the gas cell.

1. Define the gas flow volume. The maximum gas flow rate should not exceed 50 l/hour. Use a mass flow rate controller.
2. Limit the maximum allowed pressure within the cell to 3.5 bar.
 - *The gas cell is NOT delivered with a safety valve. It is the operator's duty to take safety measures which prevent the pressure inside the cell from exceeding the maximum.*
3. Set and check the temperature by means of the temperature control unit.
4. Make sure that the gas cell reaches the appropriate operating temperature. Only after the operating temperature has been reached (nominal/ actual value comparison) is a measurement reasonable.

6.2.5 Performing measurement

When performing measurement observe the order of the chapters described in the following.

6.2.5.1 Starting instrument/performance test

If AAR accessory recognition is activated (which is the case by default), the instrument/performance test starts automatically. In case of the ALPHA-G (A139/D) sampling module, however, you first have to fill in nitrogen into the gas cell. Otherwise, the tests will fail.

1	• Start OPUS spectroscopy software. • Follow the on-screen instructions. • Wait until the instrument/performance test has finished.
---	---

Table 6.4: Starting instrument/performance test

6.2.5.2 Loading measurement experiment

1. Measurement experiment file is loaded.
 - If AAR accessory recognition is activated (which is the case by default), the OPUS spectroscopy software recognizes the sampling module used. Thus, the experiment file¹ (TRANS.XPM) assigned to the sampling module is automatically loaded in OPUS. If the experiment file is not loaded automatically, perform the following steps.
2. Start OPUS. On the *Measure* menu, select the *Setup Measurement Experiment* command.
 - The *Setup Measurement Experiment* dialog opens.
3. Click the *Basic* tab.
4. Click the *Load* button and select the TRANS.XPM experiment file from the dialog that opens.
5. Click the *Accept & Exit* button.

6.2.5.3 Starting background measurement

1. Fill in reference gas (nitrogen) into the gas cell. The spectrometer status indicator and the OPUS status light must be green.
2. On the OPUS *Measure* menu select the *Measurement* command.
 - The *Measure* dialog opens.
3. Click the *Start Background Measurement* button.
4. Wait until the background measurement has finished.

6.2.5.4 Starting sample measurement

1. Fill in sample gas into the gas cell.
2. On the OPUS *Measure* menu, select the *Setup Measurement Parameters* command.
3. Click the *Check Signal* tab.
4. Close the dialog.
5. On the OPUS *Measure* menu select the *Measurement* command.
 - The *Measure* dialog opens.
6. Click the *Start sample Measurement* button.
7. Wait until the sample measurement has finished.

1. The ALPHA-G sampling module is supplied with the TRANS.XPM experiment file. This experiment file contains the default parameters set by Bruker, to be used to perform gas analysis.

6.2.6 Potential operating errors

Problem	Troubleshooting
Instrument/performance test failed	Fill in the gas cell with nitrogen.
Varying width and form of bands in the sample spectrum	Fill in the gas cell with constant pressure. In case of flow-through measurements ensure a constant flow rate (use a rotameter).

Table 6.5: Potential operating errors

6.3 ALPHA-G sampling module with A139-H1 option (heating jacket)

Measurement type:	gas analysis
Gas cell:	heatable
Usable sample material:	gases

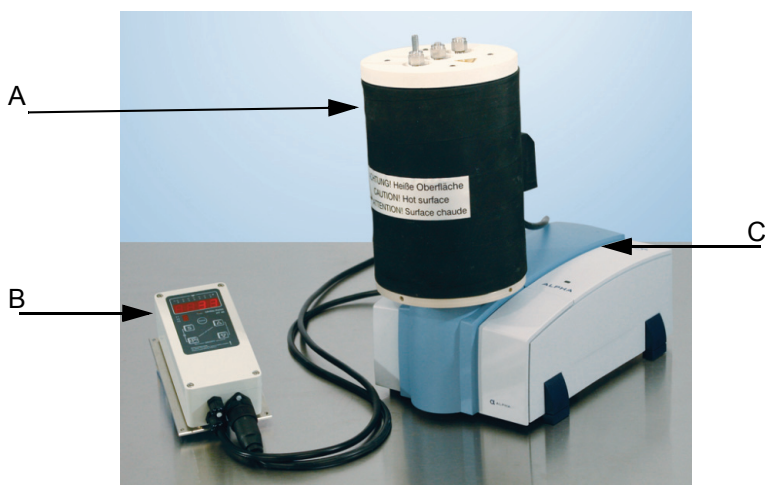


Figure 6.4: ALPHA-G sampling module with A139-H1 option (heating jacket)

	Definition
A	Heating jacket with connecting cable (1.5 m)
B	Temperature control unit
C	Basic module

Figure 6.5: ALPHA-G sampling module with A139-H1 option (heating jacket) - Components

6.3.1 Specifications

6.3.1.1 Heatable gas cell

Path length:	4.8 m
Weight:	2.4 kg
Gas cell hardware:	aluminium electroless nickel
Temperature range:	from ambient to 200°C
Pressure range:	from vacuum to 3.5 bar
Power consumption:	300 W

6.3.1.2 Heating jacket

Dimensions:	155 x 210 x 195 cm (w x d x h)
Weight:	2.4 kg

6.3.1.3 Temperature control unit

Dimensions:	100 x 160 x 70 cm (w x d x h)
Weight:	0.9 kg

6.3.2 Optical path

Identical to figure 6.2.

6.3.3 Software requirements in case of OPUS version <7

When working with an OPUS version <7 and you have purchased the sampling module at a later date than the software, the following requirements must be met:

- OVP database version 6.5.6

The OVP database must be updated (chapter 6.2.3.1).

6.3.3.1 Updating OVP data base

1. Insert the USB stick, which contains the OVP database, into the USB port of the PC.
2. Open the Windows explorer and browse to the directory of the USB stick.
3. Double click the *OVP_Database_Install.exe* file to start the installation program.
 - *The installation program first checks for previously installed components. This automatic search process may take a few seconds.*
4. Follow the on-screen instructions.

6.3.4 Functionality of heating jacket

The heating jacket heats the gas cell, and largely protects the gas cell against ambient influences such as draught and sun exposure, as well as condensation of vapor.

The heating jacket is protected with a safety fuse against overheating. The fuse disengages at approx. 240°C. After disengaging, the heating jacket cannot be operated any more and must be sent back to Bruker for repairing.

The 4.8 m gas cell is at operating temperature of 200°C after 2 hours, with a set value of 220°C. This leads to a temperature offset of approx. 20°C at this operating point.

6.3.5 Mounting heating jacket

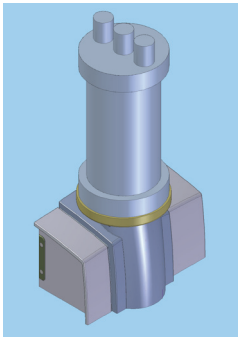
1		By default, the gas cell (4,8 m) is mounted on the sampling module.
---	---	---

Table 6.6: Mounting heating jacket

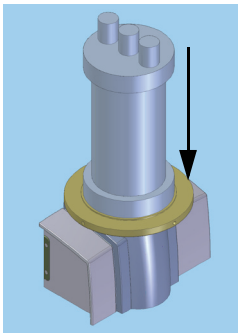
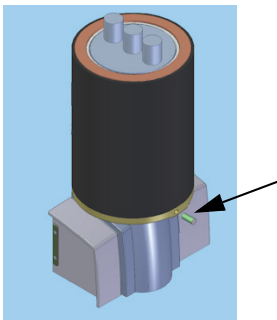
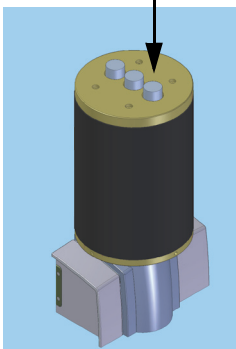
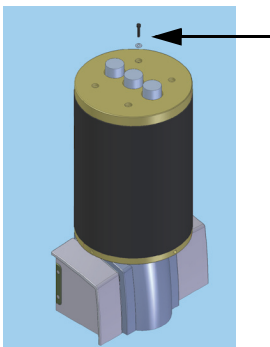
2		Carefully put the PEEK ^a ring (from top to bottom) onto the sampling module.
3		• Pull the heating jacket over the gas cell. • Tighten all hexagon sockets, located on the PEEK ring.
4		Put on ceramic lid.
5		Slightly tighten the Allen screws with washers.

Table 6.6: Mounting heating jacket

a. PEEK: polyether-etherketone, flexible thermoplastic material

6.3.6 Safety instructions

When a safe operation can no longer be ensured, shut down the heating and take measures to prevent it from being turned on again accidentally.

Safe operation is no longer possible if the heating jacket:

- shows visible damage
- does not work properly



CAUTION

Using ALPHA-G with A139-H1 option (heating jacket) at temperatures above 55°C

Risk of skin burn

➤ Wear safety gloves.

6.3.7 Temperature control unit

The automatic temperature control unit of the series HT43 works on a microprocessor basis for industrial applications. Depending on the parameters set by the manufacturer, the temperature control unit regulates released heating elements with defined sensors.



The temperature displayed on the temperature control unit refers to the temperature of the heating jacket. The temperature inside the gas cell is about 20°C lower than the one displayed on the temperature control unit.

6.3.7.1 Changing upper limit of set value

The upper limit of the set value is 200°C. As a consequence, the inner temperature of the gas cell gradually produces values between 175 and 180°C. If the upper limit of the set value is to be changed, proceed as follows:

1. On the temperature control unit, press the  key at least 7 seconds.
➤ *CONF Mode (Configuration tablet) is displayed.*
2. Press the  key to set the value from 0 to 1.
3. Click the  key to confirm.
4. Click the  key to set a value of 33 (upper set value).
5. Click the  key to confirm.

6. Use the  or  key to change the upper set value.
7. To store the value press the  key at least 3 seconds.



Details on how to change other types of control parameters are described in the manufacturer's documentation.

6.3.8 Performing measurement

When performing measurement observe the order of the chapters described in the following.

6.3.8.1 Starting instrument/performance test

If AAR accessory recognition is activated (which is the case by default), the instrument/performance test starts automatically. In case of the ALPHA-G (A139/D) sampling module, however, you first have to fill in nitrogen into the gas cell. Otherwise, the tests will fail.

1	• Start OPUS spectroscopy software. • Follow the on-screen instructions. • Wait until the instrument/performance test has finished.
---	---

Table 6.7: Starting instrument/performance test

6.3.8.2 Loading measurement experiment

1. Measurement experiment file is loaded.
 - *If AAR accessory recognition is activated (which is the case by default), the OPUS spectroscopy software recognizes the sampling module used. Thus, the experiment file¹ (TRANS.XPM) assigned to the sampling module is automatically loaded in OPUS. If the experiment file is not loaded automatically, perform the following steps.*
2. Start OPUS. On the *Measure* menu, select the *Setup Measurement Experiment* command.
 - *The Setup Measurement Experiment dialog opens.*
3. Click the *Basic* tab.
4. Click the *Load* button and select the *TRANS.XPM* experiment file from the dialog that opens.
5. Click the *Accept & Exit* button.

1. The ALPHA-G sampling module is supplied with the *TRANS.XPM* experiment file. This experiment file contains the default parameters set by Bruker, to be used to perform gas analysis.

6.3.8.3 Starting background measurement

1. Fill in reference gas (nitrogen) into the gas cell. The spectrometer status indicator and the OPUS status light must be green.
2. On the OPUS *Measure* menu select the *Measurement* command.
➤ *The Measure dialog opens.*
3. Click the *Start Background Measurement* button.
4. Wait until the background measurement has finished.

6.3.8.4 Starting sample measurement

1. Fill in sample gas into the gas cell.
2. On the OPUS *Measure* menu, select the *Setup Measurement Parameters* command.
3. Click the *Check Signal* tab.
4. Close the dialog.
5. On the OPUS *Measure* menu select the *Measurement* command.
➤ *The Measure dialog opens.*
6. Click the *Start sample Measurement* button.
7. Wait until the sample measurement has finished.

6.3.9 Potential operating errors

Problem	Troubleshooting
Instrument/performance test failed	Fill in the gas cell with nitrogen.
Varying width and form of bands in the sample spectrum	Fill in the gas cell with constant pressure. In case of flow-through measurements ensure a constant flow rate (use a rotameter).

Table 6.8: Potential operating errors

7 Quick User Guide

This quick user guide gives an overview of the OPUS spectroscopy software, and allows to learn how to perform the most relevant analytical tasks on your own without long training periods.

This chapter describes the procedures for the solution of the following practical questions:

- **Sample Measurement:** How to perform a spectroscopic measurement? How to manipulate a sample spectrum? How to save and/or print a spectrum?
- **Quality Control:** Are the delivered goods or the self-made products consistent with the specified quality criteria?
- **Identification of an unknown substance:** What kind of substance is it?
- **Quantitative Analysis:** How much of a certain component does the sample contain?

i In OPUS, you can either use the default MENTOR wizard or the distinctive menu to select the commands which allow to measure, manipulate and evaluate spectra. This chapter mainly uses the wizard variant in terms of the single operating process.

7.1 Switching on spectrometer

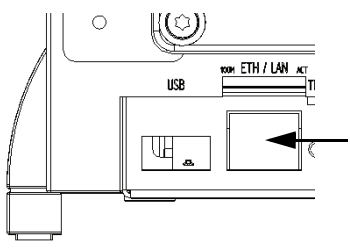
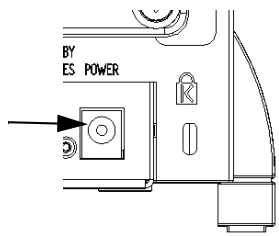
1		Connect one end of the data cable to the ETH/LAN port, located on the spectrometer rear side.
		Connect the other end of the data cable to the network/LAN port of the PC.
2		• Connect the extra low-voltage connector of the power cord to the POWER port, located on the spectrometer rear side. • Use the clamp to secure the low-voltage connector against unintentional unplugging.
3	Connect the safety plug of the power cord to the mains socket.	

Table 7.1: Switching on spectrometer

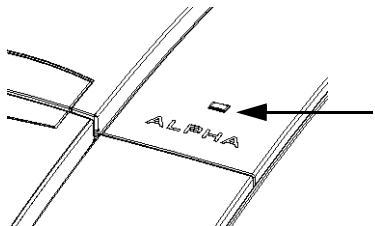
4		Wait until the spectrometer status indicator^a is green. i The spectrometer status indicator becomes green if the IR source is warmed up (after about 7 minutes).
---	---	--

Table 7.1: Switching on spectrometer

a. Further details on the spectrometer status indicator are described in chapter 10.2.1.

7.2 Starting OPUS spectroscopy software

1. Switch on PC.
2. On the Windows Start menu, click the OPUS icon.
 - *The OPUS Login dialog is displayed.*
3. Enter user ID and password.
4. Select the *ALPHA.ows* workspace.
5. Click the *Login* button.
 - *The About OPUS dialog is displayed.*
6. Click the *OK* button.
 - *The OPUS user interface is displayed.*

7.2.1 After starting OPUS spectroscopy software

The performance test¹ starts automatically, indicated by a yellow message bubble on the bottom right end of the software interface. Wait until the performance test has finished, and the OPUS status light is green.

i If the OPUS status light does not become green, this can be the result of different reasons (see chapter 10.2.2). If the performance test fails, see chapter 10.5 et seq.

1. If the instrument test has expired, the instrument test starts immediately after the performance test has finished. Both tests check whether the spectrometer achieves the specified performance. More information on both tests is given in chapter 10.5 et seq. and appendix D.

7.3 Measuring

7.3.1 Loading measurement experiment

If AAR accessory recognition is activated (which is the case by default), the OPUS spectroscopy software recognizes the sampling module used. Thus, the experiment file¹ assigned to the sampling module is automatically loaded in OPUS. If the experiment file is not loaded automatically, perform step 1 et seq.

1. Start OPUS. On the *Measure* menu, select the *Setup Measurement Experiment* command.
➤ *The Setup Measurement Experiment dialog opens.*
2. Click the *Basic* tab.
3. Click the *Load* button and select the distinctive experiment file from the dialog that opens.
4. Click the *Accept & Exit* button.

7.3.2 Editing measurement parameters

It is recommended to perform a measurement using the distinctive experiment file with the measurement parameters set by Bruker².

The OPUS spectroscopy software allows to adapt the measurement parameter settings (e.g. the *Scan Time* parameter value) to your special demands.

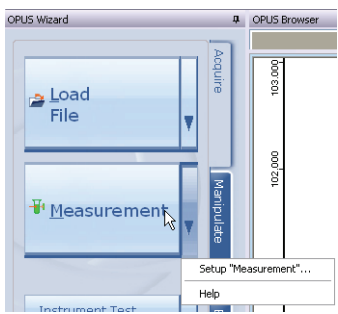
1		• In the MENTOR wizard, click the <i>Measure</i> level. • Click the arrow next to the <i>Measure</i> button.
---	---	---

Table 7.2: Setting up measurement parameters

1. The experiment file contains the default parameters for measurement, set by Bruker. The following experiment files are available for the ALPHA spectrometer: ATR_Di.XPM, ATR_Ge.XPM, ATR_ZnSe.XPM, DRIFT.XPM, TRANS.XPM.
2. If you use the measurement parameters, the spectrum is stored by default in *OPUS/MEAS*. The file name is identical to the sample name you have entered in the *Measurement* dialog. If you do not change the sample name, the file name is incremented with any further measurement.

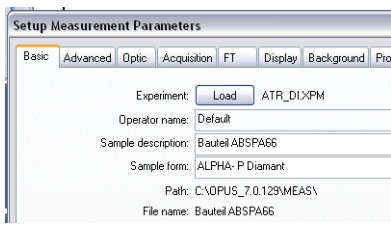
2		• On the pop-up menu, click the <i>Setup Measurement Parameters</i> command. • Change the specific measurement parameters and save the changes made. i Details on the measurement parameters are described in the OPUS/MENTOR manual.
---	---	--

Table 7.2: Setting up measurement parameters

7.3.3 Starting background measurement

Before starting sample measurement you first have to perform a background measurement, i.e. a measurement without any sample. The measurement conditions must be identical for both the background and sample measurement.

Depending on the type of sampling module used, you may have to position a reference mirror to perform the background measurement. Details are described in the different sampling module chapters.

1. On the MENTOR wizard, select the *Acquire* level and click the *Advanced Measurement* button.
 - *The Measure dialog opens.*
2. Click the *Start Background Measurement* button.
3. Wait until the background measurement has finished.
 - *The progress of the background measurement is shown in the OPUS status bar.*

7.3.4 Starting sample measurement

1. On the MENTOR wizard, select the *Acquire* level and click the *Advanced Measurement* button.
 - *The Measure dialog opens.*
2. Click the *Start Sample Measurement* button.
3. Wait until the sample measurement has finished.
 - *The progress of the sample measurement is shown in the OPUS status bar.*

i When the measurement has finished, the measured sample spectrum is displayed in the spectrum window. The MENTOR wizard view changes to the *Manipulate* level.

7.4 Manipulating spectrum

After sample measurement has finished, a spectrum postprocessing may be required to ensure that the OPUS spectroscopy software yields acceptable evaluation results, for example when you intend to perform a substance identification or a quantitative analysis.

For this purpose, OPUS provides a number of manipulation commands. In this context, *manipulating* means optimizing the measured sample spectrum.

This chapter describes the following manipulation commands:

- baseline correction
- atmospheric compensation

More manipulation commands are described in the OPUS/MENTOR manual.

7.4.1 Baseline correction

A baseline correction is always highly advisable if the spectrum baseline deviates strongly from a theoretical horizontal line. This phenomenon may occur, for example, when you measure KBr pellets.

Figure 7.1 illustrates the effect of a baseline correction. More details are described in the OPUS/MENTOR manual.

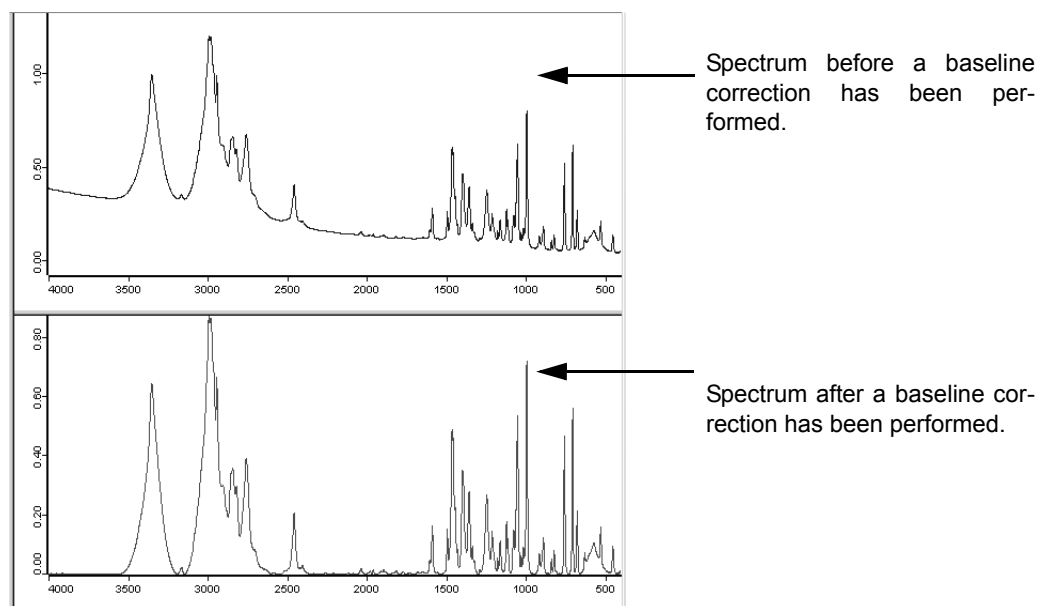


Figure 7.1: Baseline correction

7.4.1.1 Performing baseline correction

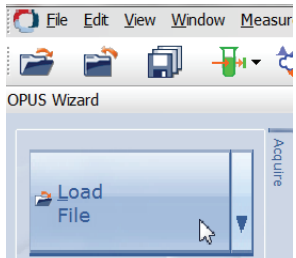
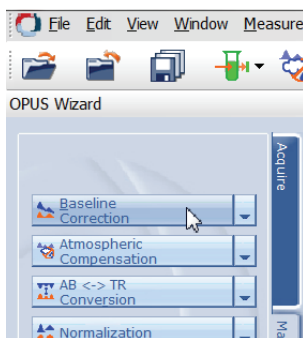
1		• In the MENTOR wizard, on the <i>Acquire</i> level click the <i>Load File</i> button. • Select the desired spectrum file from the dialog shown. ➤ <i>The spectrum is displayed in the spectrum window. The Manipulate level is shown on the MENTOR wizard.</i>
2		On the <i>Manipulate</i> button, click the <i>Baseline Correction</i> button. ➤ <i>The baseline correction is performed. The corrected spectrum is displayed in the spectrum window.</i> i To undo baseline correction, select the <i>Undo Changes</i> command in the <i>File</i> menu.

Table 7.3: Performing baseline correction

7.4.2 Atmospheric compensation

Atmospheric compensation eliminates H₂O and/or CO₂ bands from a sample spectrum. These bands result from the ambient air and occur if there is a difference in the H₂O and/or CO₂ concentration between the background measurement and sample measurement.

Figure 7.2 illustrates the effect of an atmospheric compensation. More details are described in the OPUS/MENTOR manual.

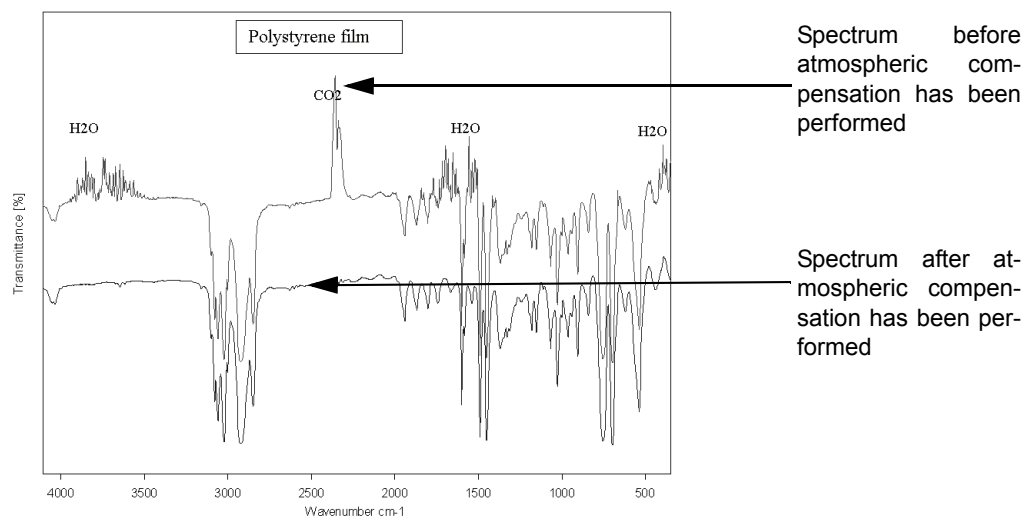


Figure 7.2: Atmospheric compensation

7.4.2.1 Performing atmospheric compensation

i Atmospheric compensation requires a spectrum file that contains an SSC (Sample Single Channel) and RSC (Reference Single Channel) data block. Otherwise, atmospheric compensation cannot be performed.

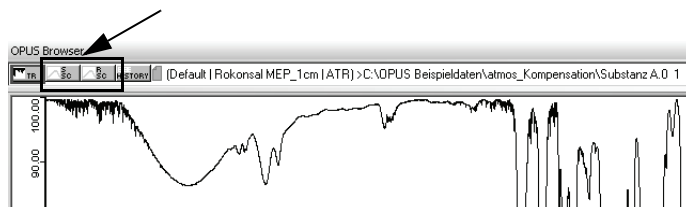


Figure 7.3: Spectrum file with SSC and RSC data block

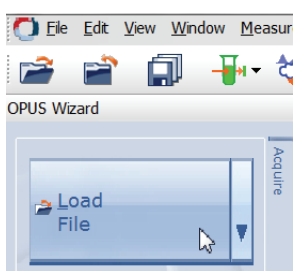
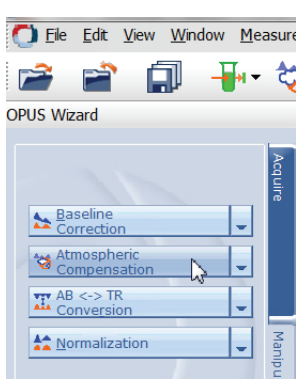
1		• In the MENTOR wizard, on the <i>Acquire</i> level click the <i>Load File</i> button. • Select the desired spectrum file from the dialog shown. ➤ <i>The spectrum is displayed in the spectrum window. The Manipulate level is shown on the MENTOR wizard.</i>
2		On the <i>Manipulate</i> button, click the <i>Atmospheric Compensation</i> button. ➤ <i>The atmospheric compensation is performed. The manipulated spectrum is displayed in the spectrum window.</i> i To undo atmospheric compensation, select the <i>Undo Changes</i> command in the <i>File</i> menu.

Table 7.4: Performing atmospheric compensation

7.4.3 Peak picking

Peak picking allows to determine the exact spectrum peak positions very quickly. The spectrum window shows the exact frequency values (x-values), at which maxima (in case of an absorption spectrum) or minima (in case of an transmission spectrum) occur.

7.4.3.1 Performing peak picking

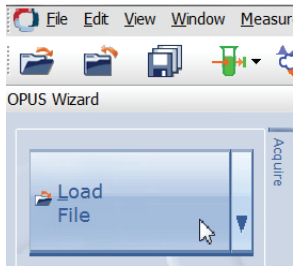
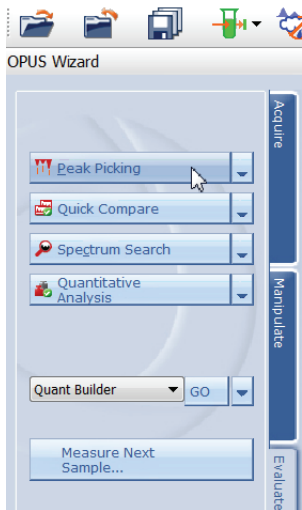
1		• In the MENTOR wizard, on the <i>Acquire</i> level click the <i>Load File</i> button. • Select the desired spectrum file from the dialog shown. ➤ <i>The spectrum is displayed in the spectrum window. The Manipulate level is shown on the MENTOR wizard.</i>
2		Go to the <i>Evaluate</i> level, and click the <i>Peak Picking</i> button. ➤ <i>Peak picking is performed.</i>

Table 7.5: Performing peak picking

3

The spectrum window shows the exact frequency values, at which maxima or minima occur.

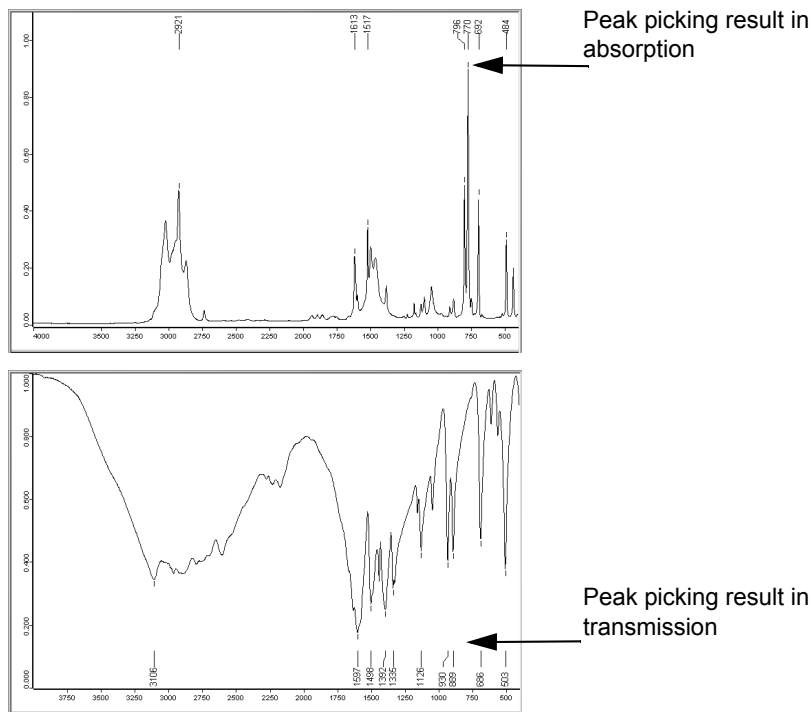


Table 7.5: Performing peak picking

7.4.3.2 Performing interactive peak picking

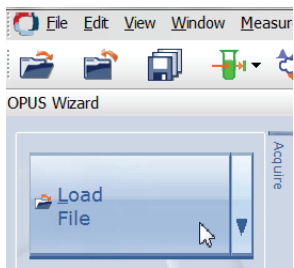
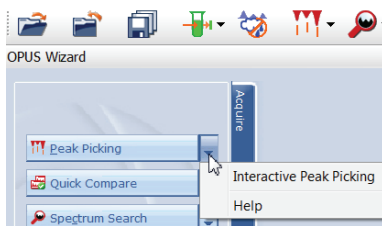
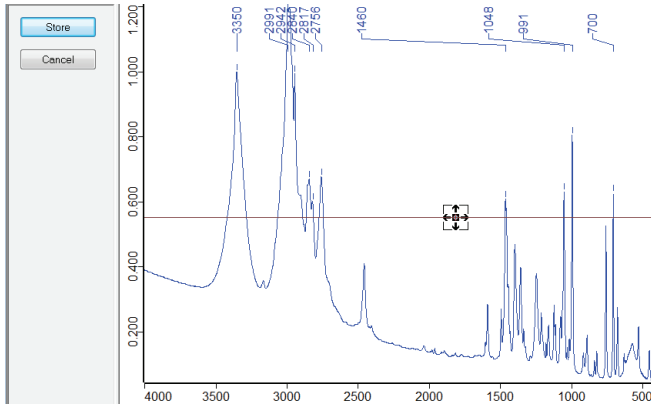
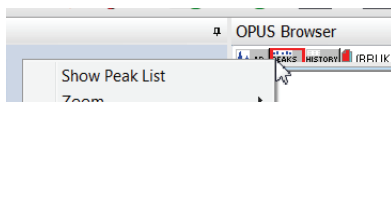
1		• In the MENTOR wizard, on the <i>Acquire</i> level click the <i>Load File</i> button. • Select the desired spectrum file from the dialog shown. ➤ The spectrum is displayed in the spectrum window. The <i>Manipulate</i> level is shown on the MENTOR wizard.
2		• Go to the <i>Evaluate</i> level, and click the arrow button next to the <i>Peak Picking</i> button. • On the pop-up menu displayed, click the <i>Interactive Peak Picking</i> command. ➤ The spectrum window is displayed with an auxiliary line.
3	• Position the cursor onto the  icon. • Press the left mouse button and move the auxiliary line upwards or downwards.  i In case of absorption spectra, the frequency values of all peaks above the auxiliary line are displayed. In case of transmission spectra, the frequency values of all peaks below the auxiliary line are displayed. • Click the button if you want to store the frequency values currently displayed. ➤ After storing the frequency values the <i>PEAKS</i> data block is added to the spectrum file in the OPUS browser. 	

Table 7.6: Performing interactive peak picking

7.4.3.3 Viewing peak picking result as report

When peak picking has finished, the result can be viewed in the form of a report.

1



- In the OPUS browser, right click the PEAKS data block of the respective spectrum file.
- From the pop-up menu displayed, select the *Show Peak List* command.
- *The report view is displayed.*

Report view of peak picking

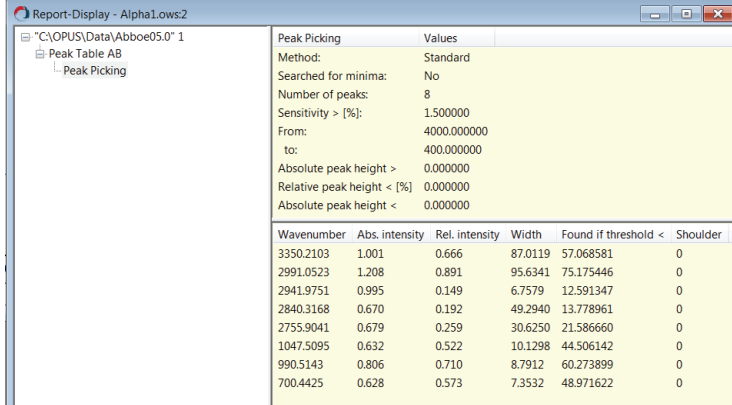
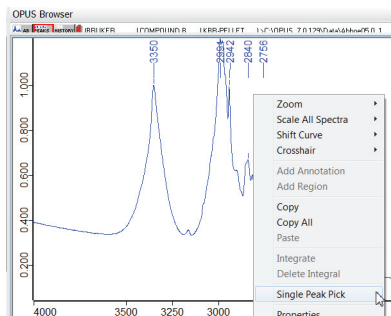


Table 7.7: Viewing peak picking result as report

7.4.3.4 Selecting single peak

The frequency value can also be displayed for one particular peak only.

1



- Right click into the spectrum window.
- From the pop-up menu displayed, select the *Single Peak Pick* command.
- *The cursor changes to* .

Table 7.8: Selecting single peak

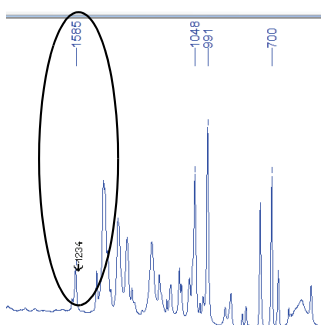
2		• Position the cursor onto the desired position. • Press the left mouse button. ➤ <i>The frequency value is displayed above the peak desired.</i>
---	---	---

Table 7.8: Selecting single peak

7.5 Evaluating spectrum

After completing a spectrum manipulation, the wizard changes to the *Evaluate* level. Now, depending on your particular analytical task (quality control, substance identification, quantitative analysis etc.) you can evaluate the sample spectrum using the corresponding evaluation command from the OPUS spectroscopy software.

The following chapters describe the most relevant evaluation commands:

- quick compare
- spectrum search
- quantitative analysis

The remaining available evaluation commands are described in the OPUS/MENTOR manual.

7.5.1 Quick compare

During quick compare, you compare a reference spectrum with a sample spectrum. Thus, you can use the *Quick Compare* evaluation command in a quality control process to check whether the delivered goods or made products are in accordance with the specified quality criteria.

Before starting quick compare, you first have to set up a suitable method.

7.5.1.1 Setting up quick compare method

To set up a quick compare method, you need to measure at least one reference spectrum or several reference spectra.

In this case, reference spectra are spectra generated by using samples which exactly meet the quality criteria defined. These samples are also called reference standards.

When selecting or measuring reference samples observe the following:

- only use pure samples as reference standard (i.e. samples without any contaminants)
- prepare the reference standards carefully
- avoid measurement errors (e.g. the ATR crystal is not completely covered with the sample)

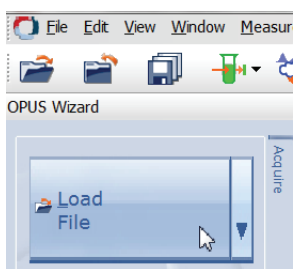
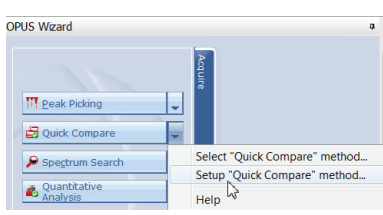
1		• In the MENTOR wizard, on the <i>Acquire</i> level click the <i>Load File</i> button. • Select the desired spectrum file from the dialog shown. <i>> The spectrum is displayed in the spectrum window. The Manipulate level is shown on the MENTOR wizard.</i>
2		• Change to the <i>Evaluate</i> level and click the arrow button right to the <i>Quick Compare</i> button. • From the pop-up menu displayed, select the <i>Setup Quick Compare method</i> command.

Table 7.9: Setting up quick compare method

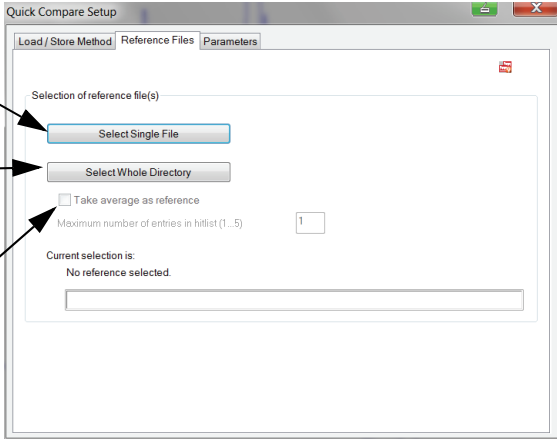
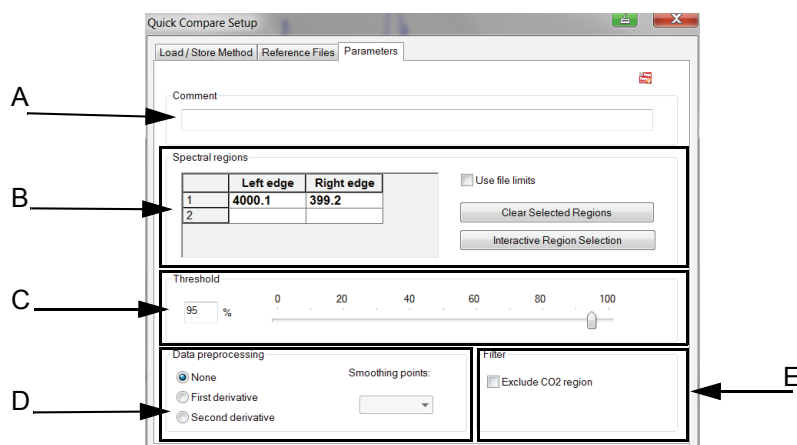
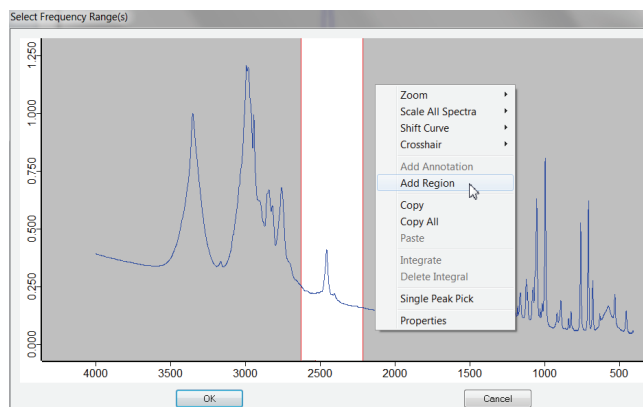
3	On the dialog displayed, click the <i>Reference Files</i> tab, and select the reference file(s) by clicking the respective button. One single reference file Several reference files which can originate from different sample material, and which are stored in one directory The average spectrum of several reference spectra which must originate from the same sample material, and which are stored in one directory   In case of a quick compare with several reference files, define the maximum number of entries to be included in the hit list. You can define up to 5 entries.
---	---

Table 7.9: Setting up quick compare method

- 4 Click the *Parameter* tab and make the settings.



- A) Entry is optional
- B) The following options are available when defining the spectral regions:
 - use file limits
 - enter values for one or more spectral regions into the table
 - interactively select one or more spectral regions; this option is recommended by Bruker
- ☞ Select the spectral region which includes the most relevant spectral information of the substance to be analyzed.
- ☞ To define further regions, right click into the *Select Frequency Range(s)* dialog.
- ☞ From the pop-up menu displayed, select the *Add Region* button.



- ☞ Position the cursor onto the boundary line of the newly added region.
- ☞ While pressing the left mouse button, move the boundary line to the position desired.
- ☞ Click the OK button to confirm the setting.

Table 7.9: Setting up quick compare method

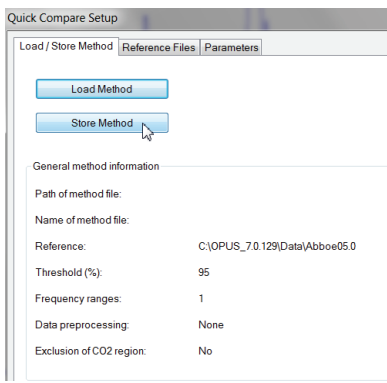
	C) The threshold value defined has an influence on the quick compare result. The threshold value determines the degree of similarity between the reference spectrum and the sample spectrum. This means, you determine how similar the reference spectrum and sample spectrum need to be so that the quick compare result is OK. As a general guideline: the higher the threshold value is, the higher the similarity between the reference and sample spectrum needs to be so that the quick compare result is OK. Either use the slider or enter a value manually to define the threshold. D) Optionally, you can select between different data pre-processing methods. If you check the <i>First</i> or <i>Second Derivative</i> option button, the <i>Smoothing points</i> selection box is enabled. The optimal number of smoothing points to be set for the specific pre-processing method has to be determined empirically. E) If you activate the checkbox, spectral regions which contain CO₂ bands are excluded during evaluation. This refers to spectral regions from 660 to 680 cm⁻¹ and 2,275 to 2,400 cm⁻¹.
5	 • Select the <i>Load/Store Method</i> tab. • Click the <i>Store Method</i> button. i Quick compare method files have the file extension *.QCM.

Table 7.9: Setting up quick compare method

7.5.1.2 Performing quick compare

When performing quick compare you have to observe the following with regard to the spectrum file(s):

- The data blocks of sample and reference files have to be of the same spectrum type, i.e. AB (absorption) or TR (transmission).
- The wavenumber range of the sample spectrum has to be at least as large as the range defined for the reference files.

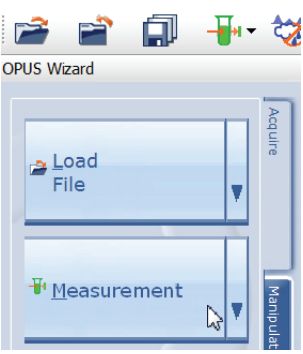
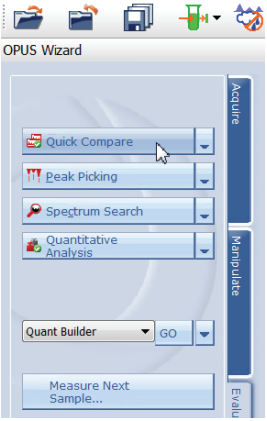
1		• In the MENTOR wizard, on the <i>Acquire</i> level click the <i>Measurement</i> button. • On the dialog displayed, click the <i>Background Single Channel</i> button, then the <i>Sample Single Channel</i> button. ➤ <i>Details on measurement are described in chapter 7.3 et seq.</i>
2		After the measurement has finished, go to the <i>Evaluate</i> level in the MENTOR wizard and click the <i>Quick Compare</i>^a button. ➤ <i>The quick compare result is displayed in the spectrum window.</i>

Table 7.10: Performing quick compare

a. Alternative: Click the arrow button next to the *Quick Compare* button. From the pop-up menu displayed, select the *Select Quick Compare Method* command. On the dialog displayed, you may load a different type of method. Then, click the *Compare* button.

7.5.1.3 Quick compare result

If quick compare has been performed, the result is stored in the QC () data block. This data block is attached to the spectrum file and displayed in the OPUS browser. To open the result view double click the QC data block.

Result view

The quick compare result is displayed as follows:

Result	Definition
 OK	The result indicates that the sample meets the quality criteria. In this case, the correlation coefficient ^a calculated is greater than the threshold defined in the method setup.
 NOT OK	The result indicates that the sample does not meet the quality criteria. In this case, the correlation coefficient calculated is less than the threshold defined in the method setup.

Table 7.11: Quick compare result view

- a. The correlation coefficient r is a measure for the similarity between the two items *reference spectrum* and *sample spectrum*. The correlation coefficient adopts a value from -1,0 (i.e. no similarity between reference spectrum and sample spectrum at all) to +1,0 (i.e. absolutely identical spectra). Then, the correlation coefficient is converted into a percentage value with the range $-1 \leq r < 0$ being mapped to 0%, and the range $0 \leq r \leq 1$ being mapped linearly to 0% ... 100%.

The quick compare result view depends on the reference standard selected.

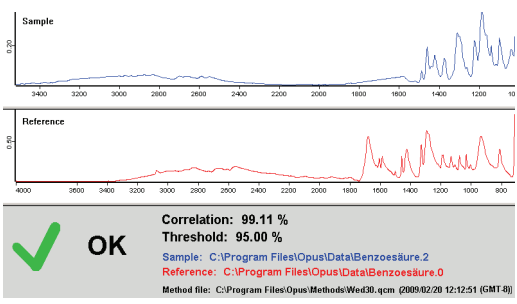
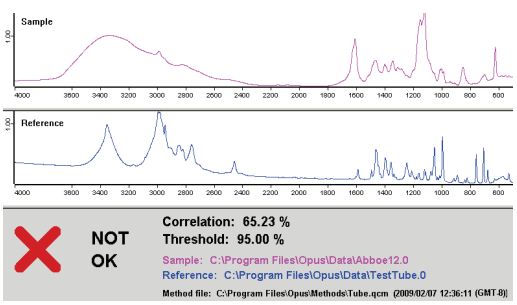
Reference standard	Result view
Single spectrum file	 OK Correlation: 99.11 % Threshold: 95.00 % Sample: C:\Program Files\Opus\Data\Benzoesäure.2 Reference: C:\Program Files\Opus\Data\Benzoesäure.0 Method file: C:\Program Files\Opus\Methods\Wed30.qcm (2009/02/20 12:12:51 (GMT+0))
	 NOT OK Correlation: 65.23 % Threshold: 95.00 % Sample: C:\Program Files\Opus\Data\Abboe12.0 Reference: C:\Program Files\Opus\Data\TestTube.0 Method file: C:\Program Files\Opus\Methods\Tube.qcm (2009/02/07 12:36:11 (GMT+0))

Table 7.12: Result view depending on the reference standard used

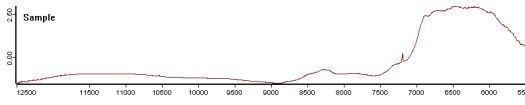
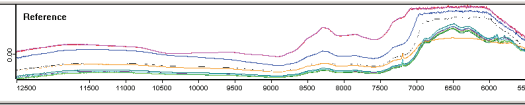
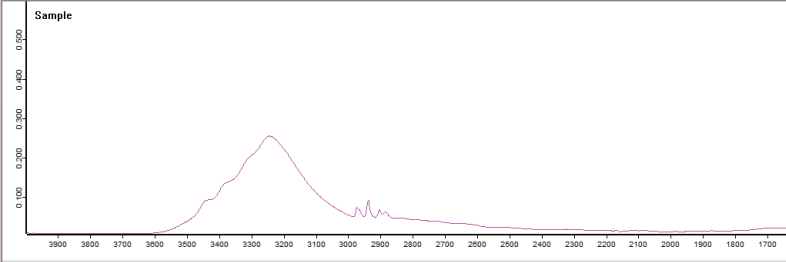
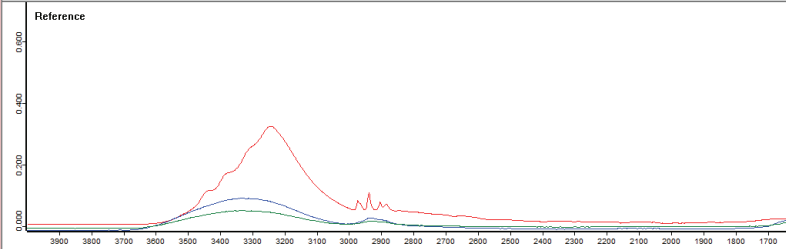
Reference standard	Result view								
Average value of several spectra	   OK Correlation: 99.49 % Threshold: 95.00 % Sample: C:\Program Files\Opus\Data\Enalapril Verla_MW.0 Reference: C:\PROGRAM FILES\OPUS\QC Method file: C:\Program Files\Opus\Methods\ENA.qcm (2009.02.20 12:44:25 (GMT 8))								
Spectra from one directory ^a	  Method file: Demo-Data-Multi1.qcm (2012/04/18 09:46:24 (GMT+2)) (C:\Data\Spektren\Demo-Daten\Spektrenvergleich) Sample: Mult1_1_test_02.0 (C:\OPUS\Data\MIR Basic Demodata\Quick Compare) <table><thead><tr><th>Reference:</th><th>Correlation:</th></tr></thead><tbody><tr><td>> Glucose.0 (C:\OPUS\Data\MIR Basic Demodata\Quick Compare)</td><td>99.73 %</td></tr><tr><td>> Rice_Starch.0 (C:\OPUS\Data\MIR Basic Demodata\Quick Compare)</td><td>87.89 %</td></tr><tr><td>> Potatoe_Starch.0 (C:\OPUS\Data\MIR Basic Demodata\Quick Compare)</td><td>87.02 %</td></tr></tbody></table>	Reference:	Correlation:	> Glucose.0 (C:\OPUS\Data\MIR Basic Demodata\Quick Compare)	99.73 %	> Rice_Starch.0 (C:\OPUS\Data\MIR Basic Demodata\Quick Compare)	87.89 %	> Potatoe_Starch.0 (C:\OPUS\Data\MIR Basic Demodata\Quick Compare)	87.02 %
Reference:	Correlation:								
> Glucose.0 (C:\OPUS\Data\MIR Basic Demodata\Quick Compare)	99.73 %								
> Rice_Starch.0 (C:\OPUS\Data\MIR Basic Demodata\Quick Compare)	87.89 %								
> Potatoe_Starch.0 (C:\OPUS\Data\MIR Basic Demodata\Quick Compare)	87.02 %								

Table 7.12: Result view depending on the reference standard used

a. If all spectrum files of a certain directory have been selected as reference, the result is not a simply *OK* or *Not OK*. In this case, only those reference spectra are marked in black which have a sufficient similarity with the sample spectrum, due to the threshold value entered and the correlation coefficient calculated. All other reference spectra, which do not have a sufficient similarity with the sample spectrum, are marked in gray.

7.5.1.4 Viewing quick compare result as report

When quick compare has finished, the result can be viewed in the form of a report.

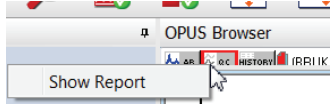
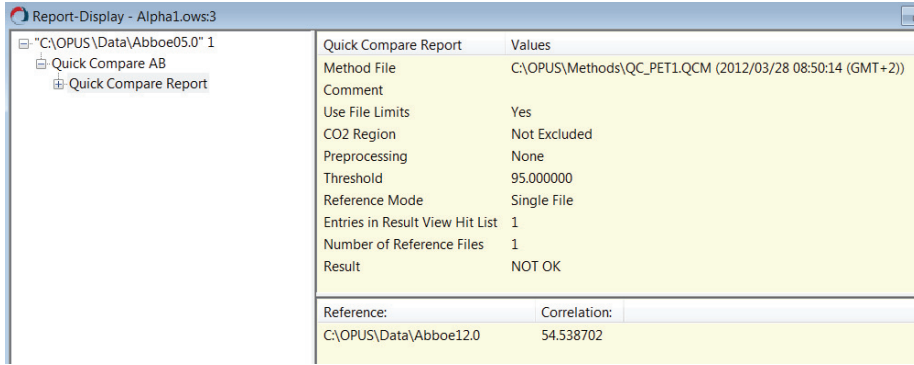
1		• In the OPUS browser, right click the QC data block of the respective spectrum file. • From the pop-up menu displayed, select the <i>Show Report</i> command. ➤ <i>The report view is displayed.</i>
Report view of quick compare 		

Table 7.13: Viewing quick compare result as report

7.5.2 Spectrum search

Spectrum search compares an unknown substance with spectra of a spectra library. Those library spectra which show a close similarity with the spectrum of the unknown substance are collected and listed in a search report. Thus, spectrum search allows to identify an unknown spectrum on the basis of its IR spectrum.

Spectrum search is only possible if at least one spectra library is available. By default, the following spectra libraries are supplied:

- DEMOLIB.SO1
- SR.IDX

The spectra libraries are stored in the <OPUS>\DATA\LIBRARY directory and contain 350 or 200 entries. Both libraries contain numerous classes of substances (e.g. polymers, pharmaceuticals etc.).

A decisive factor for a successful substance identification is the availability of spectra libraries, i.e. a spectrum search can only yield acceptable results if the selected library contains entries (i.e. spectra plus additional information) of the classes of substances that you actually analyze.

7.5.2.1 Performing spectrum search

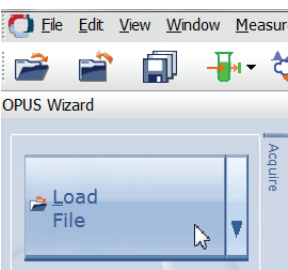
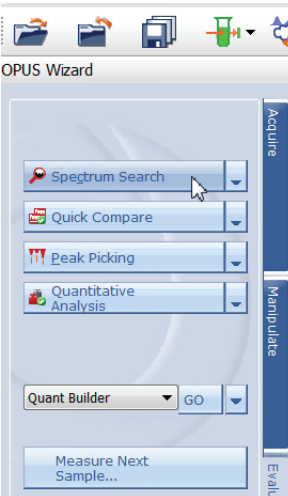
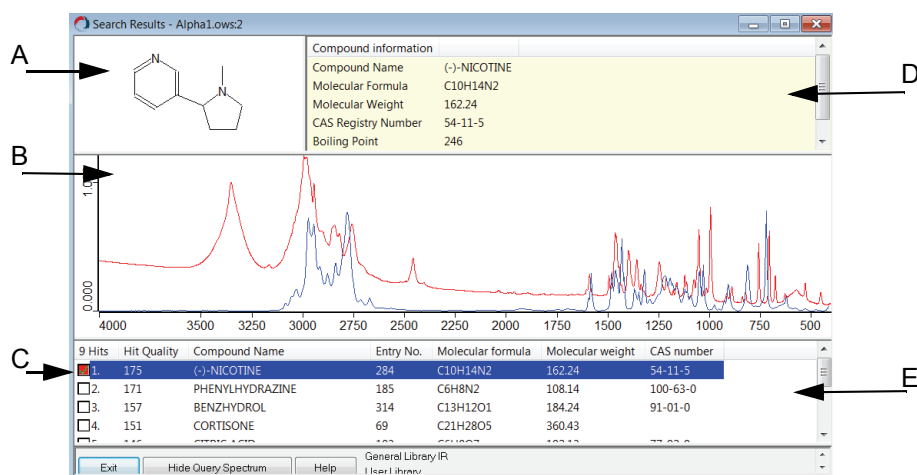
1		• In the MENTOR wizard, on the <i>Acquire</i> level click the <i>Load File</i> button. • Select the desired spectrum file from the dialog shown. ➤ <i>The spectrum is displayed in the spectrum window. The Manipulate level is shown on the MENTOR wizard.</i>
2		Change to the <i>Evaluate</i> level and click the <i>Spectrum Search</i> button. ➤ <i>Spectrum search is performed using the search parameters set by Bruker.</i> ➤ <i>The SEARCH data block is added to the spectrum file in the OPUS browser.</i>

Table 7.14: Performing spectrum search

3

The search result is displayed in the spectrum window:



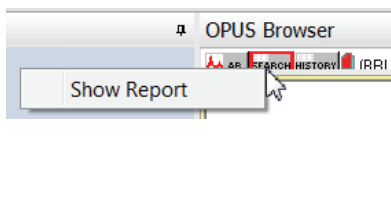
- A) Structural formula
- B) Library spectrum and query spectrum
- C) Hit quality: the value can range from 0 (i.e. no resemblance at all) to 1000 (i.e. absolute conformity of the library spectrum with the query spectrum). The hit list is sorted according the hit quality value, i.e. the hit with the highest quality value ranks first in the hit list.
- D) Substance information
- E) Hit list: hit quality value included in the hit list gives you a general idea to what extent the spectrum of the unknown substance (also called query spectrum) resembles the library spectrum.

Table 7.14: Performing spectrum search

7.5.2.2 Viewing spectrum search result as report

When spectrum search has finished, the result can be viewed in the form of a report.

1



- In the OPUS browser, right click the SEARCH data block of the respective spectrum file.
- From the pop-up menu displayed, select the *Show Report* command.
- *The report view is displayed.*

Report view of spectrum search

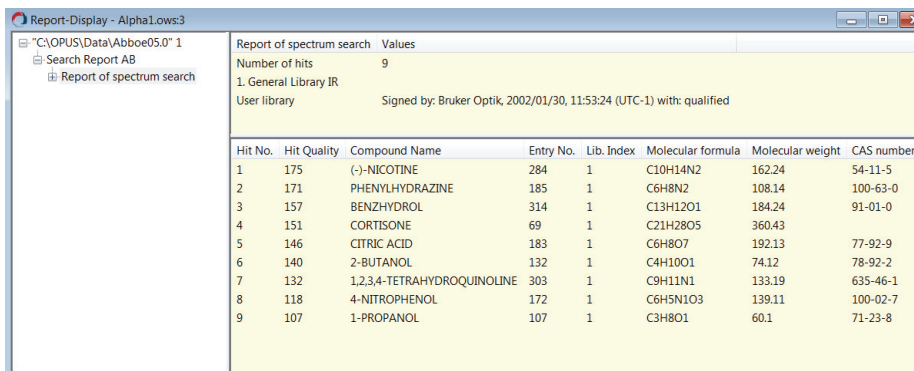


Table 7.15: Viewing spectrum search result as report

7.5.2.3 Generating user-specific library

The following commands are available in the OPUS spectroscopy software to generate a user-specific library:

- Initialize Library
- Store Spectrum in Library
- Library Editor
- Library Browser

Detailed information about these commands are described in the OPUS/MENTOR manual.

When creating a user-specific spectra library, the following aspects are very important preconditions to be able to get acceptable search results when identifying unknown substances.

- When selecting the samples for the library, make sure that they cover the whole range of the class(es) of substances you want to analyze.
- Use only pure substances, i.e. substance that are free of unwanted contaminants.
- Prepare the samples carefully (chapter 8).

- Avoid measurement errors, e.g. the ATR crystal is not covered completely with the sample.
- Before adding the measured sample spectra to the library, optimize them by using the available manipulation commands.
- Preferably, add absorption spectra¹ to the library.
- Consider what kind of additional information (e.g. boiling point) you want to include into the library, and make sure that this kind of information is available.

7.5.3 Quantitative analysis

If the sample is a mixture of several components, e.g. a tablet containing several active agents, the OPUS spectroscopy software allows to quantify one of these components. To realize quantitative analysis, the following commands are available in OPUS:

- Quant Builder - to set up a quantitative (QUANT) method
- Quantitative Analysis - to perform the actual quantitative analysis

To perform quantitative analysis observe the order described in the following chapters.

7.5.3.1 Creating a set of calibration samples

-  Ideally, the set of calibration samples should contain at least 5 samples.
-  From these calibration samples, determine the exact concentration value of the component to be analyzed using a different technique (e.g. weighing).
-  When selecting the calibration samples for the set, make sure that the concentration value of the component to be analyzed is different with each calibration sample.

i Before starting quantitative analysis, it is highly recommended to normalize all spectra, i.e. both the calibration spectra and the spectra of the sample(s). Spectra normalization is an OPUS manipulation command. Normalizing means that the spectra are scaled such that the minimum and/or maximum of the spectra assume the same value. Therefore, the spectra can be better compared with each other.

Especially in case of ATR spectra, a normalization is highly recommended. Detailed information on normalization is given in the OPUS/MENTOR manual.

1. In case of an IR spectrum, the position and intensity of the absorption bands are very substance-specific. Therefore, the IR spectrum - similar to a human fingerprint - is well-suited to identify substances.

7.5.3.2 Creating QUANT method

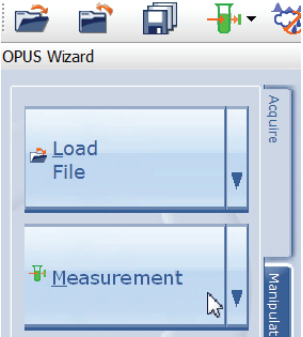
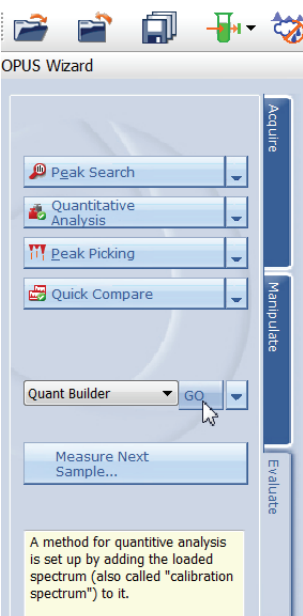
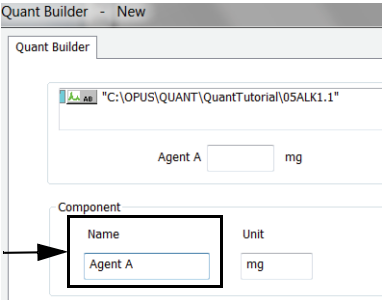
1		• In the MENTOR wizard, on the <i>Acquire</i> level click the <i>Advanced Measurement</i> button. • On the dialog displayed, click the <i>Background Single Channel</i> button, then the <i>Sample Single Channel</i> button to measure one sample from the calibration sample set. ➤ <i>Details on measurement are described in chapter 7.3 et seq.</i>
2		• After the measurement has finished, go to the <i>Evaluate</i> level in the MENTOR wizard and select the <i>Quant Builder</i> command. • Click the <i>GO</i> button. ➤ <i>The Quant Builder New dialog is displayed.</i>
3		Enter the name of the component to be analyzed. ☞ Position the cursor into the entry field, delete the default name and enter the new name.

Table 7.16: Creating QUANT method

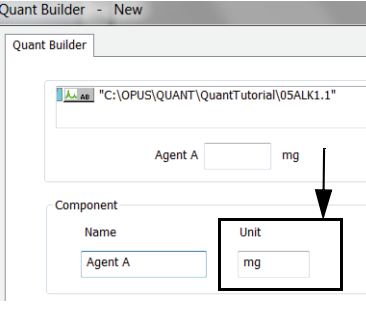
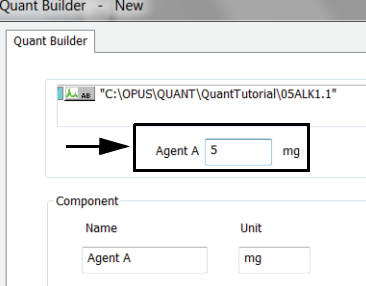
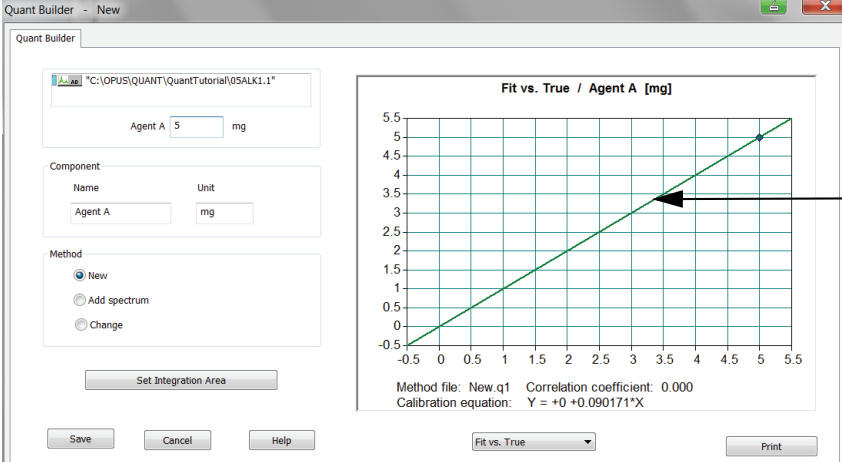
4		Specify the unit (mg, ml or %) of the concentration value. i The unit to be specified depends on the unit which you have used to determine the concentration values of the calibration samples. The result of the quantitative analysis is also indicated in this unit.
5		Enter concentration value^a of the sample. ➤ <i>OPUS calculates a calibration straight line^b. This calibration straight line is further refined as soon as the data of all calibration samples are entered.</i>
Calibration straight line: 		

Table 7.16: Creating QUANT method

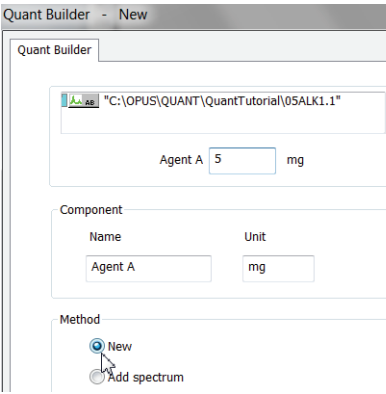
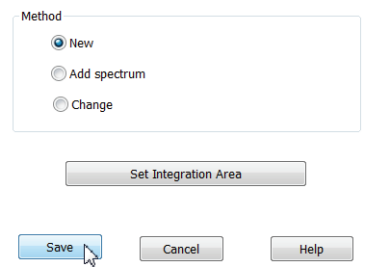
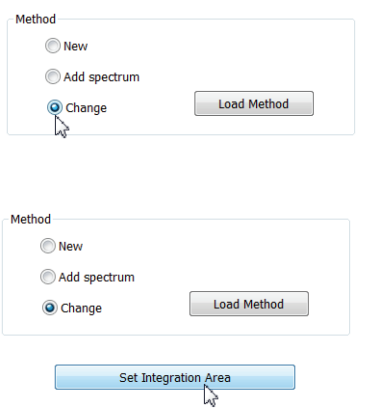
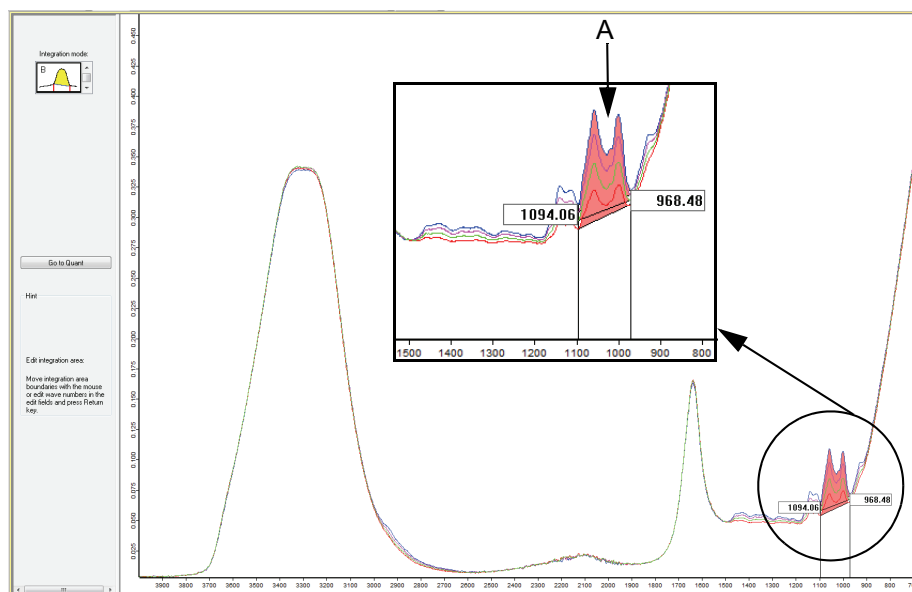
6		Activate the <i>New</i> option button. i The <i>New</i> option button must only be activated when adding the very first calibration spectrum to a new QUANT method. When adding any further calibration spectrum, activate the <i>Add spectrum</i> option button.
7		• Click the <i>Save</i> button. • Define the name and path of the method file. i The method file has the file extension <i>*.Q1</i>.
8	• In the MENTOR wizard, on the <i>Evaluate</i> level click the <i>Measure Next Sample</i> button. • Repeat the steps 1 - 5 for all the other calibration spectra to be added to the method. The name and unit of the component are set by default. • With step 6, activate the <i>Add spectrum</i> option button. • Store all calibration spectra as described with step 7. i It is recommended to set the integration area not until at least 3 or 4 calibration spectra have been added to the QUANT method. At an earlier stage, it is probably not possible to clearly see which band results from the component to be analyzed.	
9		• Click the <i>Change</i> button. • Click the <i>Load Method</i> button and load the respective method. • Click the <i>Set Integration Area</i> button.

Table 7.16: Creating QUANT method

- 10 The dialog for setting integration areas is displayed:



- A) Band selected in the spectrum. This band can be assigned to the component to be analyzed, and should not be overlapped by other sample components. The spectral intensity of this band corresponds to the concentration of the component to be analyzed with each single spectrum.
- ☞ Define the integration method
 - ☞ Interactively define the limits of the integration area. Position the cursor onto a boundary and move the area while pressing the left mouse button.
 - ☞ Click the *Go to Quant* button.

- 11 Click the *Save* button.
➤ *Quant Builder is closed.*

Table 7.16: Creating QUANT method

- a. The concentration value has to be determined first, by using a different analytical technique.
- b. The calibration straight line is the relation between the calibration spectra and the concentration values of the respective component. In case of a quantitative analysis of a sample, which contains an unknown concentration value, OPUS can calculate the quantity of this component on the basis of the sample spectrum.

7.5.4 Performing quantitative analysis

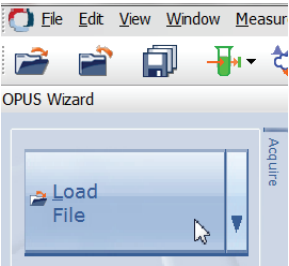
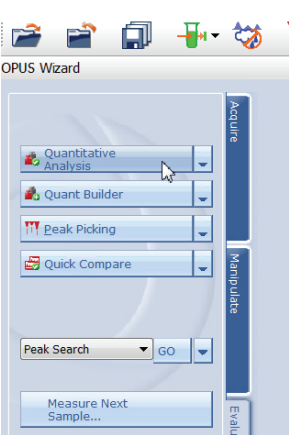
1		• In the MENTOR wizard, on the <i>Acquire</i> level click the <i>Load File</i> button. • Select the spectrum file of a sample measured before, which contains an unknown concentration value of a component, from the dialog shown. ➤ <i>The spectrum is displayed in the spectrum window. The Manipulate level is shown on the MENTOR wizard.</i>
2		• Change to the <i>Evaluate</i> level and click the <i>Quantitative Analysis</i> button. ➤ <i>The quantitative analysis is immediately performed on the basis of the QUANT method^a recently generated or used.</i>

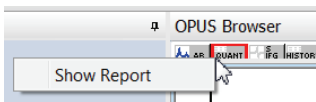
Table 7.17: Performing quantitative analysis

a. If a different kind of method is to be used, in the MENTOR wizard on the *Evaluate* level click the arrow next to the *Quant Analysis* button. From the pop-up menu displayed, click the *Select Quant Analysis* method button. On the dialog displayed, click the *Load Quant* method button and select the desired method.

7.5.5 Viewing quantitative analysis result as report

When quantitative analysis has finished, the result can be viewed in the form of a report.

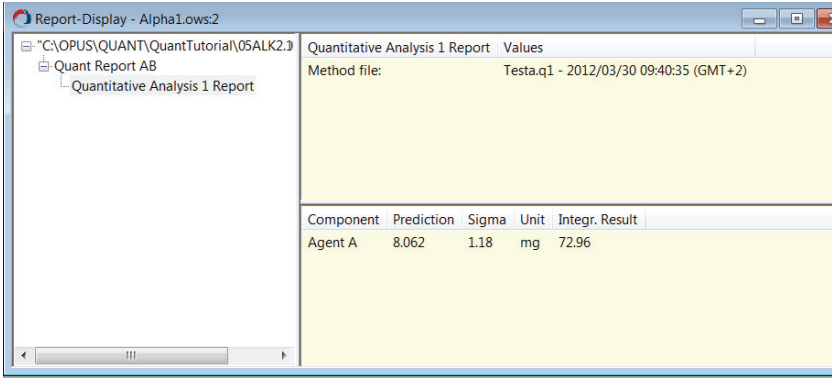
1



- In the OPUS browser, right click the QUANT data block of the respective spectrum file.
- From the pop-up menu displayed, select the *Show Report* command.

➤ *The report view is displayed.*

Report view of quantitative analysis



Component	Prediction	Sigma	Unit	Integr. Result
Agent A	8.062	1.18	mg	72.96

i The value indicated in the *Sigma* column is the standard deviation of the calculated concentration value. The sigma value is determined by the quality of the QUANT method set up. Ideally, the sigma value should be as small as possible.

Table 7.18: Viewing quantitative analysis result as report

8 Sample Preparation

Proper sample preparation is crucial to obtain good and meaningful spectra. This chapter describes the most important sample preparation methods¹.

Useful information helps you to select the most suitable sample preparation method for a given sample and the measuring technique selected.

8.1 What is to be observed?

How must the sample be like?	homogeneous (constant concentration or composition within the sample area to be analyzed)
Disadvantages of a non-homogeneous sample:	• measuring data do not represent the sample • depending on the measuring technique used, spectral artefacts are possible
What has to be considered when selecting a sample preparation method?	• aggregation state of the sample • absorptivity of the sample
What has to be considered in case of a highly absorbing sample?	• sample has to be very thin • sample has to be diluted by a solvent or powder that is not highly absorbing, otherwise the spectrum is not meaningful
What kind of solvents can be used?	• carbon tetrachloride • carbon disulphide • chloroform • cyclohexane • acetonitrile • tetrachloroethylene

Table 8.1: Sample preparation - What is to be observed

1. For more detailed information about the different sample preparation methods refer to the relevant specialist literature.

8.2 Sample preparation methods

The most common sample preparation methods are:

- No sample preparation
- KBr pellet (for solids)
- Thin film between two transparent plates (for liquids)
- Gaseous samples

i

Most of the described sample preparation methods involve the use of hygroscopic material (such as NaCl or KBr). If this type of material comes in contact with water or alcoholic solvents, it begins to dissolve or becomes cloudy and thus, impairs the measurement results.

Therefore, protect hygroscopic material from any sources of water, and even alcohol (ethanol and methanol).

8.2.1 No sample preparation

Type of samples:	films and polymers, with a thickness of less than approx. 100 µm
Tip:	a large number of solid and liquid samples can also be analyzed by using an ATR unit; this measuring method does not require any sample preparation either
Advantage of this method:	enables fast sample measurements
Disadvantage of this method:	• ATR: an absolute quantification is not possible in case of solids • Transmission: sometimes samples are too thick

Table 8.2: No sample preparation

8.2.2 KBr pellet

Type of samples:	solids
What is to be observed during sample preparation?	• grind the sample as fine as possible to minimize the infrared light scattering on the particle surface, and the portion of the reflected light • keep everything moisture free as the KBr material is hygroscopic, otherwise the pellets become cloudy • it would be best to keep the KBr material in a drying oven at a temperature of 50 to 60°C
Tip:	a common mistake is to use too much sample material; the concentration of the sample in KBr should be between 0.2% and 1% (i.e. typically a mixing ratio of 300:1)
Advantage of this method:	• meaningful IR spectrum • KBr shows no absorption in the wavenumber area of 4000 cm^{-1} to 250 cm^{-1} • this prevents the sample spectrum from containing any disturbing absorption bands
Disadvantage of this method:	• time-consuming sample preparation method • interference of water bands (3,960 to 3,480 cm^{-1} and 1,950 to 1,300 cm^{-1} and below 500 cm^{-1}) • in some cases there may be structural changes caused by high pressure applied to the KBr/sample mixture

Table 8.3: Sample preparation - KBr pellet

8.2.3 Thin film between two transparent plates

Type of samples:	liquids (e.g. oils)
What is to be observed during sample preparation?	• pipette one drop of the sample on one of the plates • to avoid air bubbles the liquid sample between the two plates has to be a uniform film • to be able to perform a quantification you have to set a defined layer thickness, use cuvettes
Tip:	• do not use the plates in case of samples that contain water as the plate material (NaCl or KBr) is extremely moisture-sensitive • keep the plates always dry (e.g. in an exicator) • clean the plates only by using chloroform or high purity acetone and polish them carefully after each use
Advantage of this method:	• easy sample preparation method • small amount of sample material required • quantification is possible
Disadvantage of this method:	• not suitable for samples which contain water • the material of the plates (NaCl or KBr) are extremely moisture-sensitive • very often there are air bubbles between the plates

Table 8.4: Sample preparation - Thin film between two transparent plates

8.2.4 Gaseous samples

Where to inject the sample?	into an evacuated gas cell
The intensity of the peak measured is influenced by:	• thickness of gas cell • pressure of gas inside the gas cell (pressure is proportional to the concentration) • absorption coefficients of the bands

Table 8.5: Sample preparation - Gaseous samples

9 Maintenance

The ALPHA spectrometer is a low-maintenance instrument, i.e. the operator can replace components with a limited service life (e.g. IR source). The following maintenance procedures are described in this chapter:

- Replacing IR source
- Replacing desiccant bags
- Replacing windows

9.1 General maintenance considerations

Perform only the maintenance procedures described in this chapter. Strictly observe the relevant safety precautions. Any failure to do so may cause property damage or personal injury. In this particular case Bruker does not assume any liability.

Maintenance procedures not described in this manual should only be performed by a Bruker service engineer. For service addresses, see the appropriate appendix.

The following precautions must be observed to ensure user and property safety:

- Disconnect power supply before performing any maintenance procedures.
- Be careful if the spectrometer covers are removed and the spectrometer is switched on to avoid contact with potentially harmful voltages.

9.2 Performing OQ¹ test by using OVP²

If you have exchanged a defective optical component, e.g. light source or laser, we recommend running the OQ test using the OPUS validation software (OVP). This test checks whether the spectrometer achieves the specified performance parameter values.

Perform the OQ test only if you have replaced a defective component. Do not perform the OQ test if you have replaced a component for the purpose of spectral range extension.

More details on how to perform an OQ test are described in the OPUS Reference manual. If the OQ test fails, read about possible causes and solutions in the *Troubleshooting* chapter.

-
1. OQ test - Operational Qualification Test: This test is to be performed once a year, or after the replacement of any kind of defective optical component. During the OQ test, the following parameters are tested: resolution, sensitivity, energy distribution, wavenumber accuracy and photometric accuracy.
 2. OVP (OPUS Validation Program) is a program used to perform validation tests (e.g. OQ and PQ).

9.3 Replacing IR source

The ALPHA spectrometer uses a highly efficient, low power, air-cooled IR source. The IR source is accessible from the spectrometer rear side and is located behind the cover plate. The light source is pre-aligned. The average IR source lifetime is specified with about 44.000 operating hours.

9.3.1 Display in OPUS spectroscopy software

Status of IR source	Display in OPUS	Message in OPUS	Measurement
Lifetime of spectrometer nearly reached	OPUS status light ^a is yellow	End of average life-time is nearly reached, spare part will be required.	still possible ^b
Defective	OPUS status light is red	Source is broken.	not possible

Table 9.1: Display in OPUS spectroscopy software

a. The OPUS status light is located in the lower right end of the OPUS interface.

b. To retrieve the status information click the yellow OPUS status light. Then, click the *Ignore* button. The status light turns green again. After intervals which always become shorter, the IR source operating hour meter will remind you again to replace the IR source.

9.3.2 Procedure

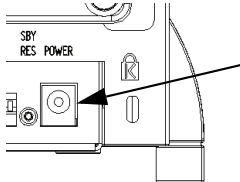
1		Switch off spectrometer. Remove the power cable from the POWER port located on the rear spectrometer side.
2		Loosen the 4 TORX screws of the rear cover plate. Use the TORX TX20 screw driver supplied.

Table 9.2: Replacing IR source

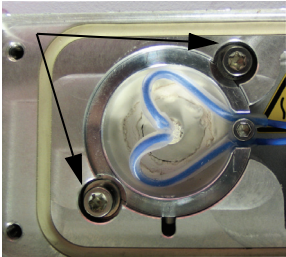
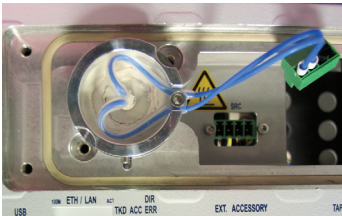
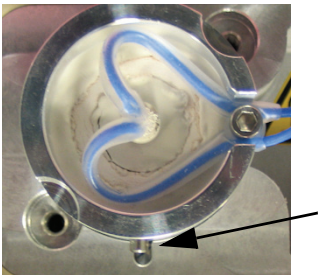
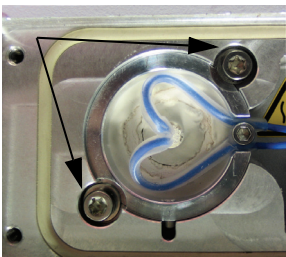
3	Remove the cover plate.	⚠ CAUTION IR source very hot Risk of skin burn. ➤ Have the IR source cooled down, before replacing it.
4		Loosen the two fixing screws of the IR source. 🔧 Use the TORX TX20 screw driver supplied
5		Strip off the green IR source plug from the SRC female connector.
6		• Remove the used IR source. • Insert the new IR source. 🔧 The alignment pin (arrow in figure) must correctly fit in the alignment groove.
7		Fasten the two TORX screws.

Table 9.2: Replacing IR source

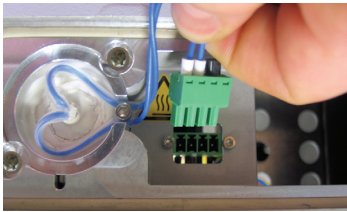
8		Insert the green IR source plug into the SRC female connector.
9		Attach the cover plate and fasten the 4 TORX screws.
10		Switch on spectrometer. ✎ Insert the power cable into the POWER port located on the rear spectrometer side.

Table 9.2: Replacing IR source

9.3.3 After replacing the IR source

After replacing the IR source, you have to perform an instrument test and reset the operating hour meter of the IR source.

9.3.3.1 Performing instrument test

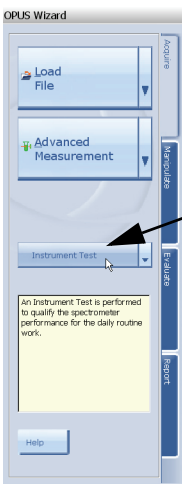
1		On the MENTOR wizard^a, click the <i>Instrument Test</i> button. ➤ The test starts and takes about 5 minutes.
2	Wait until the test has finished. ➤ If the test has passed, the OPUS status light must be green.	

Table 9.3: Performing instrument test

a. Alternatively, you can click the OPUS status light. On the dialog shown, click the icon displayed for the current measurement channel. The instrument test starts.

9.3.3.2 Resetting IR source operating hour meter

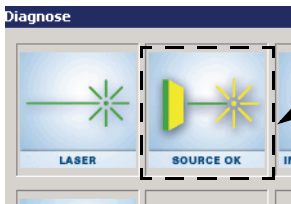
1	In OPUS, click the OPUS Status light . ➤ <i>The Instrument Status dialog opens.</i>	
2		On the <i>Instrument Status</i> dialog, click the <i>Source</i> icon. ➤ <i>Another dialog opens.</i>
3	On the dialog shown, click the <i>Reset</i> button. ➤ If the <i>Reset</i> button is deactivated, first click the <i>Help</i> button.	

Table 9.4: Resetting source operating hour meter

9.4 Replacing desiccant bags

Too high a humidity inside the spectrometer can be recognized by H₂O bands in the spectrum. To minimize this undesirable spectral effect as far as possible, the desiccant bags inside the spectrometer bind humidity.

The air inside the complete spectrometer optics unit (measuring and optics compartment) is convectively kept dry using a desiccant which is enclosed in removable small bags.

9.4.1 Replacing interval

If the desiccant is saturated and cannot further absorb any humidity, it has to be replaced. In this case, the *Humidity out of range* message is displayed in the OPUS spectroscopy software.

Besides, the OPUS status light () becomes yellow. Order new desiccant bags. Measuring is still possible even if the status light is yellow. As soon as you have received the new desiccant bags, replace the old ones.

9.4.2 Replacing procedure

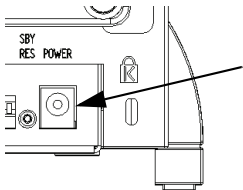
1		Switch off spectrometer.  Remove the power cable from the POWER port located on the rear spectrometer side.
2		Loosen the 4 TORX screws of the rear cover plate.  Use the TORX TX20 screw driver supplied.

Table 9.5: Replacing desiccant

3		Remove the used desiccant bags. ☞ A pair of tweezers may be helpful. i Do not cut or tear off the bags into single pieces, nor ingest the desiccant.
4	The replacement desiccant bags are protected by an extra packaging. ☞ Open this packaging and take out the desiccant bags. Alternatively, place the used bags into an oven at 130°C for at least 10 minutes. ☞ Completely insert the new desiccant bags properly.	
5		Attach the cover plate and fasten the 4 TORX screws.
6		Switch on spectrometer. ☞ Insert the power cable into the POWER port located on the rear spectrometer side.

Table 9.5: Replacing desiccant

9.5 Replacing windows

The ALPHA basic module has one beam output and one beam input port. The beam enters the sampling module by the output port, and is then directed to the basic module again via the IR input port. The ports are closed by means of exchangeable IR transparent windows.

9.5.1 Window material

Material	Chemical properties
Potassium bromide (KBr)	Soluble in water, alcohol, and glycerine NOTE Hygroscopic! ➤ Avoid any contact to humidity.
Zinc selenide (ZnSe)	Soluble in strong acids and HNO ₃ ⚠ CAUTION Toxic! ➤ Observe the safety data sheet. ➤ Do not inhale or ingest the dust of broken window material. Avoid any skin contact.

Table 9.6: ALPHA - Window material

The serial number, which is located on the rear spectrometer side, indicates what type of window material is used for the IR transparent windows.

Spectrometer type	Serial number	Window material
ALPHA Basic	1003271	KBr (hygroscopic)
	1005151	ZnSe (⚠ toxic)
ALPHA Basic HR	1010948	KBr (hygroscopic)
	1010951	ZnSe (⚠ toxic)

Table 9.7: Window material identified by serial number

9.5.2 Replacing interval

	KBr window material	ZnSe window material
Replacing interval	when opaque Opaque windows substantially reduce infrared transmittance which can be indicated by a failed performance or instrument test.	only when broken

Table 9.8: Replacing interval

9.5.3 Replacing procedure

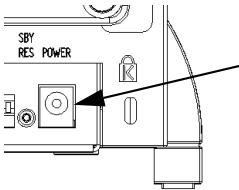
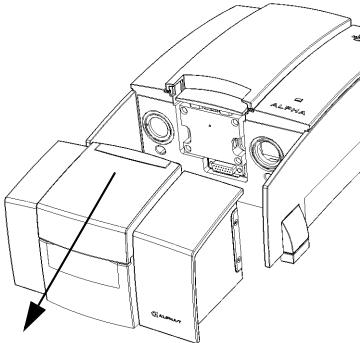
1		Switch off spectrometer. ➤ Remove the power cable from the POWER port located on the rear spectrometer side.
2		Press the lock on the basic module right down. ➤ The sampling module is released from the basic module. The lock stops at half-height.
3		Pull the sampling module towards your direction to detach the sampling module from the basic module.

Table 9.9: Replacing windows

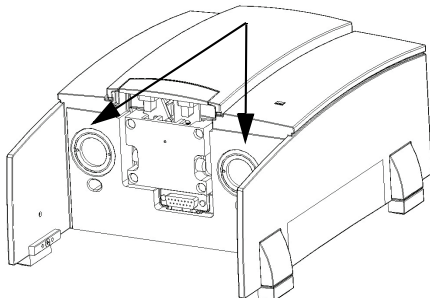
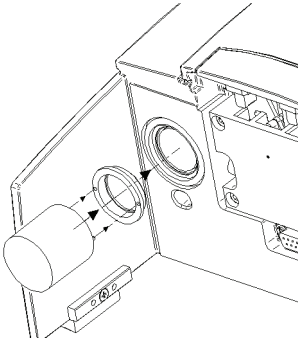
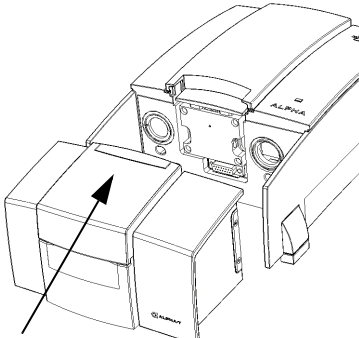
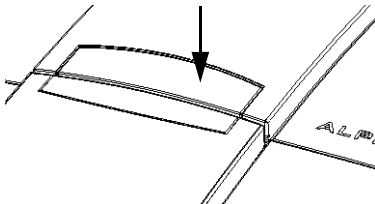
	The windows are located on the inner side of the basic module. 	
4		✎ To remove the windows use the assembly tool supplied. • Insert the two pins of the assembly tool into the respective holes on the basic module window frame. • Rotate the tool several turns counter-clockwise and pull out the tool including the window assembly (i.e. window and assembly frame). • Remove the used window from the assembly tool.
5	• Insert the two pins of the assembly tool into the respective holes of the assembly frame of the NEW window. • Put the window assembly into the holes on the basic module window frame. • Rotate the tool several turns clockwise to fasten the window. • Carefully remove the assemble tool.	
6		Push the sampling module towards the basic module.
7		Press the lock on the basic module right down and release it. ➤ The sampling module is attached to the basic module.

Table 9.9: Replacing windows

9.5.4 After replacing the windows

After replacing the windows you have to perform an instrument test.

9.5.4.1 Performing instrument test

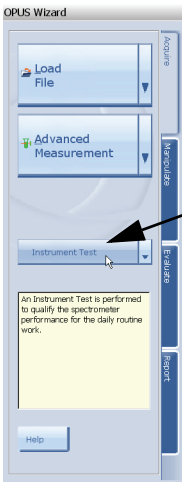
1		On the MENTOR^a wizard, click the <i>Instrument Test</i> button. ➤ The test starts and takes about 5 minutes.
2	Wait until the test has finished. ➤ If the test has passed, the OPUS status light must be green.	

Table 9.10: Performing instrument test

a. Alternatively, you can click the OPUS status light. On the dialog shown, click the icon displayed for the current measurement channel. The instrument test starts.

9.5.5 Cleaning windows

To clean the KBr windows only use a dry, lint-free cloth. Do NOT use water or solvents as the window material is hygroscopic. To clean the ZnSe windows use the cleaning solvents mentioned in chapter 4.10.1

9.6 Cleaning spectrometer housing

The outer spectrometer surface can be cleaned only by using a dry or damp cloth. Do not use detergents with organic solvents, acid or base!

10 Troubleshooting

This chapter describes possible spectrometer problems¹, their potential causes and recommended solutions spectrometer problems can be indicated by:

- spectrometer status indicator and OPUS status light
- diagnostics LEDs on the rear spectrometer side
- error messages in OPUS
- instrument/performance test

10.1 General information on fault diagnostics

A problem caused by a spectrometer component, that is either defective or not properly installed or not in operating condition, becomes apparent in several different ways. For example:

- You have started a measurement, but not any measurement result is displayed in OPUS.
 - Reason: OPUS did not start any measurement at all.
- No signal detected in OPUS.
 - Reason: Wrong OPUS parameters selected.
- You have started a validation test but OVP does not display a PQ or OQ test protocol.
 - Reason: OVP did not start the validation test at all.

To find out the concrete cause of a spectrometer problem, it is advisable to narrow down the problem in a systematic way. We recommend the following fault diagnostics procedure:

- First, check the status LEDs on the spectrometer and the OPUS status light (chapter 10.2.1 and 10.2.2). If the OPUS status light is red, click the status light. The diagnostics view opens.
- Check whether one hardware component has the status WARNING or ERROR. If yes, whether an error message is displayed for the respective component.

Check the firmware diagnostics page (chapter 10.4.1) of the respective hardware component. Check whether there is any kind of reference to the possible cause of the problem.

1. Not all failures and causes can be outlined in this chapter. If the recommended solutions do not solve the problem, contact Bruker Service (chapter 1.5).

10.1.1 Remote fault diagnostics

Remote fault diagnostics means that you send a complete spectrometer status report (Full Report) to Bruker via e-mail. This report enables a Bruker service technician to perform a first remote fault diagnostics.

The procedure for sending the report is different and depends on whether your spectrometer is connected to a network, network computer or a stand-alone computer.

10.1.1.1 If your spectrometer is connected to a network/network computer,...

With OPUS spectroscopy software version 6 or higher, it is possible to send the full report to Bruker via e-mail. Proceed as follows:

1. Click the OPUS status light (chapter 10.2.2). The *Instrument Status* dialog opens.
2. Click the *Send Report* button. Clicking this button sends the report to the *opusreports@brukeroptics.de* e-mail address.

i To be able to use this feature an e-mail program must be installed on the network computer, and an e-mail account set up.

10.1.1.2 If your spectrometer is connected to a stand-alone computer,...

Generate a full report:

1	• Measurement Menus • Measurement Status • Direct Command Entry • Messages • Diagnostics • Service • Authentication	• Open Web browser. • Enter spectrometer IP address into the address entry field. • On configuration page, click <i>Service</i>.
2	• << One Level Back • View Instrument Configuration • Full Report • Log buffer • Beep • List of commands • List of result data files	Click the <i>Full Report</i> option. The current full report is displayed.
3	• On the <i>File</i> menu of the Web browser click the <i>Save as</i> command. • Use the file extension <i>*.htm</i>. • Send original htm file to Bruker. Do not generate any screenshots or text files.	

Table 10.1: Generating full report

i Save the full report immediately after a problem or failure has occurred. Otherwise, important information will be overwritten by newer entries.

- ☞ Transfer the full report file to a network computer.
 - *Make sure that an e-mail program is installed on this network computer, and an e-mail account set up.*
- ☞ Send the full report to opusreports@brukeroptics.de as an attached file.

10.2 Retrieving spectrometer status

The spectrometer status can be retrieved by means of:

- status indicator on the spectrometer housing¹
- status light in the OPUS spectroscopy software¹
- diagnostics LEDs on the spectrometer

1. In most cases, the color of the status indicator located on the spectrometer housing corresponds to the OPUS status light. Exception: if the performance or instrument test has failed, the OPUS status light is red whereas the status indicator is yellow.

10.2.1 Status indicator on the spectrometer housing

For each spectrometer configuration the status LED is located on the front spectrometer side.

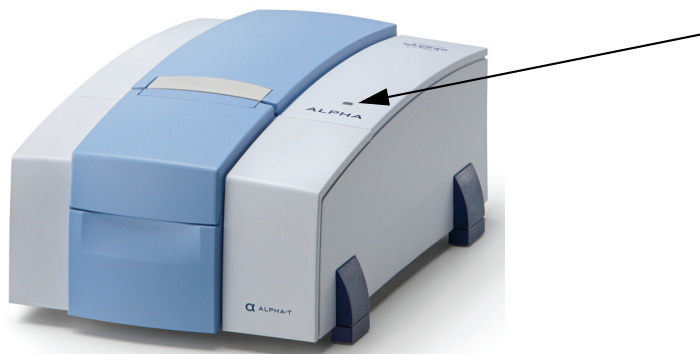


Table 10.2: Status indicator on the spectrometer housing

The status indicator can indicate different types of operating states:

Status indicator	Definition
Green (permanent)	Spectrometer operates properly
Green (flashing)	Spectrometer is in stand-by mode
Yellow	• Spectrometer is in the initialization phase • Warning is indicated (e.g. IR source has not yet reached its operating temperature, the air humidity content inside the spectrometer is too high etc.)
Red	• Spectrometer is not able to operate • Malfunction • Defective hardware component

Table 10.3: Status indicator on the spectrometer housing

10.2.1.1 Troubleshooting in case of problems indicated by the spectrometer status indicator

	Possible cause	Troubleshooting
Spectrometer status indicator is off	Spectrometer not connected properly to mains supply	Connect spectrometer to mains supply.
	No voltage applied	Check whether the proper voltage is applied at the mains outlet.
	Short circuit in the mains adapter	i A short circuit is accompanied by a "ticking" sound in the mains adapter. ☞ Disconnect power supply immediately and contact Bruker Service.
	Non of the possible causes apply	Contact Bruker Service (chapter 1.5).
Spectrometer status indicator is yellow	Spectrometer still initializes	Wait until initialization has been completed (about 5 minutes).
	IR source has not yet reached operating temperature ➤ In this case, the following message is displayed: <i>Device not ready</i>	Wait until the IR source is ready to operate. i The IR source warm-up takes about 7 minutes. As soon as the IR source has reached its operating temperature the spectrometer status indicator becomes green.
	Specified lifetime of the IR source nearly reached ➤ In this case, the following message is displayed: <i>End of average lifetime is nearly reached, spare will be required.</i>	• Order new IR source with Bruker. • After receipt replace the used IR source (chapter 9.3). i As long as the lifetime is not expired, measuring is still possible. To turn the OPUS status light green again, click the <i>Ignore</i> button on the <i>Instrument Status Message</i> dialog (table 10.15 on page 187).
	Performance or instrument test failed	Details are described in chapter 10.5 et seq.

Table 10.4: Troubleshooting in case of problems indicated by the spectrometer status indicator

	Possible cause	Troubleshooting
	Instrument test validity period is expired. ➤ In this case, the following message is displayed in the yellow information bubble (table 10.15 on page 187): <i>An instrument test is required or one of the instrument components does not meet its specifications</i>	☞ Click the OPUS status light. ➤ <i>The Instrument Status dialog opens. The measurement channel must read EXPIRED.</i> Close the dialog and start a new instrument test: ☞ On the MENTOR wizard, click the <i>Instrument Test</i> button.
	Air humidity inside the spectrometer is too high ➤ In this case, the following message is displayed: <i>Humidity out of range.</i>	Replace desiccant (chapter 9.4).
	Laser wavenumber must be recalibrated ➤ In this case, a respective message is displayed in a yellow information bubble (table 10.15 on page 187).	In the information bubble, click the <i>Calibrate</i> button.
	Too high a temperature inside the spectrometer due to too high an ambient temperature ➤ In this case, the following message is displayed: <i>Temperature out of range.</i>	Operate the spectrometer only within the specified temperature range between 18 and 35°C
	Non of the possible causes apply	Contact Bruker Service.

Table 10.4: Troubleshooting in case of problems indicated by the spectrometer status indicator

	Possible cause	Troubleshooting
Spectrometer status indicator is red	Defective hardware component	Detect the defective component: ☞ Click the OPUS status light. ➤ <i>The Instrument Status dialog opens.</i> i If the laser or the interferometer, electronics, automation or the detector is defective, contact Bruker Service. If the IR source is defective, replace the IR source (chapter 9.3).
	The sampling module is not properly attached to the basic module.	Attach the sampling module properly (chapter 2.2).
	ATR module: ATR-crystal plate not installed properly	Install the ATR-crystal plate properly (chapter 4.5).
	Non of the possible causes apply	Contact Bruker Service.

Table 10.4: Troubleshooting in case of problems indicated by the spectrometer status indicator

10.2.2 Status light in the OPUS spectroscopy software

The status light in the OPUS spectroscopy software is located on the lower right end of the interface. It indicates the status based on the currently active channel or measurement experiment loaded.

The OPUS status light can indicate different types of operating states.

LED	Definition
	Gray: no spectrometer connected
	Green: spectrometer is connected and works properly
	Yellow: warning (e.g. service life of a spectrometer component comes to an end, instrument test has expired etc.) i Measuring is still possible.
	Red: error (e.g. spectrometer malfunction, defective spectrometer component, instrument test failed etc.) i Measuring is not possible.

Table 10.5: OPUS status light

10.2.2.1 Troubleshooting in case of problems indicated by the OPUS status light

	Possible cause	Troubleshooting
Spectrometer status indicator is yellow	IR source has not yet reached operating temperature ➤ In this case, the following message is displayed: <i>Device not ready</i>	Wait until the IR source is ready to operate. i The IR source warm-up takes about 7 minutes. As soon as the IR source has reached its operating temperature the spectrometer status indicator becomes green.
	Specified lifetime of the IR source nearly reached ➤ In this case, the following message is displayed: <i>End of average lifetime is nearly reached, spare will be required.</i>	• Order new IR source with Bruker. • After receipt replace the used IR source (chapter 9.3). i As long as the lifetime is not expired, measuring is still possible. To turn the OPUS status light green again, click the <i>Ignore</i> button on the <i>Instrument Status Message</i> dialog (table 10.15 on page 187).
	Instrument test validity period is expired ➤ In this case, the following message is displayed in the yellow information bubble (table 10.15 on page 187): <i>An instrument test is required or one of the instrument components does not meet its specifications.</i>	 Click the OPUS status light. ➤ <i>The Instrument Status dialog opens. The measurement channel must read EXPIRED.</i> Close the dialog and start a new instrument test:  On the MENTOR wizard, click the <i>Instrument Test</i> button.
	Air humidity inside the spectrometer is too high ➤ In this case, the following message is displayed: <i>Humidity out of range.</i>	Replace desiccant (chapter 9.4).

Table 10.6: Troubleshooting in case of problems indicated by the OPUS status light

	Possible cause	Troubleshooting
	Laser wavenumber must be recalibrated ➤ In this case, a respective message is displayed in a yellow information bubble (table 10.15 on page 187).	In the information bubble, click the <i>Calibrate</i> button.
	Non of the possible causes apply	Contact Bruker Service.
Spectrometer status indicator is red	Performance or instrument test failed	Details are described in chapter 10.5 et seq.
	Defective hardware component	Detect the defective component: ☞ Click the OPUS status light. ➤ <i>The Instrument Status dialog opens.</i> i If the laser or the interferometer, electronics, automation or the detector is defective, contact Bruker Service. If the IR source is defective, replace the IR source (chapter 9.3).
	Non of the possible causes apply	Contact Bruker Service.
Spectrometer status indicator is gray	Spectrometer not connected to PC	Connect spectrometer to PC (see Installation Instructions).
	Wrong cable type used to connect spectrometer to PC	Only use the cross-over, Cat5-cable supplied.
	RJ45-plug(s) of the data cable not properly connected to PC and/or spectrometer.	Check the data cable cord connection at spectrometer and PC.

Table 10.6: Troubleshooting in case of problems indicated by the OPUS status light

	Possible cause	Troubleshooting
	Not any or wrong spectrometer IP address specified in OPUS	Specify correct spectrometer IP address in OPUS: ☞ On the <i>OPUS Validation</i> menu, select the <i>Optic Setup and Service</i> command. ➤ <i>The Optic Setup and Service dialog opens.</i> ☞ In the <i>Configuration</i> drop-down list, select the <i>ALPHA</i> spectrometer. ☞ In the <i>Optical Bench URL</i> entry field, enter the spectrometer standard IP address (10.10.0.1).
	The factory-set standard IP address of the spectrometer (10.10.0.1) has been changed	Reassign the standard IP address to the spectrometer: ☞ Set the DIP-switch on the spectrometer rear side. See <i>Installation Instructions</i>. Reset spectrometer: ☞ Press the <i>SBY/RES</i> key, located on the spectrometer rear side, more than 6 seconds.
	Spectrometer is in stand-by mode. ➤ The spectrometer status indicator flashes green.	Deactivate stand-by mode: ☞ Press the <i>SBY/RES</i> key, located on the spectrometer rear side.

Table 10.6: Troubleshooting in case of problems indicated by the OPUS status light

10.2.3 Diagnostics LEDs on the spectrometer

The diagnostics LEDs are located on the rear spectrometer side.

	LED	Definition
A	TKD	Green: ➤ interferometer mirror is within the data acquisition range Light green: ➤ data acquisition runs Black: ➤ no data acquisition The TKD abbreviation means <i>Take Data</i> .
B	ACC	Yellow: ➤ the optics is accessed by the network Light yellow: ➤ data transfer runs The ACC abbreviation means <i>Access</i> .
C	DIR/ERR	Slightly red flashing: ➤ normal operation mode, scanner moves Permanent bright red: ➤ interferometer problem, e.g. missing laser signal; no data acquisition possible The DIR abbreviation means <i>Direction</i> , ERR means <i>Error</i> .

Table 10.7: Definition of diagnostics LEDs on spectrometer

10.2.3.1 Possible error causes and troubleshooting

	Possible cause	Troubleshooting
All LEDs off	Spectrometer not connected properly to mains supply	Connect spectrometer to mains supply.
	No voltage applied	Check whether the proper voltage is applied at the mains outlet.
	Short circuit in the mains adapter	i A short circuit is accompanied by a "ticking" sound in the mains adapter. ✎ Disconnect power supply immediately and contact Bruker Service.
Yellow ACC LED does not light	Spectrometer not connected PC	Connect spectrometer to PC.
	Wrong cable type used to connect spectrometer to PC	Only use the cross-over, Cat5-cable supplied.
	RJ45-plug(s) of the data cable not properly connected to PC and/or spectrometer.	Check the data cable cord connection at spectrometer and PC.
	Not any or wrong spectrometer IP address specified in OPUS	Specify correct spectrometer IP address in OPUS: ✎ On the <i>OPUS Validation</i> menu, select the <i>Optic Setup and Service</i> command. ➤ <i>The Optic Setup and Service dialog opens.</i> ✎ In the <i>Configuration</i> drop-down list, select the <i>ALPHA</i> spectrometer. ✎ In the <i>Optical Bench URL</i> entry field, enter the spectrometer standard IP address (10.10.0.1).

Table 10.8: Troubleshooting - LEDs on spectrometer rear side

	Possible cause	Troubleshooting
	Factory-set standard IP address of the spectrometer (10.10.0.1) has been changed	Reassign the standard IP address to the spectrometer:  Set the DIP-switch on the spectrometer rear side. See <i>Installation Instructions</i> . Reset spectrometer:  Press the SBY/RES key, located on the spectrometer rear side, more than 6 seconds.
	Spectrometer is in stand-by mode ➤ The spectrometer status indicator flashes green.	Deactivate stand-by mode:  Press the SBY/RES key, located on the spectrometer rear side.
DIR ERR LED is permanently red	Spectrometer problem caused by the laser and/or interferometer	Contact Bruker Service.
	Strong vibrations or other strong mechanical impacts on the spectrometer, which impair the spectrometer operation	Put the spectrometer on a vibration-free surface. If the problem still persists:  Contact Bruker Service.

Table 10.8: Troubleshooting - LEDs on spectrometer rear side

10.3 Spectrometer status report

In OPUS, it is possible to save a complete report about the current spectrometer status. This report can be sent to Bruker service for remote fault diagnostics. Proceed as follows:

- Click the OPUS status light ().
➤ The *Instrument Status* dialog opens.
- To send the report click the *Send Report* button.
➤ The report is sent by e-mail to opusreports@brukeroptics.



This function requires an e-mail program installed on your computer and an e-mail account to be set up. In addition, your spectrometer needs to be connected to a network computer.

10.4 *Instrument Status* dialog in OPUS

The *Instrument Status* dialog in OPUS shows the status of each hardware component, and the status of the instrument test for the measurement channel currently used.

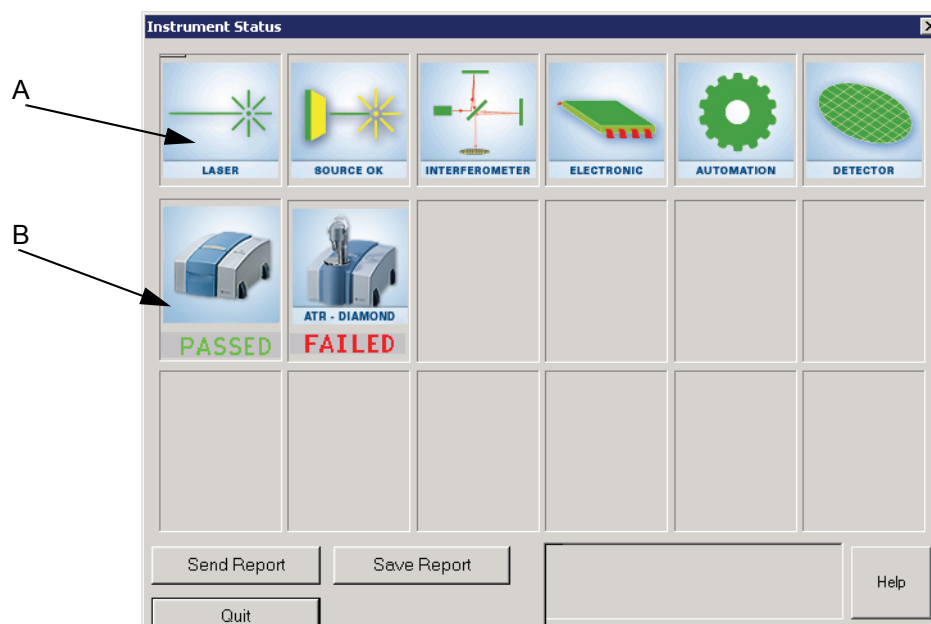


Figure 10.1: OPUS *Instrument Status* dialog

- A) Status of hardware components, e.g. IR source, laser etc. is displayed. The status can be as follows:

Status	Definition
	OK (green): component is ok
	Warning (yellow): the exact meaning depends on the specific component; in case of the source a warning means: • source lifetime nearly reached (measuring is still possible) • source warms up (measuring not possible)
	Error (red): Component is defective (measuring not possible)

Table 10.9: Hardware status

- B) The second row of icons refers to the current measurement channel¹ used, and indicates the result of the last instrument test performed. The results can be as follows:

Status	Definition
	INACTIVE (yellow): the single tests of the particular test category are disabled
	PASSED (green): instrument test configured or passed, test is still valid

Table 10.10: Status of measuring channel

1. The measurement channel shown in table 10.10 is an example. The icon can vary and depends on the spectrometer type connected.

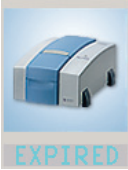
Status	Definition
	EXPIRED (light blue): instrument test validity period has expired
	FAILED (red): last instrument test has failed

Table 10.10: Status of measuring channel

10.4.1 Diagnostics of hardware components

To perform a fault diagnostics for a particular hardware component proceed as follows:

1. On the *Instrument Status* dialog, click the icon which indicates an error or warning.
➤ The *Instrument Status Message* dialog opens.
2. Click the *Service Info* button.
➤ The firmware diagnostics page of the respective spectrometer hardware component opens. This page contains all relevant information about the current operating state of the respective hardware component.

i You can send the diagnostics pages of all hardware spectrometer components as full report to Bruker service for remote fault diagnostics (see chapter 10.1.1).

10.5 Diagnostics via instrument or performance test

The instrument or performance test checks whether the spectrometer achieves the performance specified. The difference of these two types of test procedures are as follows:

	Instrument test	Performance test
The test procedure includes the following single tests:	• signal-to-noise test • deviation from 100% line • wavenumber accuracy test	• signal-to-noise test • deviation from 100% line
Number of test measurements:	11	1
Test duration:	about 5 minutes	about 1 minute
When is the test performed?	• automatically when the PC and ALPHA communicate with each other for the very first time • automatically when you use a sampling module or an ATR crystal plate for the very first time • manually, i.e. you have to start the instrument test yourself after the defined instrument test validity period has expired i The factory-set validity period is 7 days.	• automatically after OPUS has been opened again • automatically after ALPHA has been accessed again • automatically after the sampling module or the ATR crystal plate has been substituted by another sampling module or another ATR crystal plate

Table 10.11: Instrument or performance test features

10.5.1 Performance test does not start automatically

Possible cause	Troubleshooting
The sampling module is not properly attached to the basic module.	Attach the sampling module properly to the basic module (chapter 2.2).
The ATR-crystal plate is not installed properly.	Install the ATR-crystal plate properly (chapter 4.5).

Table 10.12: Performance test does not start automatically

10.5.2 Performance test failed

The performance test procedure includes the following tests:

- Signal-to-noise test
- 100% line test

If the performance test failed, an OVP - PQ Test Protocol (as pdf file) is displayed. The PDF files with the PQ test protocol are saved in the <OPUS\Validation\Reports> directory.

OVP - PQ Test Protocol

Company:	Bruker Optik GmbH		
Operator:	Default		
Instrument Type:	Alpha Sample Compartment RT-DLaTGS		
Optics Configuration:	Sample Compartment with: MIR, KBr, RT-DLaTGS [Internal Pos.1]		
Accessory:	None		
Instrument Serial Number:	0123456789		
Instrument Firmware Version:	0.091a Mar 29 2010		
OPUS/DB Version:	20100322 / 7.0		
Overall Test Result:	FAIL		
Test expires:	Test failed		
Test Date/Time:	2010/04/04 13:58:14		
Test Spectra Path:	C:\OPUS_ALPHA\Validation\Data\20100404\135814		
Comment:	AAR_TEST		

Signal to Noise Test			
Minimum Signal to Noise Limit:	1000	Signal to Noise measured:	1674

100% Line Test			
Maximum 100% Line Deviation:	0.5	Measured 100% Line Deviation:	0.96

Overall Test Result = FAIL

Figure 10.2: Test protocol of a failed performance test

10.5.2.1 Possible causes and troubleshooting

Possible cause	Troubleshooting
As the performance test performs only one single measurement, this test may fail under certain circumstances. This is not a spectrometer problem or malfunction.	In this case, we recommend to perform an instrument test which is going to be passed. - Click the OPUS status light. - On the dialog that opens, click the icon displayed for the current measurement channel. - Wait until the test is completed (duration: about 5 minutes). If the problem still persists and the instrument test fails as well, see chapter 10.5.3.
During the performance test, the spectrometer was exposed to strong vibrations.	The spectrometer must not be exposed to strong vibrations. Perform an instrument test as follows: - Click the OPUS status light. - On the dialog that opens, click the icon displayed for the current measurement channel. - Wait until the test is completed (duration: about 5 minutes).
The spectrometer operates in too low an ambient temperature, so that the IR source warm-up phase of 7 minutes is not sufficient.	Wait until the IR source has reached its operating temperature. Then, perform an instrument test as follows: - Click the OPUS status light. - On the dialog that opens, click the icon displayed for the current measurement channel. - Wait until the test is completed (duration: about 5 minutes).
Transmission module: there is a sample or another object in the sample compartment.	Remove the sample or object. Then, perform an instrument test as follows: - Click the OPUS status light. - On the dialog that opens, click the icon displayed for the current measurement channel. - Wait until the test is completed (duration: about 5 minutes).
ATR module: ATR crystal is dirty or there is a sample on the ATR crystal.	Remove the sample or clean the ATR crystal (chapter 4.10). Then, perform an instrument test as follows: - Click the OPUS status light. - On the dialog that opens, click the icon displayed for the current measurement channel. - Wait until the test is completed (duration: about 5 minutes).

Table 10.13: Performance test failed

10.5.3 Instrument test failed

The instrument test procedure includes the following tests:

- Signal-to-noise test
- 100% line test
- Wavenumber accuracy test

If the performance test failed, an OVP - PQ Test Protocol (as pdf file) is displayed. The PDF files with the PQ test protocol are saved in the <OPUS\Validation\Reports> directory.

OVP - PQ Test Protocol			
Company:	Bruker Optik GmbH		
Operator:	Default		
Instrument Type:	Alpha Sample Compartment RT-DLaTGS		
Optics Configuration:	Sample Compartment with: MIR, KBr, RT-DLaTGS [Internal Pos.1]		
Accessory:	None		
Instrument Serial Number:	0123456789		
Instrument Firmware Version:	0.087 Jan 25 2007		
OPUS/DB Version:	20070129 / 6.5		
Overall Test Result	FAIL		
Test expires:	Test failed		
Test Date/Time:	2007/02/02 09:00:55		
Test Spectra Path:	C:\OPUS_ALPHA\Validation\Data\20070202\090055		
Comment:			
Signal to Noise Test			
Minimum Signal to Noise Limit:	1000	Signal to Noise measured:	1799
100% Line Test			
Maximum 100% Line Deviation:	0.5	Measured 100% Line Deviation:	0.24
Wavenumber Accuracy Test - Polystyrene			
Sample Material:	Polystyrene		
Specified Peak:	1601.00 cm ⁻¹	Maximum Deviation:	0.50 cm ⁻¹
Measured Peak:	1601.80 cm ⁻¹	Measured Deviation:	0.80 cm ⁻¹
Overall Test Result = FAIL			
Date and Signature		Date and Signature	

Figure 10.3: Test protocol of a failed instrument test

10.5.3.1 Possible causes and troubleshooting

	Possible cause	Troubleshooting
All three tests failed	Defective hardware component	Detect the defective component: - Click the OPUS status light. - The <i>Instrument Status</i> dialog opens. i If the laser or the interferometer, electronics, automation or the detector is defective, contact Bruker Service. If the IR source is defective, replace the IR source (chapter 9.3).
- All three tests failed - Only the S/N or 100% line test failed	Transmission module: an object (e.g. sample) in the spectrometer sample compartment obstructs the IR beam	Remove the object and repeat the instrument test: - Click the OPUS status light. - On the dialog that opens, click the icon displayed for the current measurement channel.
	ATR module: ATR-crystal plate not installed properly	Install the ATR-crystal plate properly (chapter 4.5).
	Dirty ATR crystal, or sample material on crystal	Clean the crystal (chapter 4.10).
	Damaged ATR crystal	- Order a new crystal plate with Bruker. - After receipt, replace the damaged crystal plate (chapter 4.5).
	Opaque KBr windows, which causes the infrared transparency to be substantially reduced	- Order new KBr windows with Bruker. - After receipt, replace the opaque windows (chapter 9.5).
	Non of the possible causes apply	Contact Bruker Service (chapter 1.5).
Only the wavenumber accuracy test failed	Laser wavenumber must be recalibrated	- On the <i>Validation</i> menu, select the <i>Setup Instrument Test</i> command. - On the <i>Setup OVP</i> dialog, click the <i>Measure LWN</i> button.

Table 10.14: Instrument test failed

	Possible cause	Troubleshooting
	Non of the possible causes apply	Contact Bruker Service (chapter 1.5).

Table 10.14: Instrument test failed

10.6 Spectrometer problem indicated by an error message in OPUS

Error messages which refer to the spectrometer can be displayed in OPUS as follows:

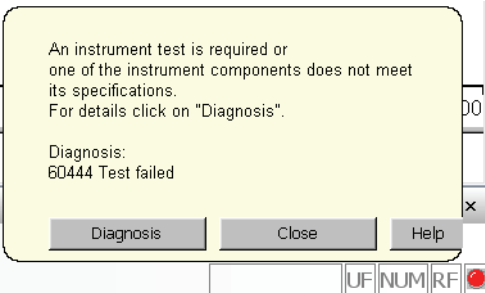
Instrument Status Message dialog^a:	
Yellow information bubble^b:	
Message window:	

Table 10.15: Display options in case of OPUS error messages

- The dialog opens if you have clicked a hardware component icon in the *Instrument Status* dialog (chapter 10.4).
- The information bubble is displayed on the lower right end of the OPUS interface.

i In case of a spectrometer problem indicated by an error message, the OPUS status light is yellow or red.

10.6.1 Troubleshooting in case of problems indicated by OPUS error messages

Error message in OPUS	Possible cause	Troubleshooting
<i>An instrument calibration is required.</i>	Factory setting, displayed every 6 months in the form of a yellow information bubble	On the information bubble, click the <i>Calibrate</i> button. i The purpose of a regular recalibration is to ensure that the measurements performed are based on a correct laser wavenumber.
<i>Device not ready.</i>	IR source has not yet reached operating temperature	Wait until the IR source is ready to operate. i The IR source warm-up takes about 7 minutes. As soon as the IR source has reached its operating temperature the spectrometer status indicator becomes green.
<i>End of average lifetime is nearly reached, spare part will be required.</i>	Specified lifetime of the IR source nearly reached	• Order new IR source with Bruker. • After receipt replace the used IR source (chapter 9.5). i As long as the lifetime is not expired, measuring is still possible. To turn the OPUS status light green again, click the <i>Ignore</i> button on the <i>Instrument Status Message</i> dialog (table 10.15 on page 187)
<i>Front sample not connected.</i>	ATR crystal plate not or not properly installed	Install ATR crystal plate properly (chapter 4.5).
<i>Humidity out of range.</i>	Air humidity inside the spectrometer is too high	Replace desiccant (chapter 9.4).
<i>No accessory connected.</i>	Sampling module not properly attached to basic module	Properly attach the sampling module to the basic module (chapter 2.2).
<i>Source is broken.</i>	Defective IR source	• Order new IR source with Bruker. • After receipt replace the defective IR source (chapter 9.3).

Table 10.16: Error messages in OPUS

Error message in OPUS	Possible cause	Troubleshooting
<i>Source is not connected.</i>	Plug of the IR source not or not connected properly to the female connector	Check whether the plug is connected properly (chapter 9.3).
<i>Standby mode activated.</i>	Spectrometer is in stand-by mode ➤ Spectrometer status indicator is green.	Deactivate stand-by mode: ☞ Press the SBY/RES key on the spectrometer rear side.
<i>Temperature out of range.</i>	Too high a temperature inside the spectrometer due to too high an ambient temperature	Operate the spectrometer only within the specified temperature range between 18 and 35°C
<i>Wrong accessory inserted, unable to run OVP tests. Cannot run OVP tests, please check OVP Setup.</i>	The instrument test is started by the OPUS planer command. But the type of sampling module defined in the OPUS planer is not identical to the one currently attached to the basic module.	The sampling module defined in the OPUS planer must be identical to the one currently attached to the basic module.
	The instrument test is started via the OPUS <i>Validation</i> menu. But the test channel defined in OPUS is not identical to the currently attached sampling module.	Select the correct test channel: ☞ On the <i>Validation</i> menu, select the <i>Setup Instrument Test</i> command. ☞ In the dialog displayed, select the correct test channel.
If an error message is displayed which is not listed above:		Contact Bruker Service.

Table 10.16: Error messages in OPUS

10.7 Checking communication between spectrometer and PC

1. Switch on spectrometer.
2. Wait about 1 minute to allow the spectrometer to boot.
 - *The spectrometer is ready to operate when the spectrometer status indicator is green.*
3. Start the internet browser.
 - *The internet browser must not be off-line. In case of the Microsoft Internet Explorer the off-line mode is activated if the Offline Mode command on the File menu of the browser is checked.*
 - *Ensure that the internet browser does not use a proxy server, or at least not in case of addresses of direct access in the 10.10.x.x.-range. If you use the Microsoft Internet Explorer, you can check this by selecting the Internet Options command on the Extra browser menu. Click the Connections tab and click the Settings button in the LAN-Settings group field.*
4. Enter the spectrometer IP address into the browser address field as follows: *http://10.10.0.1/diag.htm*.
5. Click the *Enter* button.
 - *Now, the Internet Explorer should display the firmware diagnostics page for the ALPHA spectrometer (figure 10.4). If the Internet Explorer shows a blank page, this indicates that the PC cannot access the spectrometer. A wrong spectrometer IP address may be the cause of the problem.*
6. Close the Internet Explorer.

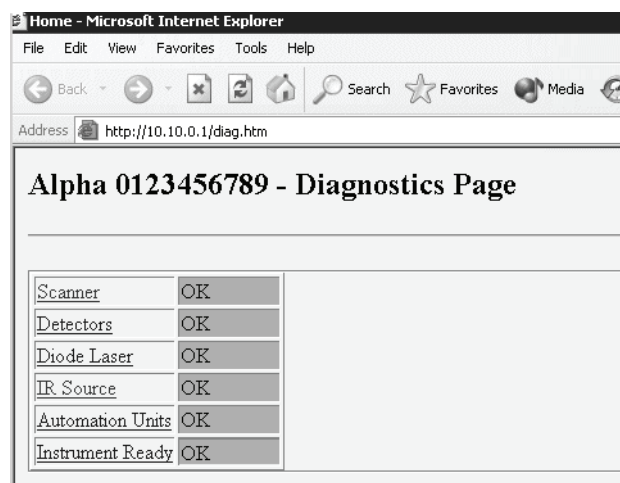


Figure 10.4: Firmware diagnostics page for the ALPHA spectrometer

A Specifications

A.1 General

Parameter	Specification
Dimension	Transmission module: 22 x 33 x 26 cm (w x d x h) ATR module: 22 x 31 x 14 cm (w x d x h)
EMC regulations	Complies with EN/IEC 61326/01.2006 (<i>Electrical equipment for measurement, control and laboratory use - EMC requirements</i>)
Environmental conditions ^a	• Operational temperature range: 18 - 35°C • Temperature variation: max. 1°C/ per hour and max. 2°C per day • Humidity (non-condensing): less than 80% (relative humidity) • Installation site: in a closed room, max. 2000m above sea level • Pollution degree: 2, complies with 61010-1 or IEC 60664-1
Overvoltage category ^a	II, complies with 61010-1 or IEC 60664-1
Power consumption of spectrometer	Standard configuration (with external power supply unit): Maximum: 48 W
Power supply	Standard configuration (with external power supply): • AC Input: 100 - 240 V, 2,5 A; 50 - 60 Hz • DC Output: 24V DC $\pm$ 10%, 3,75 A • safety extra low voltage circuit (i.e. either double or reinforced isolated from primary supply circuits) with functional earthing at the negative terminal in the external power supply
Safety regulations	• Complies with EN/IEC 61010-1/08.2002 (<i>Safety requirements for electrical equipment for measurement, control and laboratory use</i>) • Complies with EN 60825-1/10.2003 (<i>Safety requirements for laser equipment</i>)
Weight	7 kg

Table 11.1: Specifications - General

a. The values indicated are target values which have an effect on the performance of the entire configuration. In case of non-observance the functioning and safety of the configuration may be affected adversely.

A.2 Performance

Parameter	Specification
Photometric accuracy	0.1% T
Resolution	better than 2 cm ⁻¹ ; optionally better than 0.8 cm ⁻¹
Spectral range	• 375 - 7,500 cm⁻¹, with standard KBr beam splitter • 500 - 6,000 cm⁻¹, with optional high-humidity ZnSe optics
Wavenumber accuracy	Better than 0.05 cm ⁻¹ @ 1,576 cm ⁻¹
Wavenumber reproducibility	Better than 0.01 cm ⁻¹

Table 11.2: Specifications - Performance

A.3 Optics

Parameter	Specification
Beamsplitter	Multi-layer coating on Quarz
Design	Sealed optics housing, convective desiccating
Detector	High resolution DTGS detector
Interferometer	Mechanical ROCKSOLID interferometer, permanently aligned
Scanner	Mechanical frictionless bearing (no compressed air required), 4 software-selectable mirror velocities
IR source	Air-cooled, 12V, 20W

Table 11.3: Specifications - Optics

A.4 Electronics

Parameter	Specification
A/D converter	100 kHz with 24 bit dynamic range
Connectors	Ethernet
Laser diode	The spectrometer is a CLASS 1 LASER PRODUCT and contains a laser diode (850 nm).

Table 11.4: Specifications - Electronics

A.5 ATR crystal types

Crystal type	To be used for	Spectral range [cm ⁻¹]	Refraction index
Diamond	- all kinds of sample material - hard and sharp-edged samples	7500 - 350	n = 2.43
Germanium (Ge)	samples with a high absorptivity, especially carbon-filled samples (e.g. O-rings, polymer)	5500 - 550	n = 4.01
Zinc selenide (ZnSe)	- all kinds of sample material - exceptions: acids (pH -value <5), bases (pH-value >8) and sharp-edged samples	>500	n = 2.43

Table 11.5: Specifications - Crystal types

A.6 Window material

	Size	Material	Chemical properties
Standard	25 x 4 mm	Potassium bromide (KBr)	Soluble in water, alcohol, and glycerine NOTE Hygroscopic! ➤ Avoid any contact to humidity.
Option	25 x 4 mm	Zinc selenide (ZnSe)	Soluble in strong acids and HNO ₃ CAUTION Toxic! ➤ Observe the safety data sheet.

Table 11.6: ALPHA - Window material

B Spare Parts

Part No.	Description
1005855	IR source with housing, pre-adjusted
1011885	Desiccant bags
1005847	2x replacement window (KBr), frame-mounted
1006041	2x replacement window (ZnSe), frame-mounted

Table B.1: Spare parts

C Firmware Update

The spectrometer firmware needs to be updated in order to make new features available. The update is performed by using the FCONF program (Firmware Configuration Tool). This program performs all the necessary actions automatically.

The FCONF program allows:

- updating the firmware
- restoring a previous firmware version
- backing up the current firmware version
- initializing the firmware (for service purposes only!)
- running a custom script (for service purposes only!)

Typically, firmware updates are delivered on CD or by e-mail. The delivered firmware update performs all the actions required to properly substitute the existing firmware version.

i Before the firmware update, restoration or initialization starts, backup copies of the previous firmware version are generated automatically, and stored in the *Backup* program folder.

C.1 Firmware update on CD

1. Start the FCONF program directly from CD.
2. Double click the *fconf.exe* file.
3. Proceed as described in chapter C.3.

C.2 Firmware update via E-mail

1. Store the delivered files into a temporary directory.
2. Start the FCONF program by double-clicking the *fconf.exe* file
3. Proceed as described in chapter C.3.

The following dialog opens:

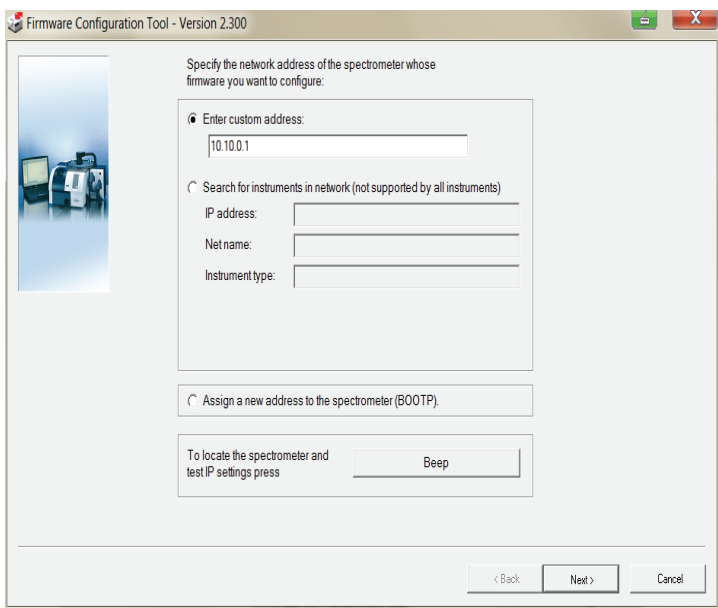


Figure E.1: Program FCONF - Firmware Update

C.3 Performing firmware update

1	Specify the network address of the spectrometer whose firmware you want to configure: ☒ Enter custom address: 10.10.0.1 ☐ Search for instruments in network (not supported by all instruments) IP address: Net name: Instrument type: ☐ Assign a new address to the spectrometer (BOOTP). To locate the spectrometer and test IP settings press Beep	• Activate the <i>Enter custom address</i> option button. • Enter the IP address in dotted notation. i In case of a stand-alone spectrometer configuration, the default IP address is 10.10.0.1. If the spectrometer is directly integrated into a network, it must have a different IP address. In this case, use the IP address labeled on the spectrometer rear side. The operating company is obliged to write the IP address on the label provided on the spectrometer rear side.
2	• Check whether the spectrometer desired is recognized by the firmware. • Click the <i>Beep</i> button. If the spectrometer beeps three times in rapid succession (once in case of ALPHA), it has been recognized by the firmware. • Click the <i>Next</i> button.	

Table C.1: Updating firmware

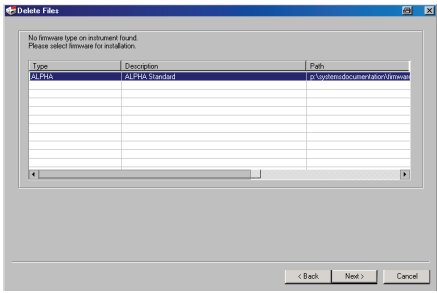
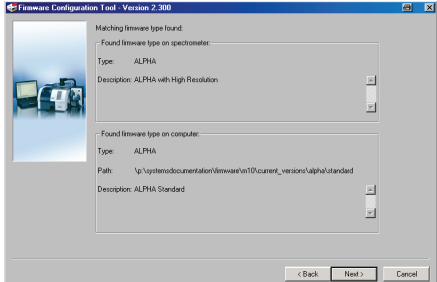
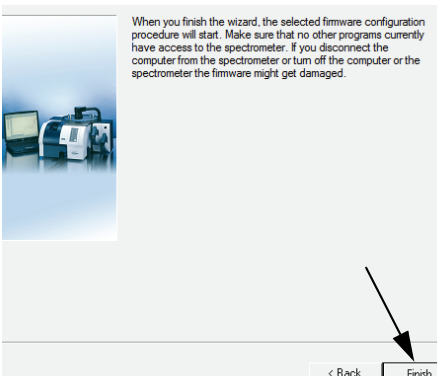
3	Select the desired firmware configuration procedure. ☒ Update firmware update.fcs ☐ Restore previous firmware ☐ Backup current firmware ☐ Initialize firmware ☐ Run custom script Description UPDATE SCRIPT Updates the current firmware. Creates a backup of the current firmware before performing the update. Resets the spectrometer.	• Activate <i>Update firmware</i> option button. • Click the <i>Next</i> button.
4		i In case of firmware-versions < 1.3 the firmware type has to be selected manually. • Select <i>Firmware Type</i> (selected field is highlighted blue). • Click the <i>Next</i> button.
5		i For firmware-versions ≥ 1.3 the firmware type is selected automatically. • Click the <i>Next</i> button
6	Select the run folder. The Firmware Configuration Tool will store information about the current run (backup data, etc.) in this folder. Run folder: c:\programdata\bruker\conf\run2 Browse...	• Either accept the default directory setting or define a different path by clicking the <i>Browse</i> button. • Click the <i>Next</i> button.
7	When you finish the wizard, the selected firmware configuration procedure will start. Make sure that no other programs currently have access to the spectrometer. If you disconnect the computer from the spectrometer or turn off the computer or the spectrometer the firmware might get damaged. 	Click the <i>Finish</i> button to start the update. i The update procedure may take several minutes, depending on the available bandwidth and the amount of files to be updated.

Table C.1: Updating firmware

During firmware update

- During firmware update, a log window is displayed showing all actions performed by the FCONF program.
- The log-file is stored in the same directory as the backup files.
- If an error occurs during the update procedure, the FCONF program terminates the procedure and recommends to restore the previous firmware version.

After firmware update has finished

- If the firmware update has finished, the FCONF program resets the spectrometer. The log window reads: *Resetting the spectrometer... done*.
- After a successful spectrometer initialization, the firmware version is displayed in the log window.

C.4 Restoring a previous firmware version

Restoring a previous firmware version is only possible if a firmware update has been performed first.

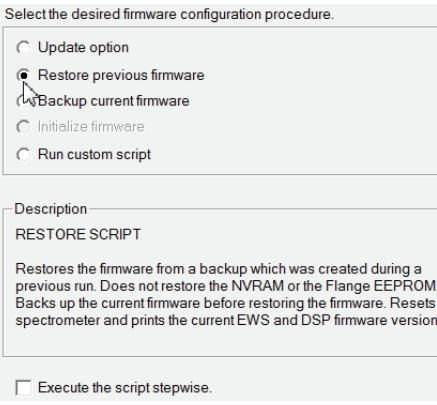
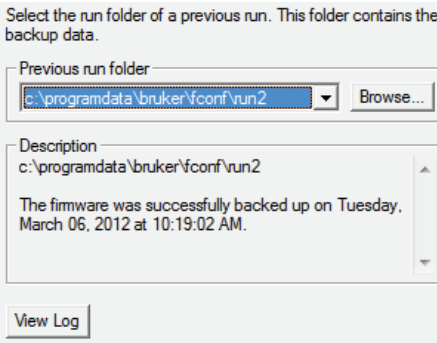
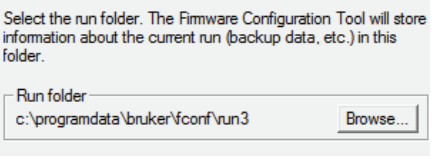
1		• Activate the <i>Restore previous firmware</i> option button. • Click the <i>Next</i> button.
2		• Select the directory containing the backup information of the last firmware version (<i>Previous run folder</i>). By default, this directory is displayed automatically. i If you click the <i>View Log</i> button, a log window is displayed. The window contains detailed information about the last update including errors, warnings or other irregularities. • Click the <i>Next</i> button.
3		Specify a directory for the backup files generated. i It is recommended to accept the directory set by the FCONF program. • Click the <i>Next</i> button.
4	In the next dialog, click the <i>Finish</i> button. ➤ The previous firmware version is going to be restored.	

Table C.2: Restoring previous firmware version

C.5 Backing up the current firmware version

1	Select the desired firmware configuration procedure. ☐ Update firmware ☐ Restore previous firmware ☒ Backup current firmware ☐ Initialize firmware ☐ Run custom script Description BACKUP SCRIPT Creates a backup of the current firmware (including the NVRAM and Flange EEPROM). ☐ Execute the script stepwise.	• Activate the <i>Backup current firmware</i> option button. • Click the <i>Next</i> button.
2	Select the run folder. The Firmware Configuration Tool will store information about the current run (backup data, etc.) in this folder. Run folder c:\programdata\bruker\fcconf\run3 Browse...	• Specify a directory for the backup files generated. i It is recommended to accept the directory set by the FCONF program. • Click the <i>Next</i> button.
3	In the next dialog, click the <i>Finish</i> button. ➤ The backup starts.	

Table C.3: Backup current firmware version

D Glossary

Aperture The aperture wheel contains accurately etched holes to admit light into the spectrometer.

ATR measuring ATR is the abbreviation of Attenuated Total Reflection and is also known as Internal Reflection Spectroscopy (IRS).

Principle:

- IR light is directed through an internal reflection element (ATR crystal) which requires a high refraction index.
- Due to the internal reflection in the ATR crystal the IR light is repeatedly reflected on the crystal surface. The number of reflections depends on the refraction index as well as on the crystal length and thickness, as well as on the incidence angle of the IR light.
- If a sample gets in contact with the crystal, it absorbs IR light at each reflection point of the crystal. Therefore, a good contact between sample and crystal has to be ensured.
- The IR light absorbed by the sample is missing in the reflected beam. Thus, the reflected beam changes its intensity, which is recorded as absorption spectrum, depending on the wavenumber.

The sample needs to have a lower refraction index than the crystal to achieve total internal reflection.

Advantages of ATR measuring method:

- Non-destructive analysis method, i.e. the sample has not to be destructed for analysis
- No sample preparation required
- Only a small number of sample quantity required
- Suited for samples which are actually too thick or highly absorbing for transmission measurement
- Due to the multiple reflection of the IR light, the sample absorbs more light than in case of transmission measurement

Disadvantages of ATR measuring method:

- Falsified measurement result due to possible contaminations caused by previous sample material, solvents or fingerprints.
- In case of multi-layer materials only the first layer will be spectroscopically analyzed.
- The quality of the measurement result substantially depends on the contact pressure between sample and ATR crystal during measurement.

ATR spectra	In case of ATR spectra the intensities of the spectral features are of lower absorbance than the corresponding features in a transmission spectrum, especially, in the high wavenumber (short wavelength) region of the spectrum. The intensity is related to the penetration depth of the evanescent wave into the sample. This depth depends on the refractive index of the crystal and sample, and on the wavelength of the IR radiation. The relatively thin depth of penetration of the IR beam into the sample creates the main benefit of ATR sampling. This is in contrast to traditional FT-IR sampling by transmission where the sample must be diluted with IR transparent salt, pressed into a pellet or pressed to a thin film, prior to analysis to prevent totally absorbing IR bands.
Beamsplitter	The beamsplitter splits the incident beam into two separate light waves. Generally, the beamsplitter is a half-mirrored substrate that reflects and transmits approximately equal portions of the incident radiation.
Calibration	Calibration comprises the entirety of working steps to define the relation between analyte concentrations and the corresponding standard measurement values. Under specified conditions the analyte concentrations are indicated by an instrument, measuring system, or values which have been evaluated when measuring substances or reference material.
Collimated light	Light for which all flux lines are parallel.
Cross-over cable	The cross-over cable includes a RJ45 plug on each cable end. In one plug, however, the pairs leading the signal are swapped (crossed). That means, one plug swaps the transmission data with the received data to enable data exchange when one terminal device is directly connected to another. If the transmission and received data were not swapped at one cable end, a connection could not be accomplished.
Detector	A detector converts incoming light into an electrical signal.
Diffuse reflection	In case of diffuse reflection the light scatters evenly at all angles, from one particular point of reflection. Diffuse reflection techniques are used to analyze rough-surfaced solid samples.
DRIFTS	<u>D</u>iffuse <u>r</u>eflectance <u>i</u>nfrared <u>f</u>ourier <u>t</u>ransform <u>s</u>pectroscopy is an analyzing technique where infrared light hits on a rough or mat surface. The infrared light penetrates the sample surface, and part of the light is reflected over all angles (diffusely reflected) due to the rough sample surface.

Ethernet	Ethernet is the most widely installed Local Area Network (LAN). A LAN is a network of interconnected workstations sharing the resources of a single processor or server within a relatively small geographic area (i.e. an office building). Ethernet is a set of hardware and signaling standards used for LANs. The most commonly installed systems are 10/100/1000 BaseT.
FIR	Far InfraRed; covers the spectral range from 400 to about 5cm^{-1} wave numbers. It is about vibrational frequencies of both backbone vibrations of large molecules, as well as fundamental vibrations of molecules that include heavy atoms (e.g. inorganic or organometallic compounds) and pure rotational vibrations.
Fourier Transform	Fourier Transform (FR) spectroscopy is a mathematical transformation method used to convert an interferogram into an infrared spectrum. Essentially, Fourier Transform decomposes or separates a waveform or function into sinusoids of different frequencies. All these different frequencies together sum to the original waveform. Fourier Transform identifies or distinguishes the different frequency sinusoids and their respective amplitudes.
Frequency	Frequency is the number of occurrences of a repeating event per unit time. In equations, frequency is denoted by the Greek letter ν. The following equation applies to the frequency of a wave ν: $\nu = \frac{c}{\lambda}$, with c being the phase velocity of the wave in the respective medium, and λ (lambda) being the wavelength. The wavelength is no timely but local parameter. Any kind of time-restricted vibration process, even in the form of a sinusoid, always represents an overlapping of several frequencies.
FT-IR	The <u>F</u>ourier <u>T</u>ransform <u>I</u>nfra<u>R</u>ed Spectroscopy is a method to obtain infrared spectra of a sample using an interferometer. A Fourier transform is performed on the resulting interferogram to calculate the spectrum. In case of an FT-IR spectrometer infrared light emitted from a source is directed to an interferometer which modulates the light. Leaving the interferometer the light passes through the sample compartment (and also the sample) and is focused onto the detector. The signal measured by the detector is called the interferogram.

Fuse	Fuses and circuit protectors protect electrical devices and components from overcurrents and short circuits. This is achieved automatically by the melting of an internal melting element through which a fault current flows. Both components are rated so as to reliably interrupt current flow when it reaches a predictable magnitude for a fixed time period. For all practical purposes, a fuse or protector is invisible to a properly operating circuit. A fuse switches or blows when a high short-circuit or fault current causes an irreversible, physical separation of the melting element. As the weakest element in a circuit, the fuse/protector resistivity increases, eventually causing the melting point of the melting element to be reached. The element then changes state from a solid to a liquid (or even gas) within a defined time period.
Gateway address	A gateway is a kind of connecting point between different networks, which controls data traffic of the respective network. A gateway address is the address of a particular gateway within a network.
Instrument test	The instrument test checks whether the spectrometer achieves the specified performance. The instrument test contains the following single tests: signal-to-noise, deviation from 100% line and wavenumber accuracy. The instrument test is performed when: • a communication connection between spectrometer and computer has been established • the sampling module or ATR crystal plate has been exchanged • the test validity period has been expired (manual start required) The instrument test lasts about 5 minutes. Exactly 11 measurements are performed during the instrument test, which has a validity period of 7 days (factory-set).

Interferogram The interferogram is a plot of detector signal as a function of optical path length difference. Spectrometers are equipped with a broadband light source, which yields a continuous infinite number of wavelengths. The interferogram is the continuous sum, i.e. the integral of all the interference patterns produced by each wavelength. This results in the intensity curve as function of the optical retardation.

At the zero path difference of the interferometer ($\Delta x=0$) all wavelengths undergo constructive interference and sum to a maximum signal. As the optical retardation increases different wavelengths undergo constructive and destructive interference at different points, and the intensity therefore changes with retardation. For a broadband source, however, all the interference patterns will never be simultaneously in phase except at the point of zero path difference, and the maximum signal occurs only at this point. This maximum in the signal is referred to as the centerburst. Dispersive elements detected in the optical path can cause chirping in the centerburst.

Interferometer An interferometer detects interferences, that means overlappings of light waves. Inside the interferometer a light beam is splitted into two light waves (beams) by the beam splitter. These two light waves pass through optical paths of different lengths, are reflected by additional mirrors, and finally recombined.

If you change the optical path of one of the two light waves, e.g. by moving one mirror, the phases of the two light waves displace against each other. When recombining the light waves, interferences occur.

The result is an interference pattern (stripes or rings) which has been written by the light waves. This pattern is determined by the difference of the optical path lengths which the single light waves have passed before being recombined.

IP address An IP address is the network address of a workstation or network. IP addresses consist of 4 number fields separated by dots. Each number field represents 1 byte. Values can be between 0 and 255. The numbers on the left of the string define the network, the numbers on the right define the individual workstation or Network Interface Card (NIC).

IR source In the mid and near infrared an IR source is used which emits infrared light.

KBr pellet The pellet is prepared by grinding the sample, diluting it in KBr and pressing it into a transparent disc. The KBr pellet is directly placed into the infrared beam for analysis.

Laser	The laser is a coherent source of monochromatic radiation. It is an essential part of the interferometer system. Due to laser light interference the position of the movable mirror, and the data sampling positions are determined.
Micrometer (µm)	Micrometer is a length unit of 10^{-6} meter, which is equal to a thousandth of a millimeter.
Optical path	The optical path is the distance of the light passing through the spectrometer. The distance between two points in the light beam is calculated in longitudinal direction, and multiplied by the refraction index of the medium.
OQ test	The OQ (<u>O</u>perational <u>Q</u>ualification) is a test category used to validate the spectrometer. This test checks the current instrument performance and compares it to the one specified. The OQ test consists of several single tests and is performed within a defined time interval. Generally, the OQ test protocol runs during final testing before an instrument is delivered to the customer, after each major repair, exchange of optical components which may influence the instrument performance, and on a regular yearly or semi-yearly basis after maintenance.
Performance test	The performance test checks whether the spectrometer achieves the specified performance. The performance test contains the following single tests: signal-to-noise and deviation from 100% line. In case of the ALPHA spectrometer the performance test starts when: • the OPUS spectroscopy software has been started • the sampling module has been exchanged • the ATR crystal plate has been exchanged The performance test lasts about 1 minute. Exactly 1 measurement is performed during the performance test.
PQ test	The PQ (<u>P</u> erformance <u>Q</u> ualification) is a test category used to validate the spectrometer. The test evaluates the performance and proper function of the spectrometer. The PQ test consists of several single tests and is performed within a defined time interval. Generally, the test is intended to be performed once a day.
Reflectance	Reflectance describes the ratio of the radiation energy reflected from a surface to the radiation energy incident on the surface. It is the percentage or part of the incident light (= 100%), which an object reflects again.
Reflection	In case of reflection light incidents on a sample and is converted to radiation energy. This kind of energy is subsequently reflected by the sample surface at a defined wavelength.

Relative humidity	Relative humidity is the amount of water vapor in the air, which can be between 0 and 100%.
Resolution	Resolution in FT-IR is determined by the maximum optical path difference reached in the measurement of an interferogram. The spectral resolution corresponds to the minimum possible spectral distance between 2 adjacent spectral lines which can be resolved by the spectrometer. It is produced by the instrumental spectral line shape and the line width of the sample's absorption. The instrumental line shape is reciprocally related to the travel distance of the movable mirror in the interferometer, and also influenced by the apodization function and divergence in the interferometer. The higher the spectral resolution (smaller values), the higher the noise in the spectrum.
Sample	A sample is a special kind of substance which is put into the spectrometer sample position and measured.
Signal-to-noise ratio	The signal-to-noise ratio describes the quality of a wanted signal which has been superimposed by noise. It outlines the ratio between the average performance of a wanted signal emitted by a signal source, and the average performance of noise.
Spectrometer	Spectrometer derives from the Latin word <i>specere</i> meaning image, and the Greek word <i>metron</i> meaning to measure. A spectrometer uses some sort of mechanical or electrical detection device to obtain the infrared spectrum of a sample. It contains a source of infrared radiation, a sample compartment to allow the radiation to interact with a sample, a detector for radiation, and units to determine and display the intensity of radiation.
Spectrum	A spectrum is an image of radiation energy dispersed into its wavelength constituents or a two-dimensional plot of radiation energy, or radiation energy ratio versus wavelength. It includes a set of absorption, reflection or transmission values which have been measured by an IR spectrometer based on defined wavelengths. The features in an infrared spectrum correlate with the presence of functional groups of molecules of a sample.
Subnet mask	The subnet mask is a network mask used to partition network addresses for efficiency and security. Subnet masks work by 'masking' less significant address bits on all workstations in the sub-network.

Transmission Transmission describes the portion of transmitted infrared light. One part of the incident light directly passes through the sample without any interactions with electrons or molecules in the sample. Another part of light is absorbed and continuously re-emitted to the same direction from which the incident light comes.

Transmission measurement In IR spectroscopy, transmission is the classical analysis method for samples. Depending on the sample material, sample preparation can be very complex in case of transmission measurement.

Principle:

- IR light transmits a sample. The sample absorbs specific wavelengths of the IR light. This kind of absorption of certain wavelengths enables to identify the sample.
- If samples are evaluated for quantitative analysis, the intensity ratio of the IR light before and after passing the sample is measured within a specific wavelength.

Advantages:

- Almost all kinds of samples can be analyzed.
- Measurement represents the entire sample.

Disadvantages:

- Depending on the sample material a complex and sometimes difficult sample preparation is required. A successful transmission measurement substantially depends on how carefully the sample has been prepared.
- Depending on the sample preparation method required, measuring in transmission may sometimes destroy the sample.
- Too much a sample quantity can cause total absorption.
- Additional accessories (e.g. liquid and gas cells) are required.

Transmittance Transmittance is the ratio between the beam energy transmitted through a sample and the beam energy incident on the surface of the sample.

USB The Universal Serial Bus (USB) is a serial bus system which allows to connect peripherals to a host computer. Devices or storage media equipped with USB can be hot plugged, and the properties of peripherals can be automatically detected.

Validation	In general, validation evidently documents that, by the use of specific laboratory investigation methods, a process or system (spectrometer) meets the previously defined requirements (acceptance criteria) in reproducible manner and for its intended analytical use. During spectroscopic validation the x-axis (frequency) and y-axis (transmission, reflection) are verified by transmission or reflectance standards which may be included in the internal spectrometer validation unit. Validation can be performed at individual time intervals by the application software.
Wavelength	The wavelength λ is the distance between two maxima on a sinusoidal wave, i.e. the distance traversed by one period of an electromagnetic wave. The wavelength depends on the refractive index of the medium which the electromagnetic wave travels through.
Wavenumber	The wavenumber $\tilde{\nu}$ is the reciprocal of the wavelength λ. The wavenumber is defined as $\tilde{\nu} = \frac{1}{\lambda_0}$ with the unit of cm^{-1}, that means the number of waves per each centimeter.

E Addresses

Bruker has an international network of branch offices and representations to ensure a customer service throughout the world. The following list includes the addresses of Bruker headquarters.

For a complete list with the addresses and telephone numbers of the Bruker branch offices and representations worldwide, refer to the internet:

<http://www.brukeroptics.com/offices.html>

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EC-DECLARATION OF CONFORMITY

The undersigned, representing the following manufacturer

Manufacturer: BRUKER OPTIK GMBH
Address: Rudolf-Plank-Straße 27, 76275 Ettlingen, Germany

herewith declares that the product

Product identification:	ALPHA
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is in conformity with the provisions of the following EC directive(s)
(including all applicable amendments)

Reference no.	Title
2004/108/EC	Directive of the European Parliament and of the Council of 15 December 2004 (Electromagnetic Interference Directive) former 89/336/EEG
2006/95/EC	Directive of the European Parliament and of the Council of 12 December 2006 (Low Voltage Directive) former 73/23/EEG

and that the standards and / or technical specifications referenced overleaf have been applied.

Ettlingen

(Place)

January 16, 2012

(Date)

(Signature)

Dr. Arno Simon, Development Manager

(Name and function of the signatory empowered to bind the manufacturer or his authorized representative)



References of standards and/or technical specifications applied for this declaration of conformity, or parts thereof :

Harmonized standards:

No.	Issue	Title
EN 61326-1:2006 Emission: Class B Immunity: Industrial Locations	October 2006	Electrical equipment for measurement, control and laboratory use - EMC requirements
EN 61000-3-2:2006 +A1:2009+A2:2009	March 2010	Electromagnetic compatibility Part 3-2: Limits - Limits for harmonic current emissions (equipment input current $\leq 16A$)
EN 61000-3-3:2008	June 2009	Electromagnetic compatibility; Part 3-3: Limits - Limitation of voltage fluctuations and flicker in low-voltage supply systems for equipment with rated current $\leq 16A$
EN 61010-1:2001 (2 nd Edition)	August 2002	Safety requirements for electrical equipment for measurement, control and laboratory use Part 1: General requirements
EN 60825-1:2007	May 2008	Safety of laser products Part 1: Equipment classification, requirements and user's guide

Other standards and/or technical specifications:

No.	Issue	Title	Parts (1)

Other technical solutions, the details of which are included in the technical documentation or the technical construction file:

.....

Other references or information required by the applicable EC directive(s):

.....

(1) Where appropriate, the applicable parts or clauses of the standard or the technical specification shall be referenced.